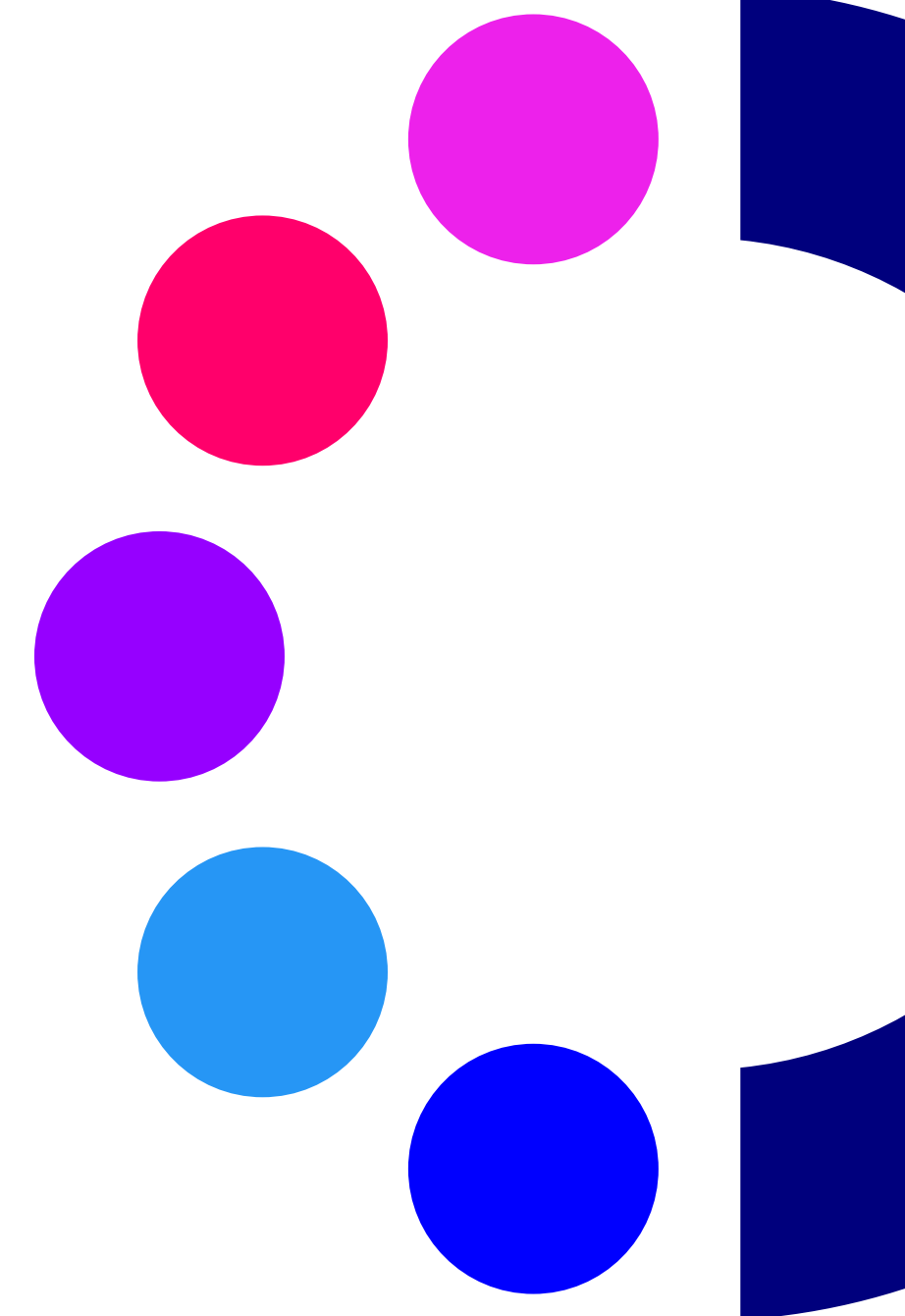


Real Time Performance Diagnosis and Machine Learning

Partha Kanuparth, Alolita Sharma (AWS)
All Things Open 2019



About

- Partha: Principal Engineer, AWS
- Alolita: Principal OS Technologist, AWS
- This is the work of a team:

Performance Diagnosis: Adithya Chandra, Joydeep Sinha, Karthik Kumarguru, Palash Hedau, Prachi Kumar, Balaji Kannan, Bhaven Avalani, Kiran Dinakurthi, Partha Kanuparth

Machine Learning: Jiang Lai, Jin Lei, Kaituo Li, Steve Liu, Jack Mazanec, Mihir Soni, Chris Swierczewski, Yaliang Wu, Jing Zhang, Sean Zheng, Partha Kanuparth

This talk

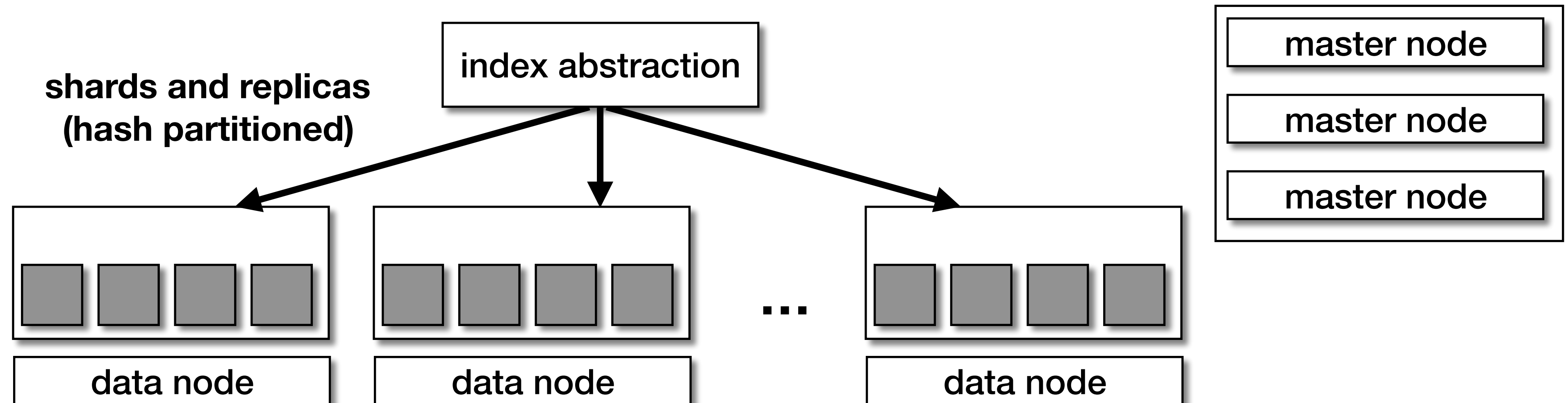
- Two themes focused on upcoming Open Distro features:
 - Systems Performance Diagnosis (support for ML-based insights)
 - Systems for ML: Anomaly Detection for data analytics

Elasticsearch Intro

- eventually consistent data store
- no ACID semantics
- focus on real-time guarantees (data freshness)
- mainly log analytics: write-heavy, append-only; canned queries

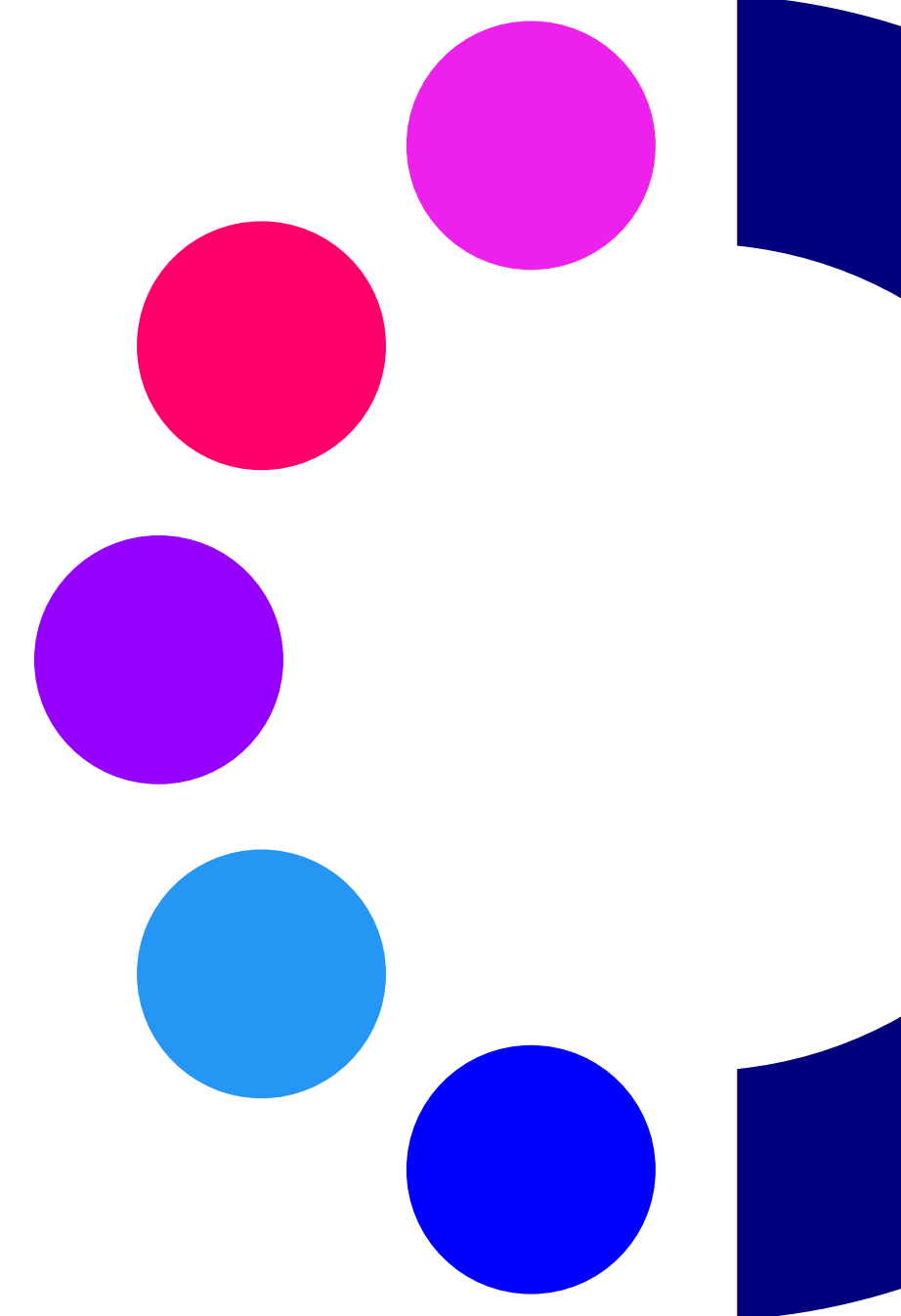
APIs:

- ingest: doc and bulk
- query
- data and cluster management



PART-I:

Real Time Performance Diagnosis in Distributed Databases

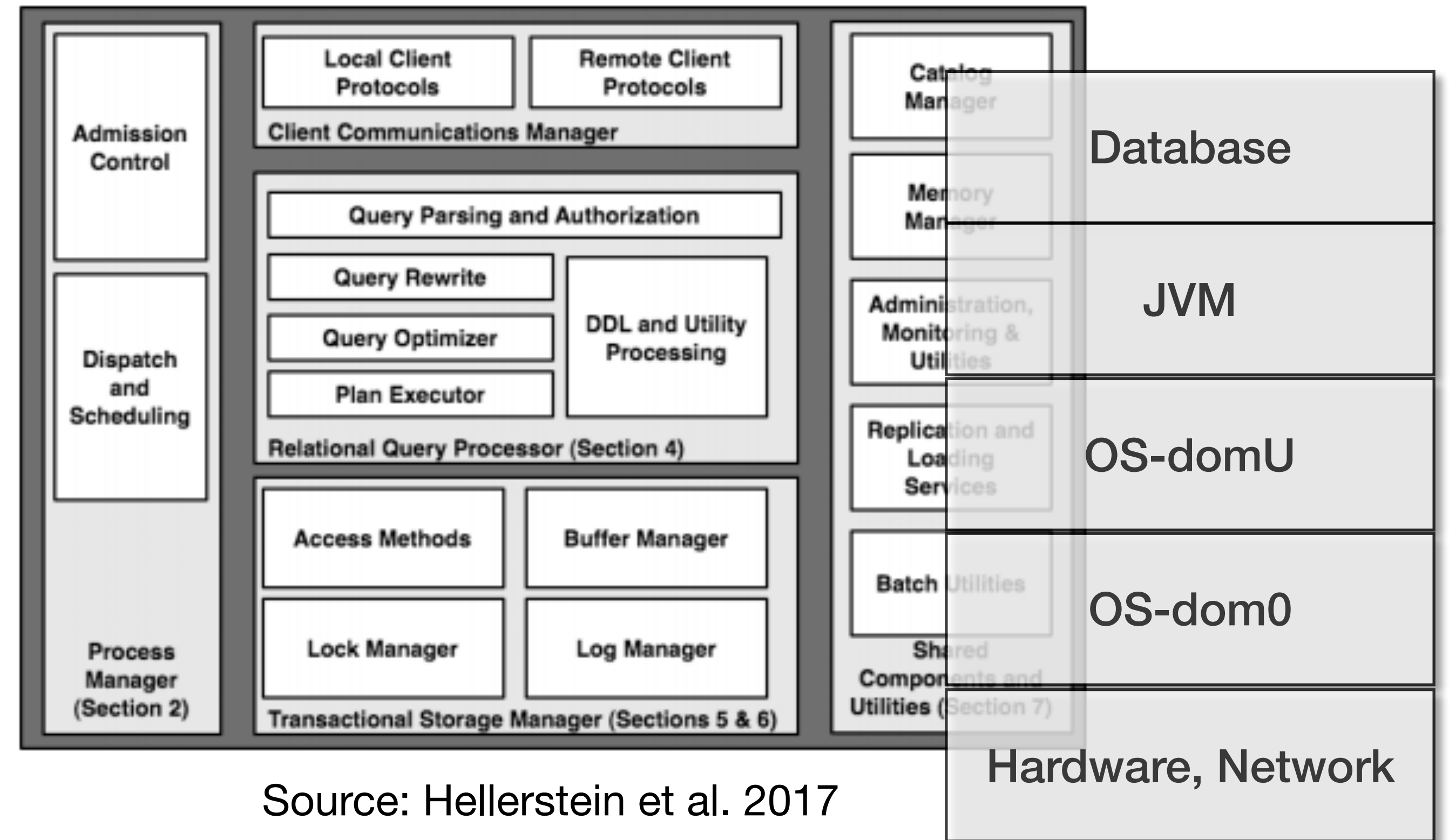


This talk

- Observability
- Root cause analysis

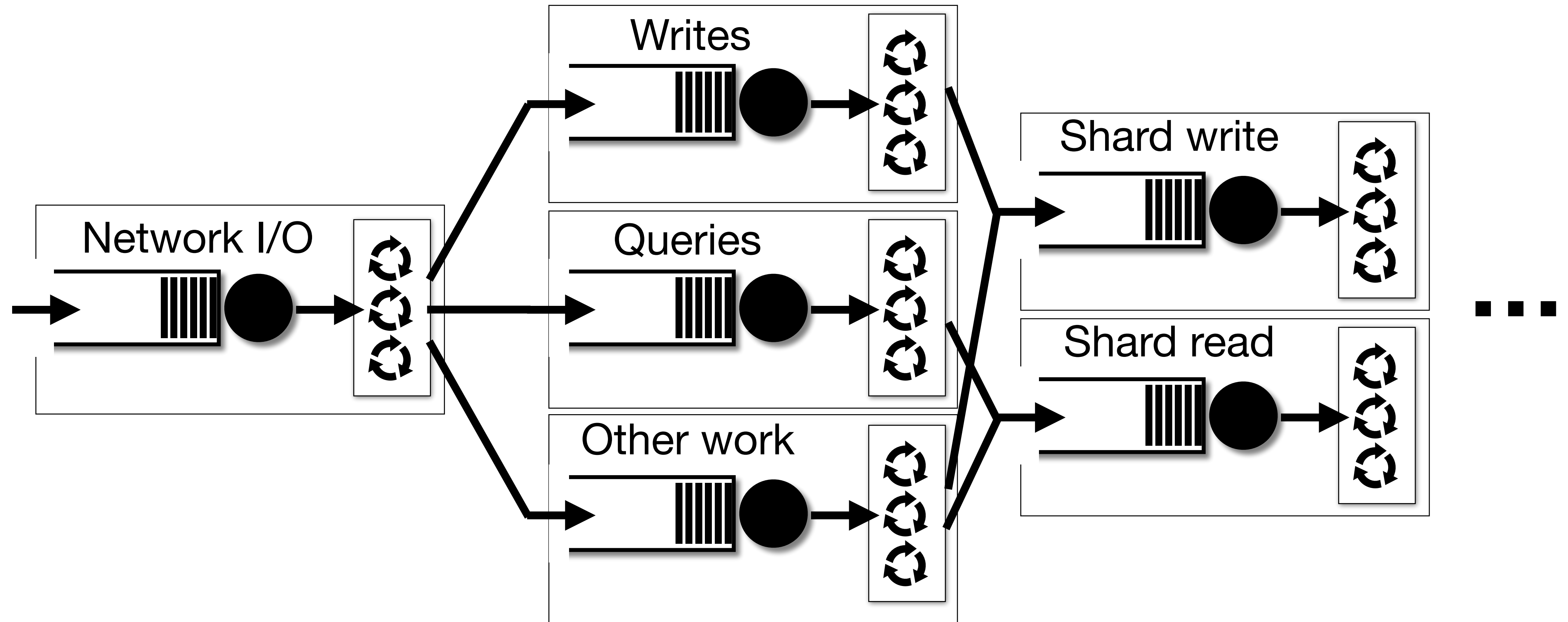
Database diagnosability

- Complex systems!
- Not built for performance *diagnosability*
- Is there a problem?
- Why is my query slow?
- Why is my ingestion slower?
- How to (re-)provision cluster?



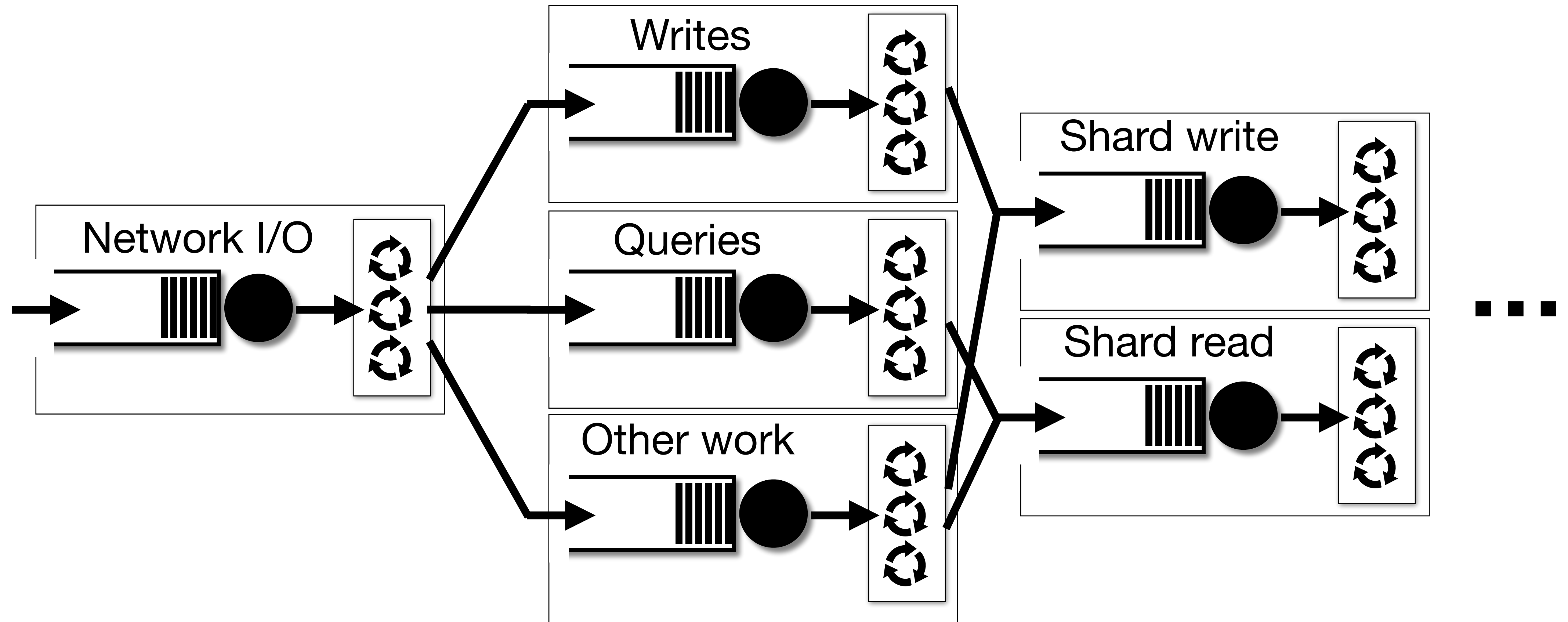
Low-level architecture

- Staged and functional (e.g., Elasticsearch): enables scalability



Low-level architecture

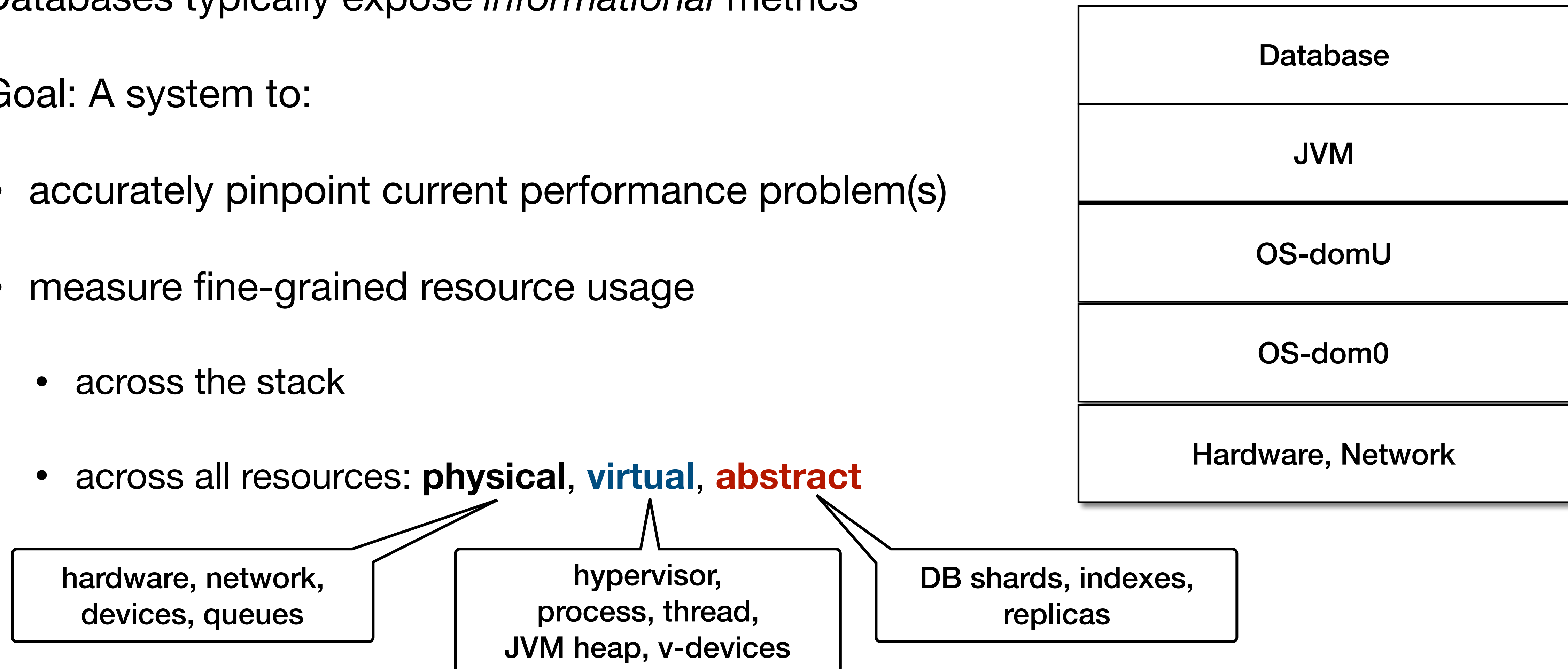
- Staged and functional (e.g., Elasticsearch): enables scalability



each thread executes atomic units of work from queue

Real Time Performance Diagnosis

- Databases typically expose *informational* metrics
- Goal: A system to:
 - accurately pinpoint current performance problem(s)
 - measure fine-grained resource usage
 - across the stack
 - across all resources: **physical**, **virtual**, **abstract**



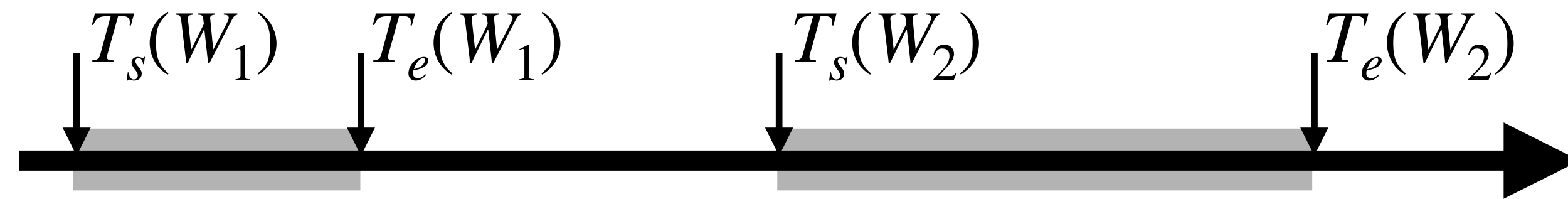
Observability

- Goal: Measure fine-grained resource usage of DB work
- High-level idea: construct event and utilization timelines; join

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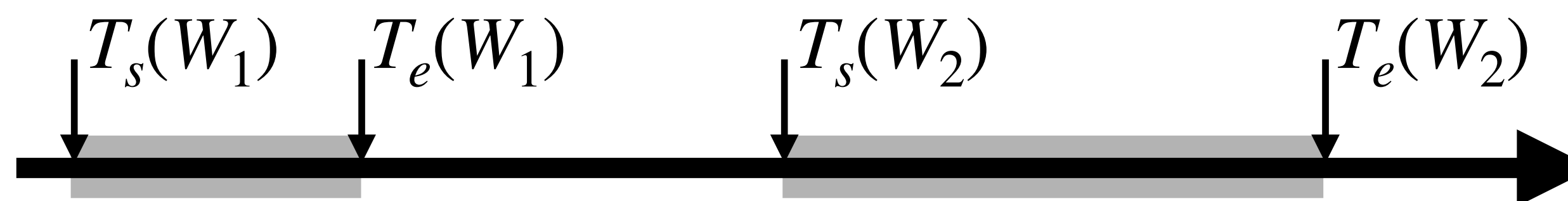
Thread i :
Events for units of work
 $\{W_1, W_2\}$: e.g., queries



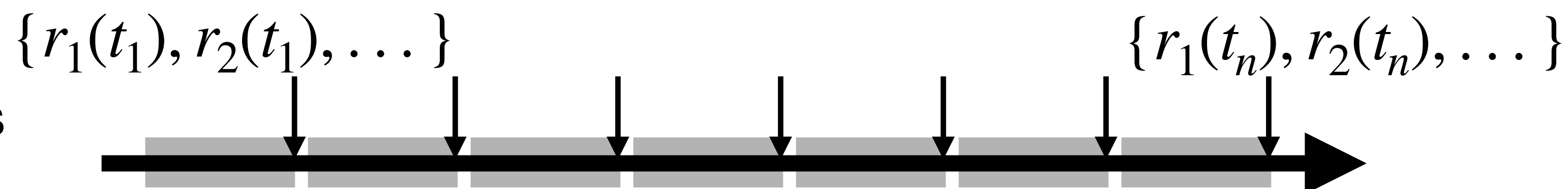
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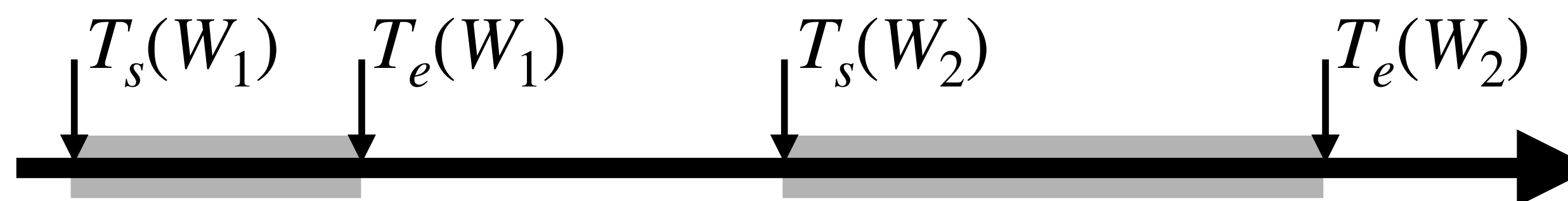
Thread i :
resource measurements
(e.g., CPU util.)



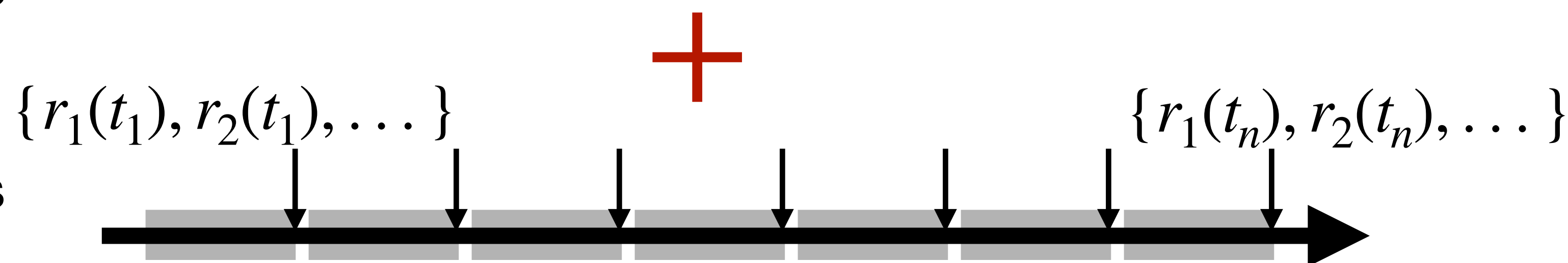
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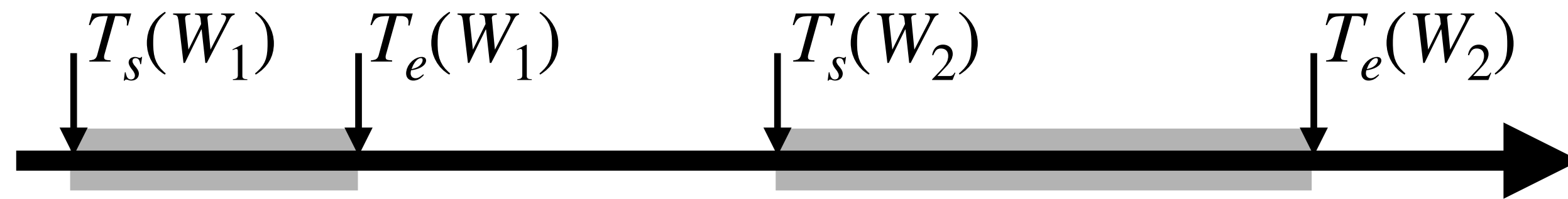
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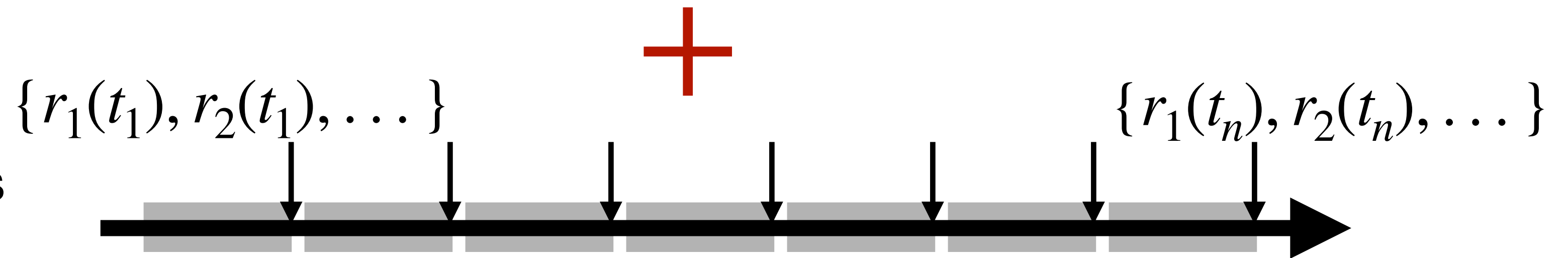
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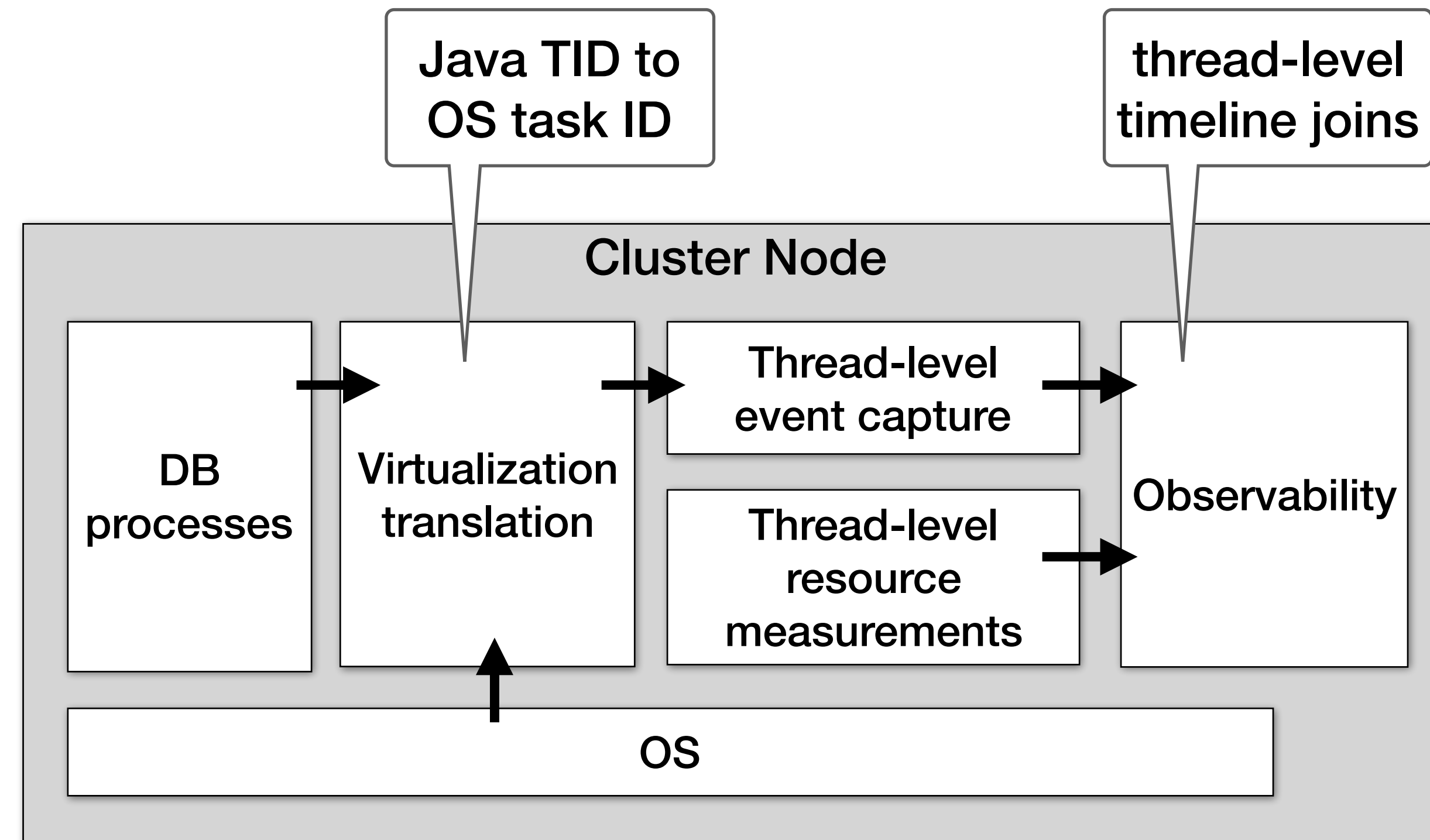
Thread i :
resource measurements
(e.g., CPU util.)



Unit of work W_i :
resource measurements

$$W_i \rightarrow \{r_1(t) \dots\}, \{r_2(t) \dots\}, \dots$$

Logical organization



- Agent: Process isolation from Elasticsearch

Metrics API

- Output of Observability: returns last 5s of system view

`GetMetric ( metric, dimensions... )`

e.g., CPU
utilization

e.g., index name,
shard ID, node ID

CPU utilization	Shard ID	Index name	Node ID

Metrics API

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`GetMetric ( metric, dimensions... )`

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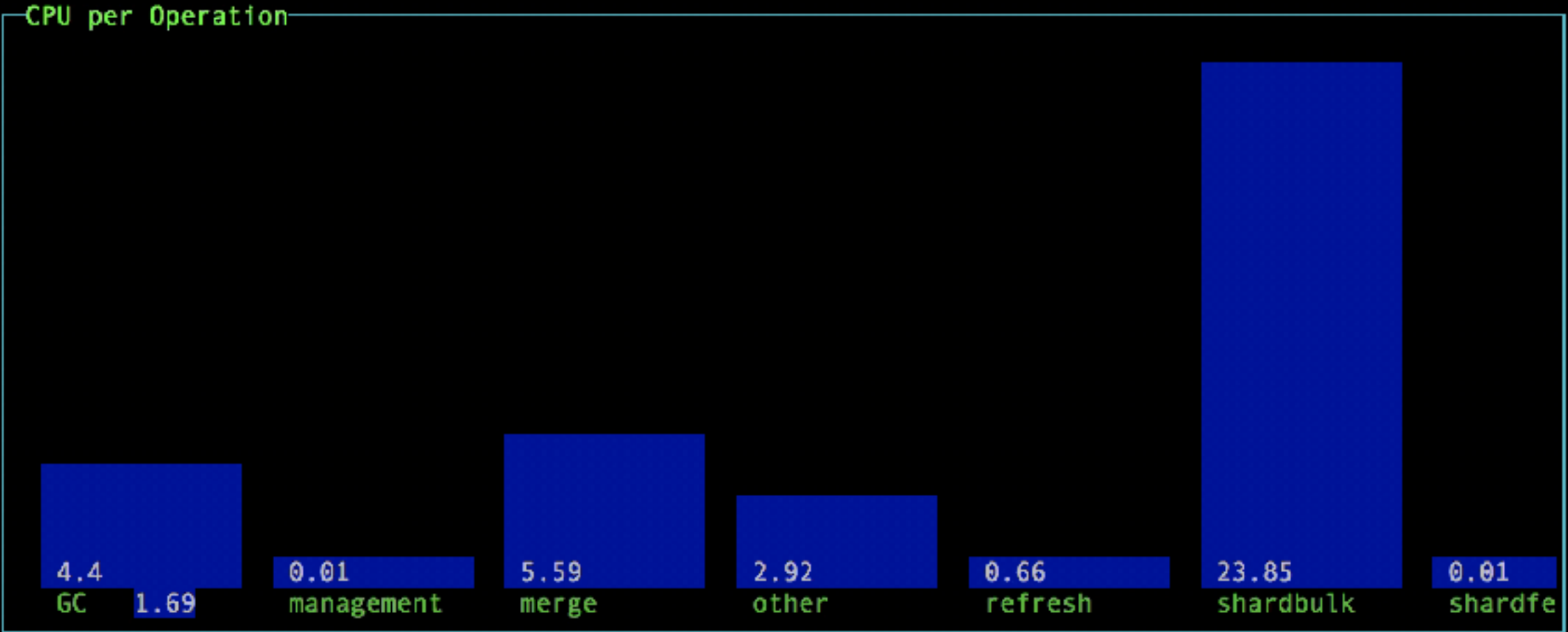
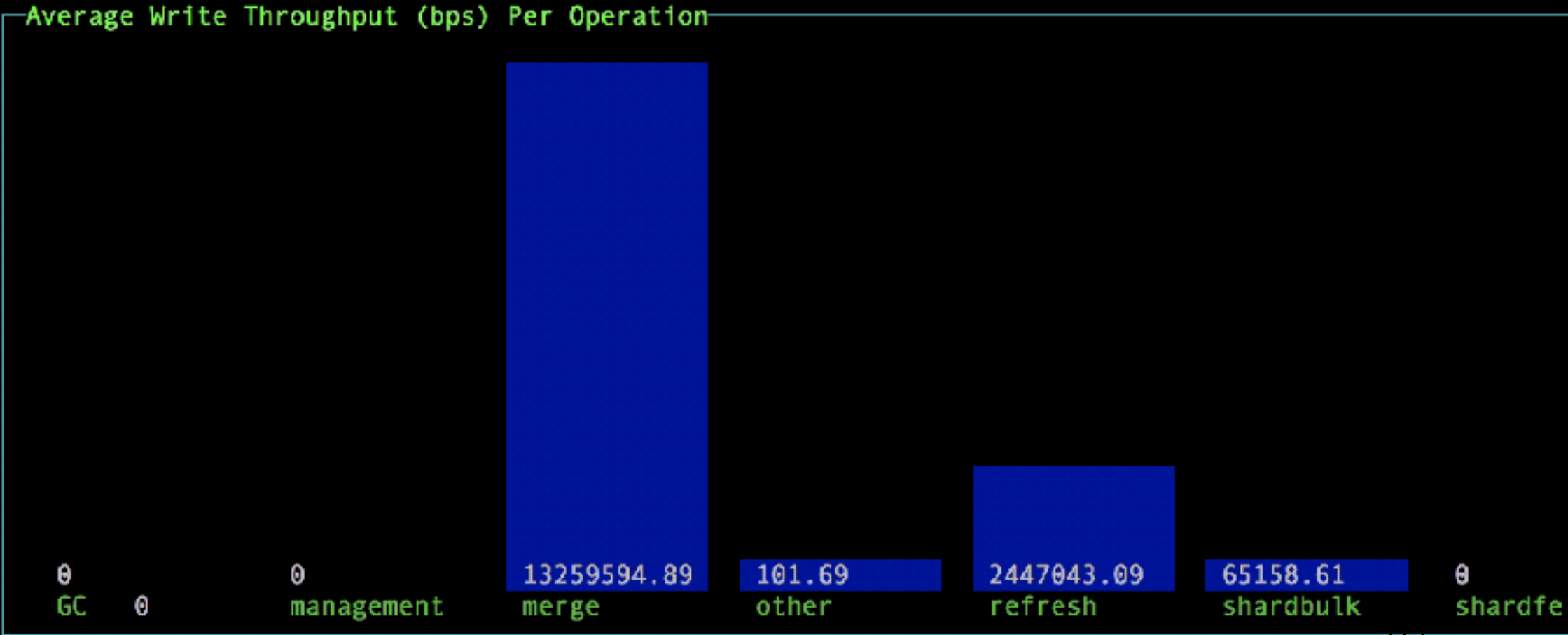
CPU utilization	Shard ID	Index name	Node ID

“get paging activity and disk I/O for bulk activity on a node”
“get CPU and heap allocations for search query”
“get GC CPU and page faults for a node”

dotJrE

Resource Metrics						
operation	index	shard	cpu	avgReadThroughputBps	avgWriteThroughputBps	node
merge	nyc_taxis	10	0.81	0	19162542.8	LjWleUdqSZKExhOk7ia
merge	nyc_taxis	6	0.76	0	13795881.71	LjWleUdqSZKExhOk7ia
shardbulk	nyc_taxis	24	0.76	0	17436410.87	LjWleUdqSZKExhOk7ia
merge	nyc_taxis	4	0.73	0	14057907.06	LjWleUdqSZKExhOk7ia
shardbulk	nyc_taxis	26	0.71	0	17128893.59	LjWleUdqSZKExhOk7ia
merge	nyc_taxis	2	0.71	0	11450013.28	LjWleUdqSZKExhOk7ia
merge	nyc_taxis	8	0.68	0	13728315.43	LjWleUdqSZKExhOk7ia
shardbulk	nyc_taxis	28	0.65	0	14366378.8	LjWleUdqSZKExhOk7ia
merge	nyc_taxis	16	0.65	0	10982184.01	LjWleUdqSZKExhOk7ia
merge	nyc_taxis	18	0.63	0	12109447.97	LjWleUdqSZKExhOk7ia
merge	nyc_taxis	14	0.63	0	10790466.85	LjWleUdqSZKExhOk7ia
shardbulk	nyc_taxis	61	0.54	0	3771860.57	hzauoB43SLCoLLUDsF6
shardbulk	nyc_taxis	15	0.52	0	3660108.69	hzauoB43SLCoLLUDsF6
shardbulk	nyc_taxis	11	0.51	0	3576194.27	hzauoB43SLCoLLUDsF6
shardbulk	nyc_taxis	5	0.5	0	3551458.45	hzauoB43SLCoLLUDsF6
shardbulk	nyc_taxis	41	0.5	0	3520545.15	hzauoB43SLCoLLUDsF6
shardbulk	nyc_taxis	35	0.5	0	3529262.39	hzauoB43SLCoLLUDsF6

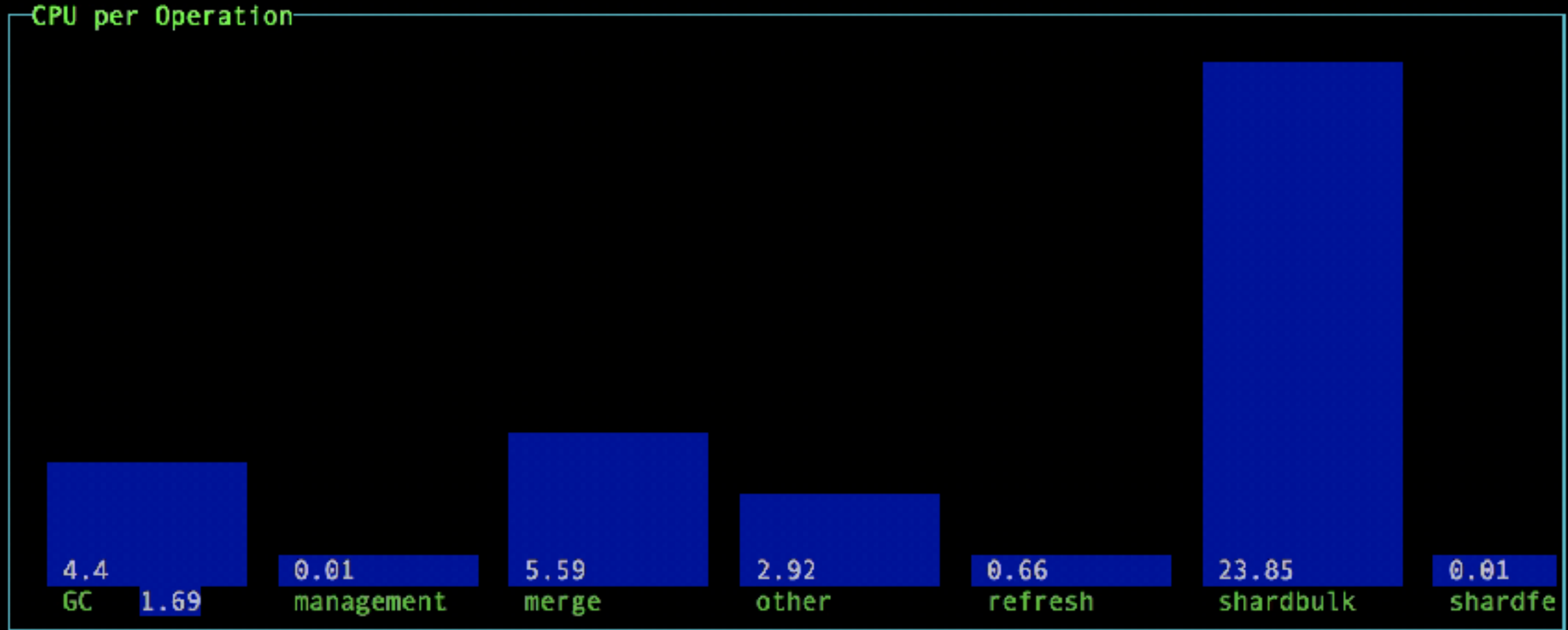
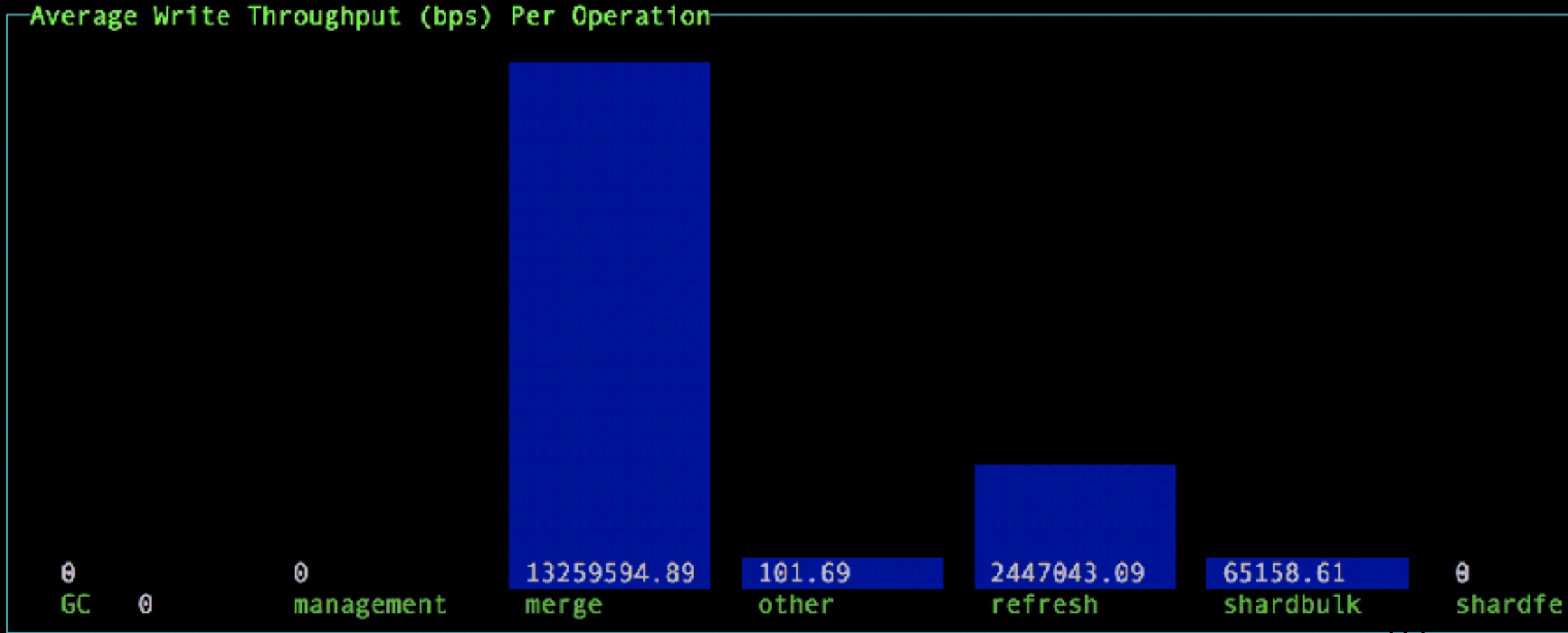
Workload					
operation	status	indices	latency	itemCount	node
bulk	200	0	302.77	897711	hzauoB43SLC
bulk	200	0	274.93	381878	LjWleUdqSZK
search	200	admin	171	0	hzauoB43SLC
search	200	nyc_taxis	245	0	hzauoB43SLC
search	200	nyc_taxis1	450	0	hzauoB43SLC
search	200	sonested	1465.5	0	hzauoB43SLC
search	200	sonested	1876	0	LjWleUdqSZK



dotJrE

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search	200	sonested	1465.5	0	hzauoB43SLC
search	200	sonested	1876	0	LjWleUdqSZK



Root Cause Analysis

- Observability emits hundreds of fine-grained metrics
- RCA distills down to current problem(s)
- What is a root cause?

$$\text{root_cause}_i = f_{rca}(\text{symptom}_k, \text{root_cause}_j)$$

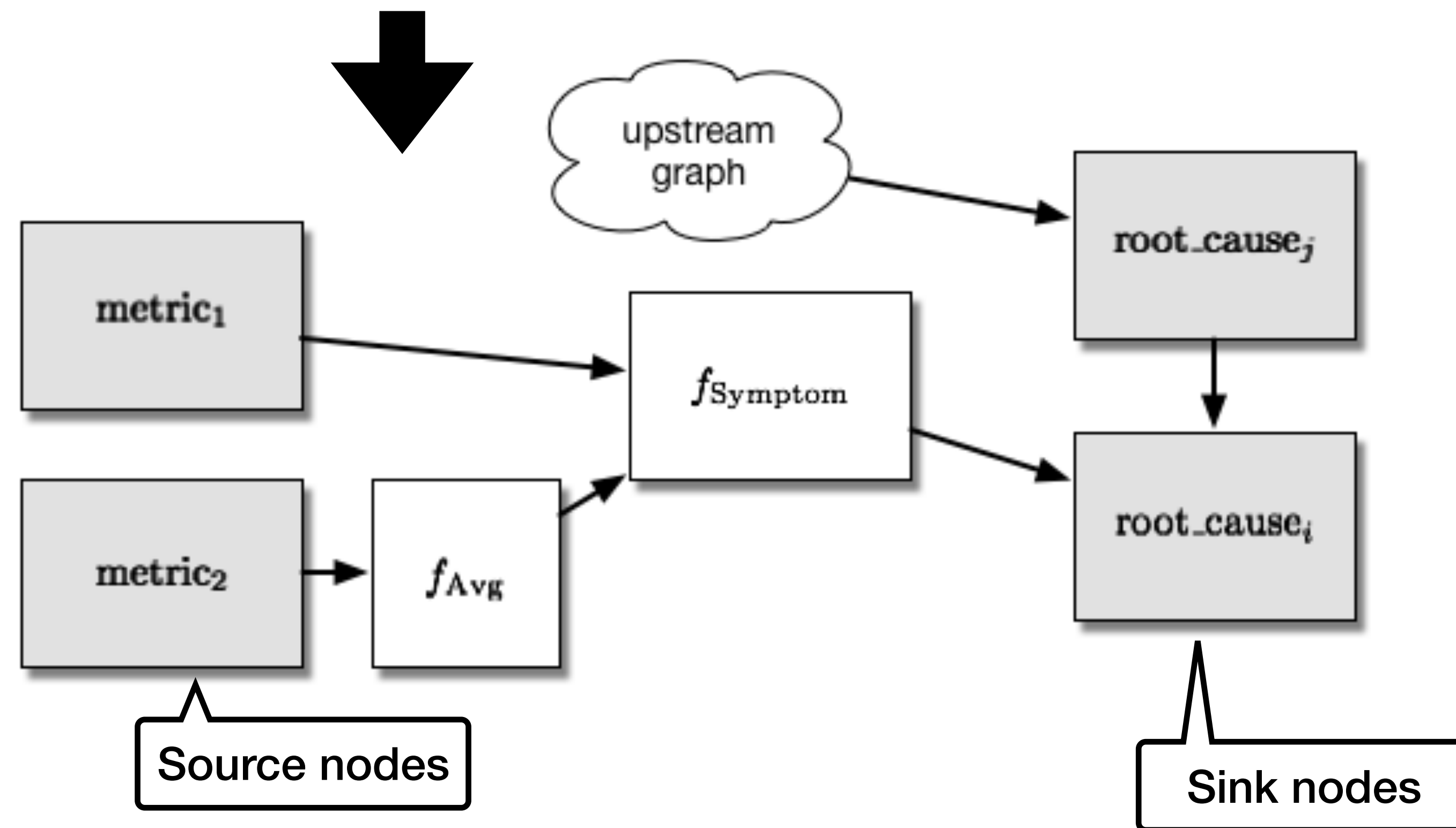
$$\text{symptom}_k = f_{\text{symptom}}(\text{metric}_1, f_{\text{avg}}(\text{metric}_2))$$

RCA as data flow graph

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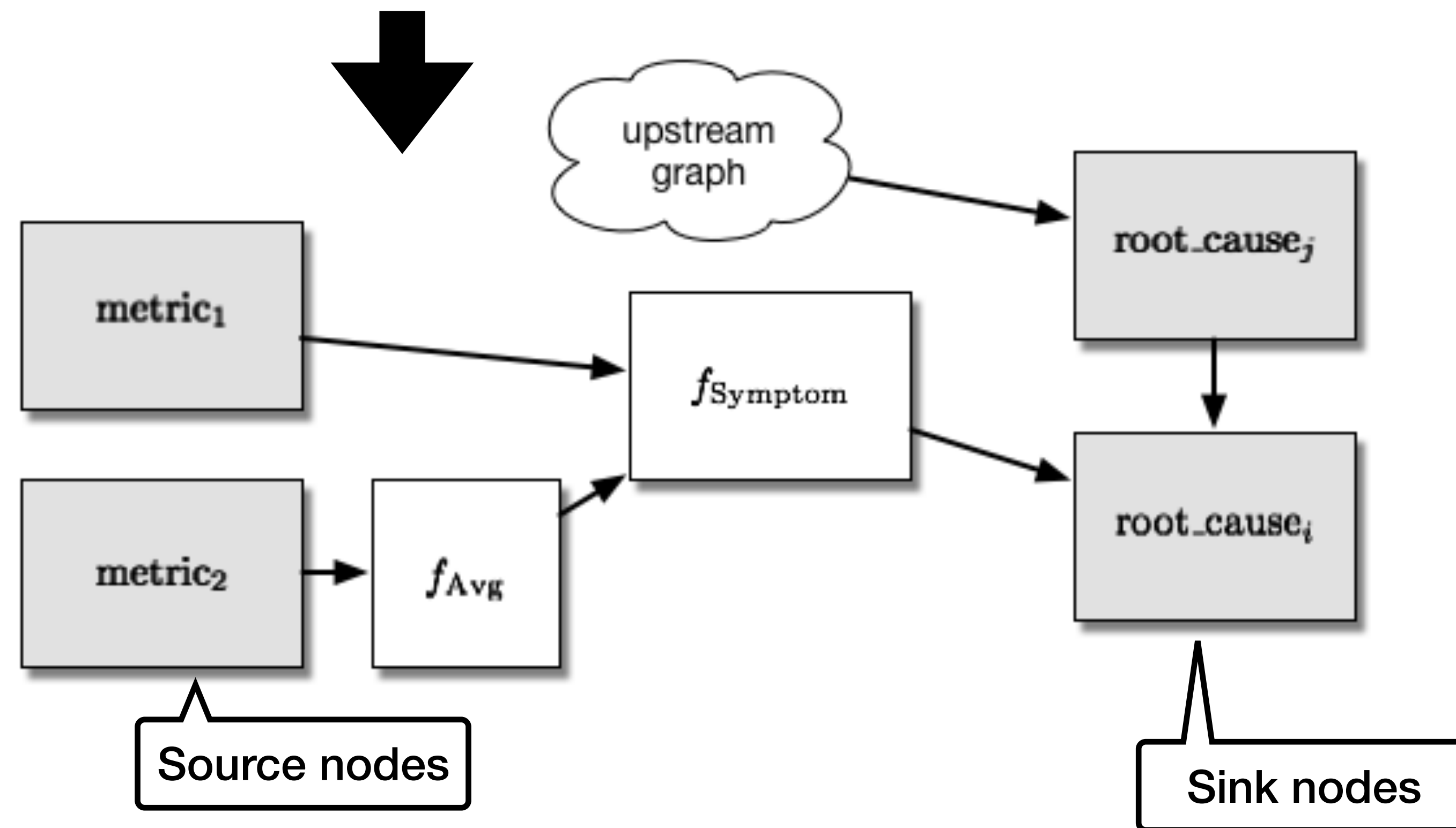
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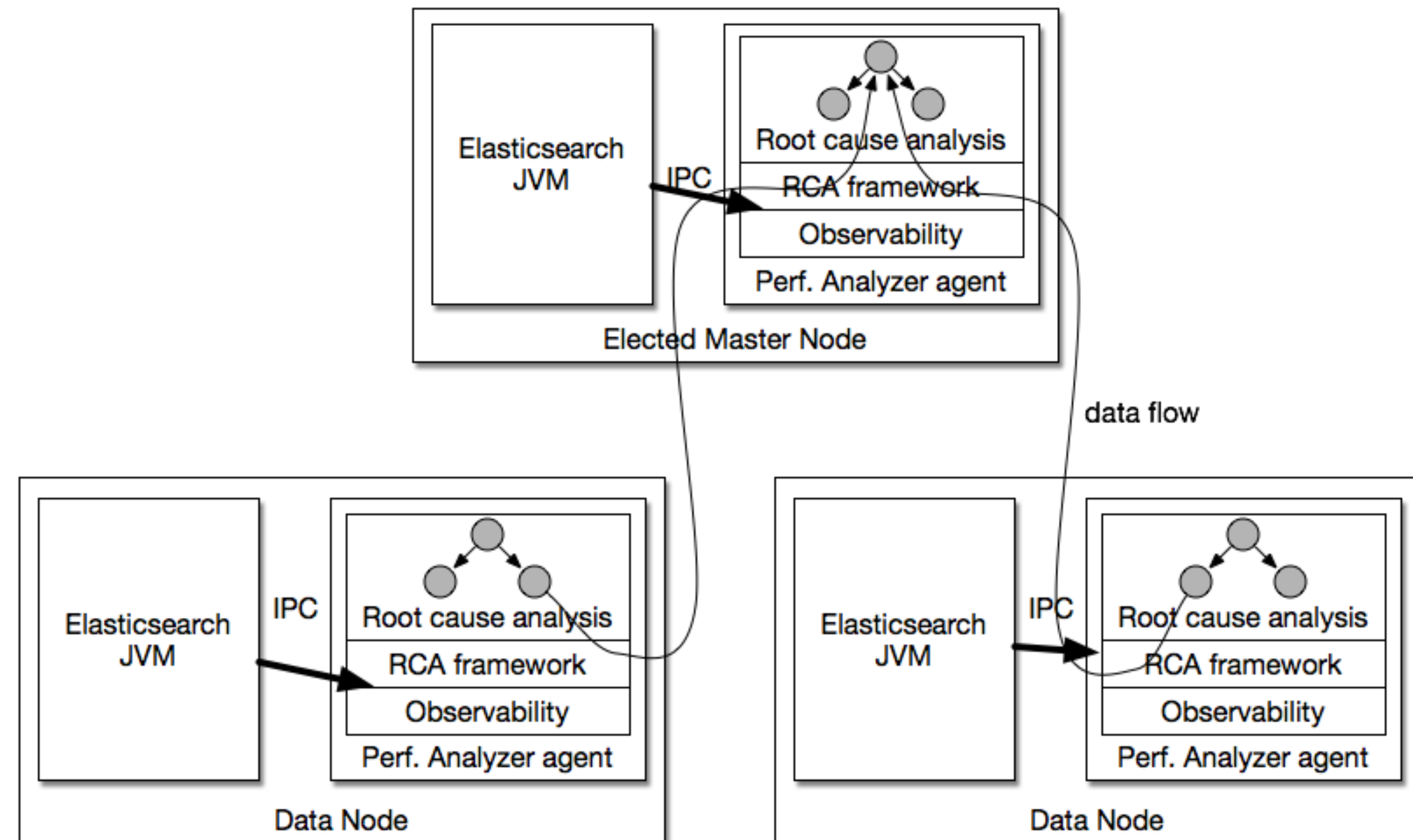


- Root cause can span multiple cluster nodes
- RCA node is 1-1 with cluster node
- Graph of all RCAs is a DAG

RCA framework

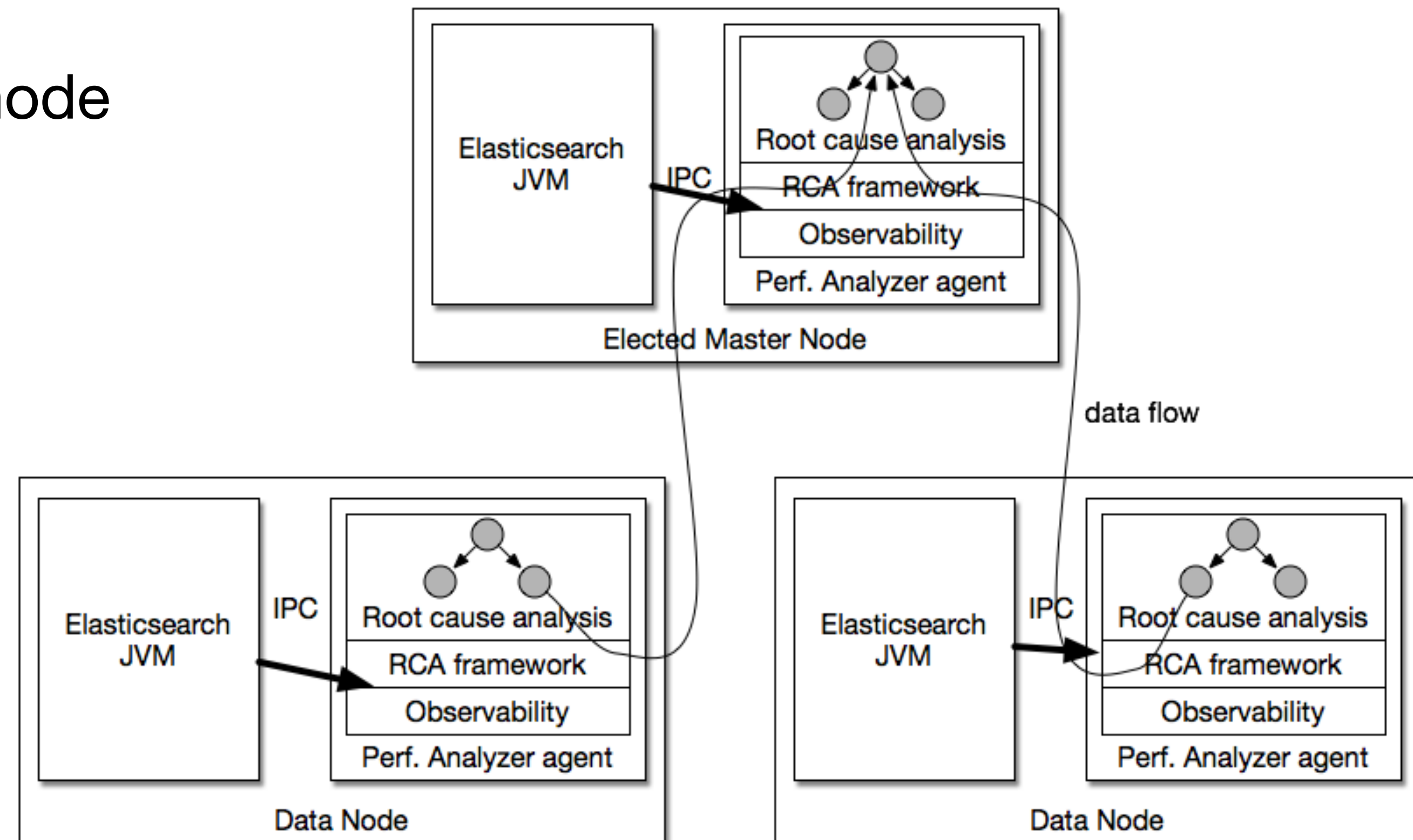
RCA framework

- Implements real time streaming runtime for the RCA graph



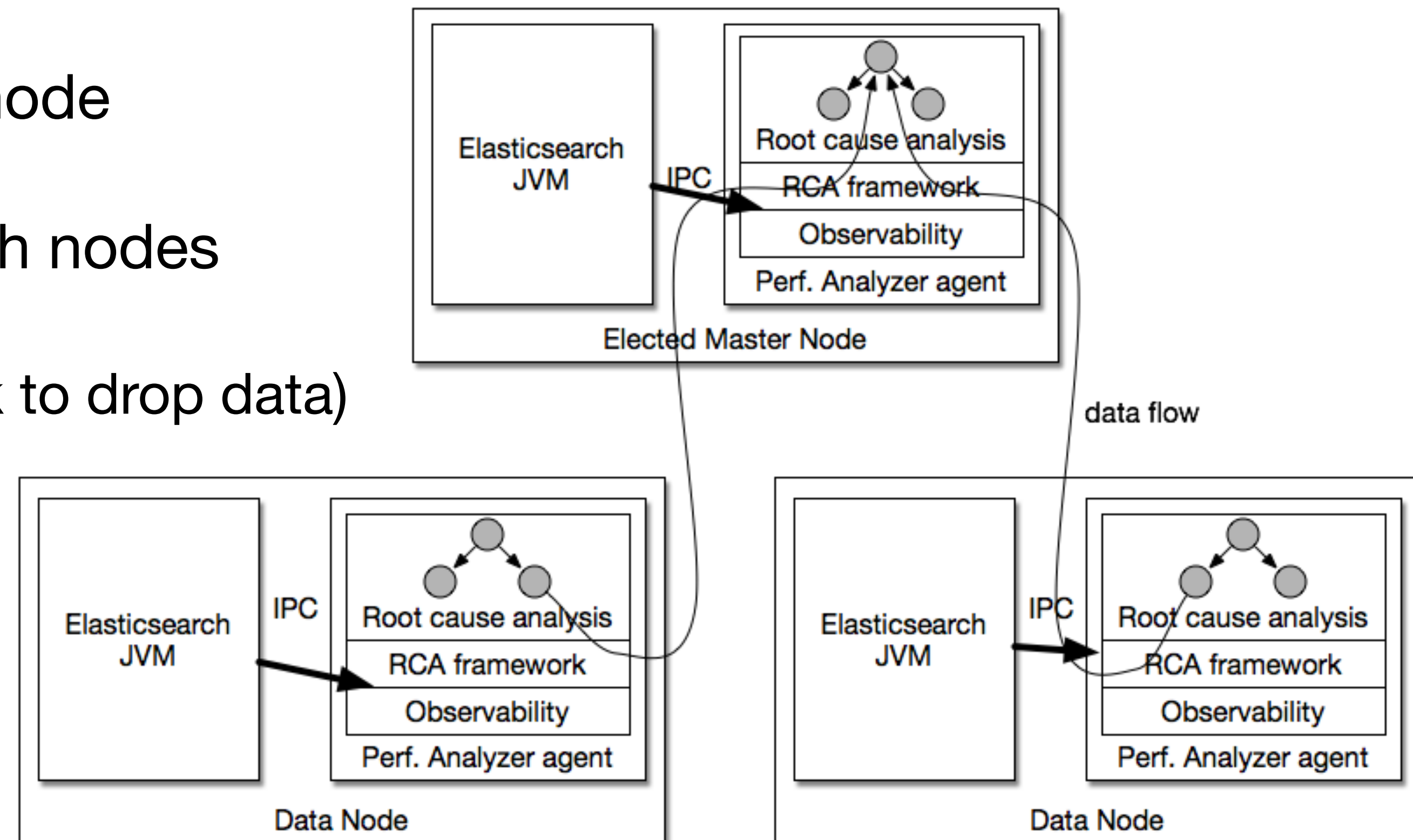
RCA framework

- Implements real time streaming runtime for the RCA graph
- *schedules* and *executes* each graph node



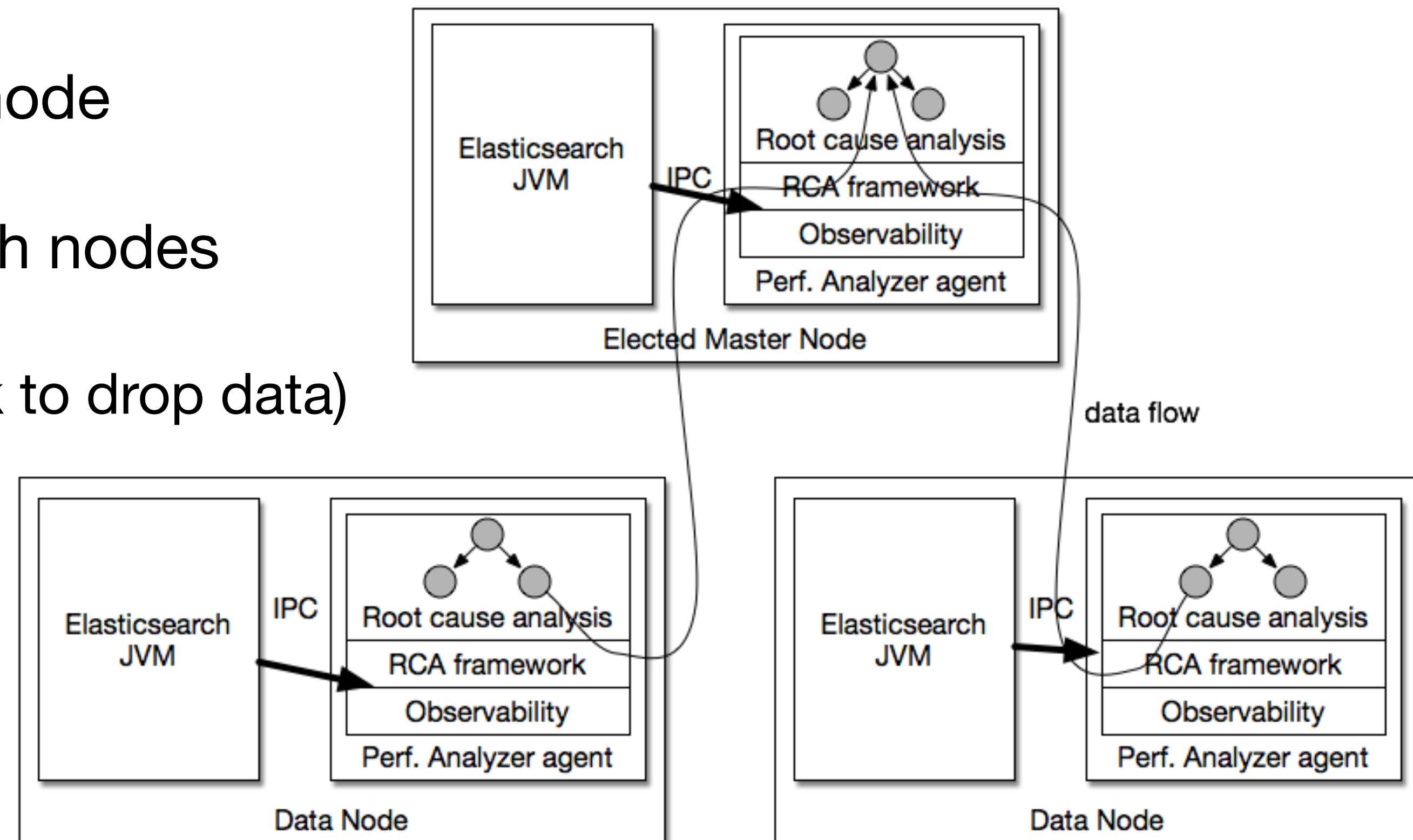
RCA framework

- Implements real time streaming runtime for the RCA graph
- *schedules* and *executes* each graph node
- *passes* opaque objects between graph nodes
 - ordered and at-most-once delivery (ok to drop data)
 - staleness and loss: via timestamps



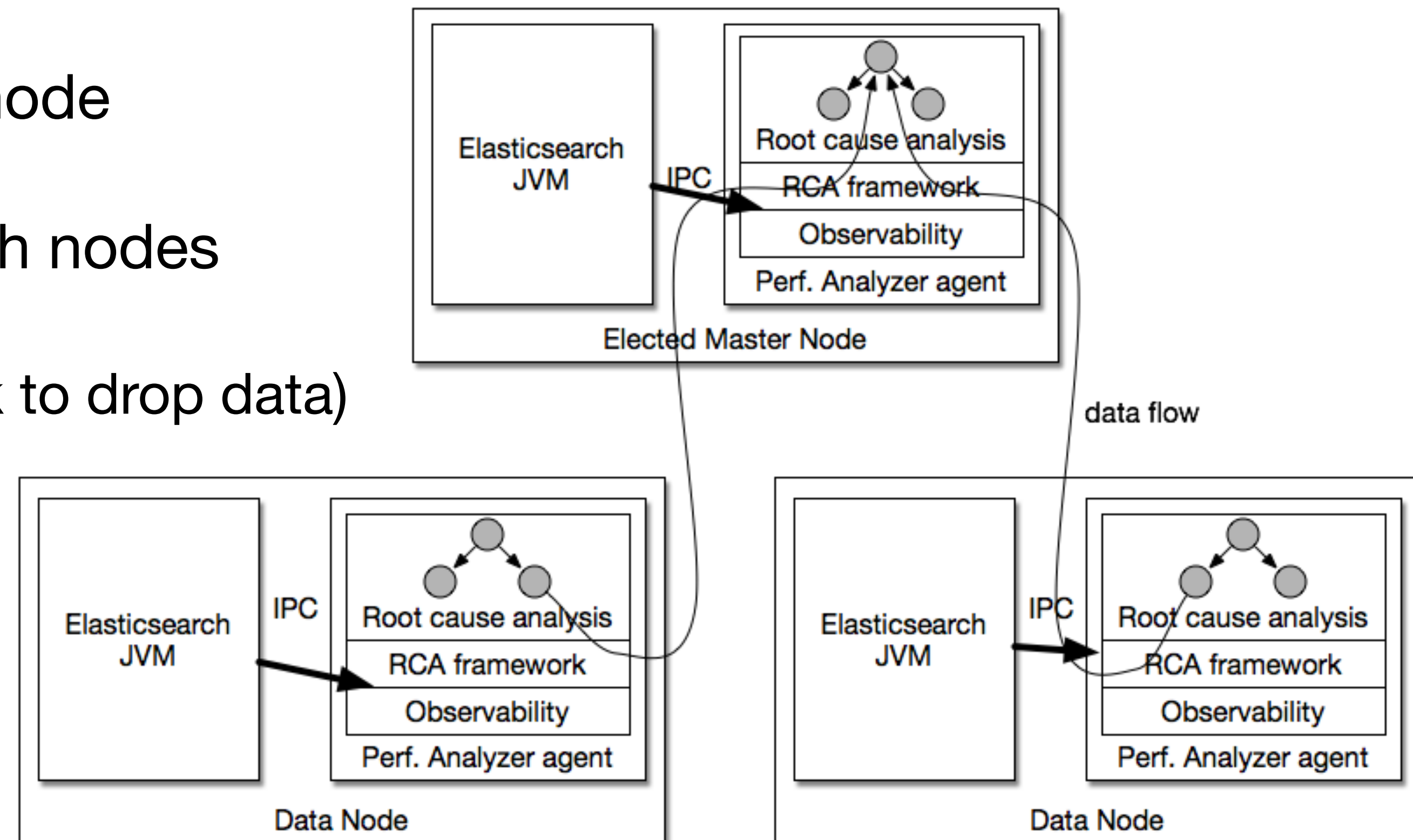
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RCA framework

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- edge setup: nodes subscribe
- framework de-dups compute



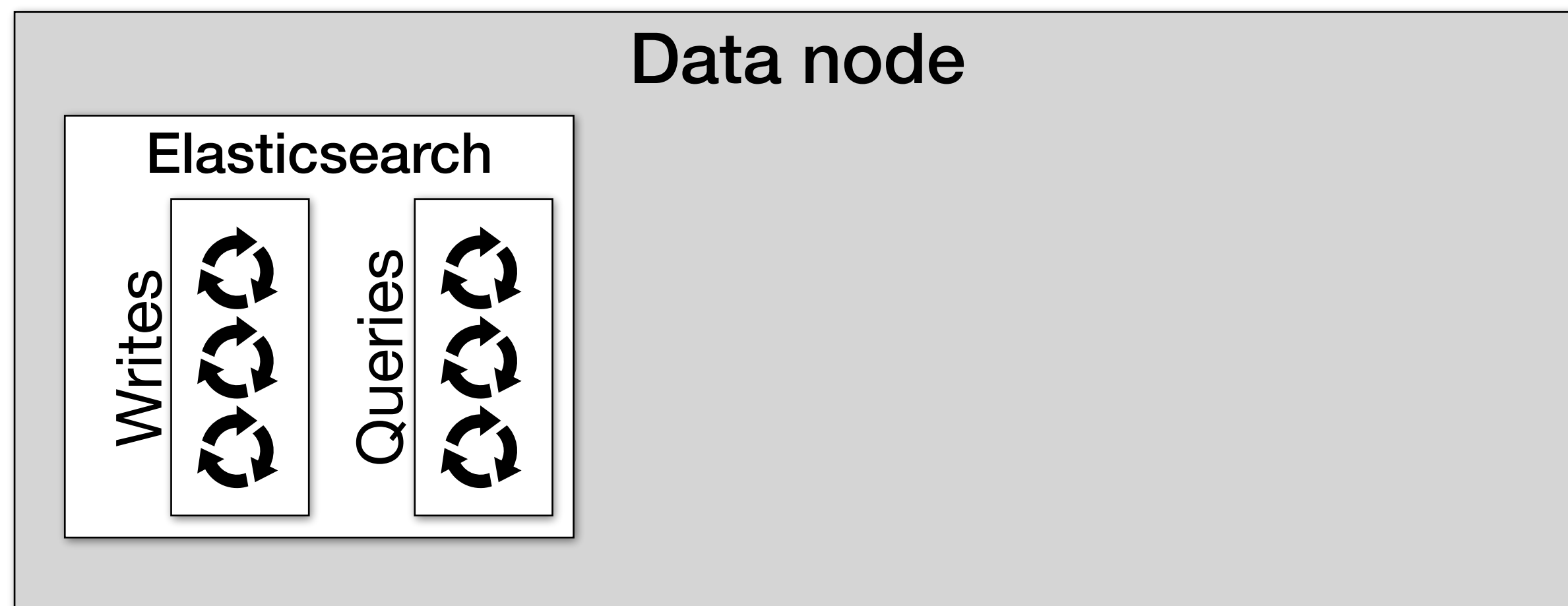
RCA API

- Output of RCA: returns last 5s of diagnoses

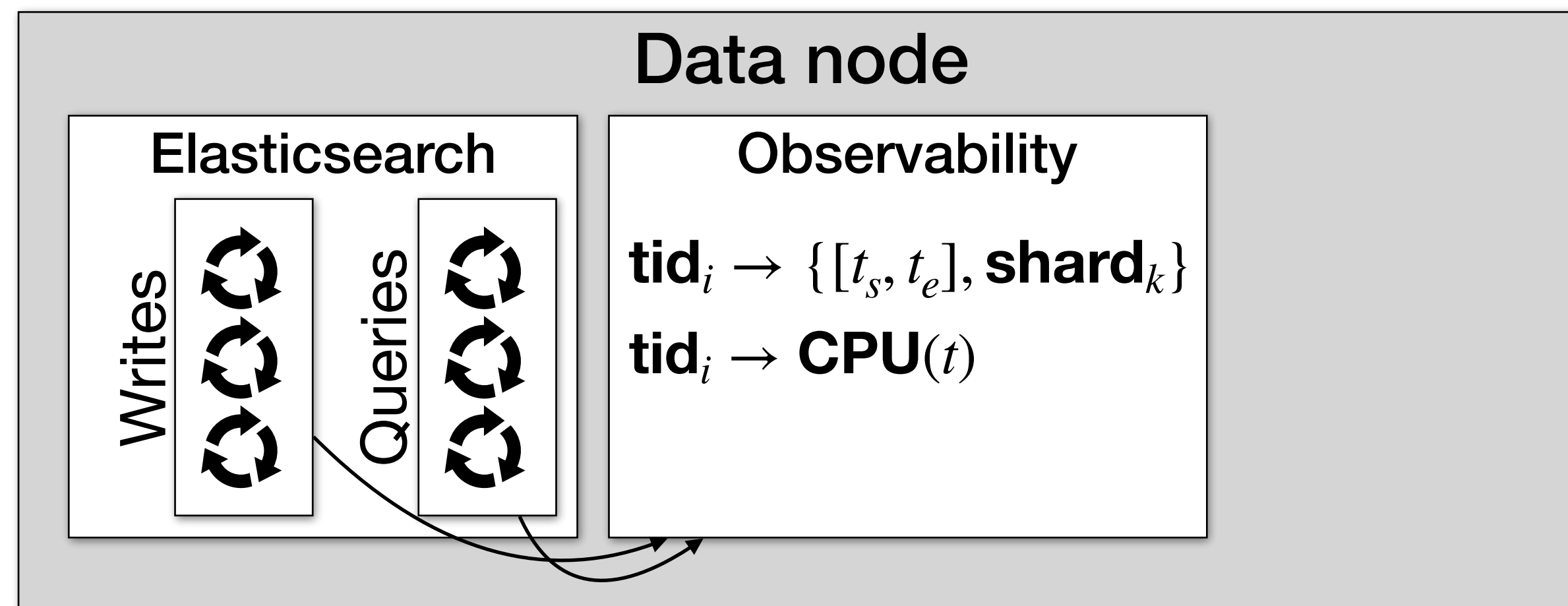
`GetRCAs ()`

RCA name	Time since first diagnosis	Context (e.g., metrics, thresholds ...)

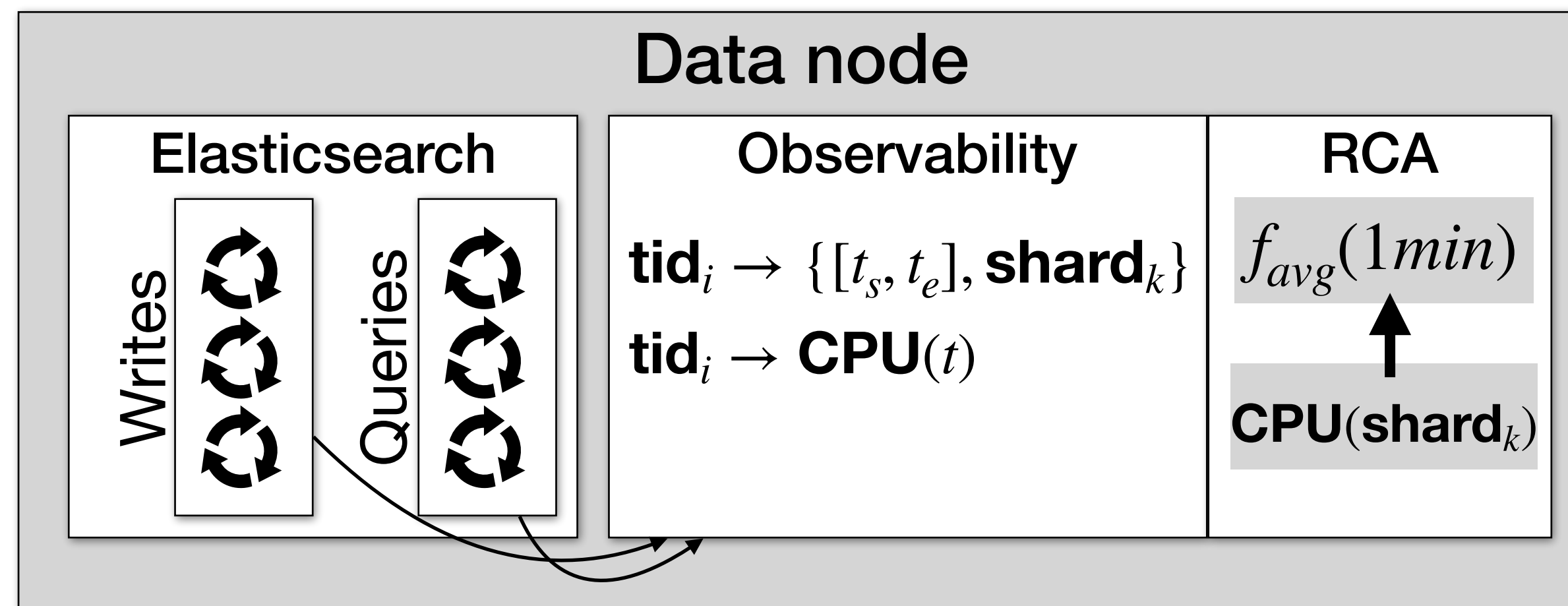
Example RCA: High-CPU Shards



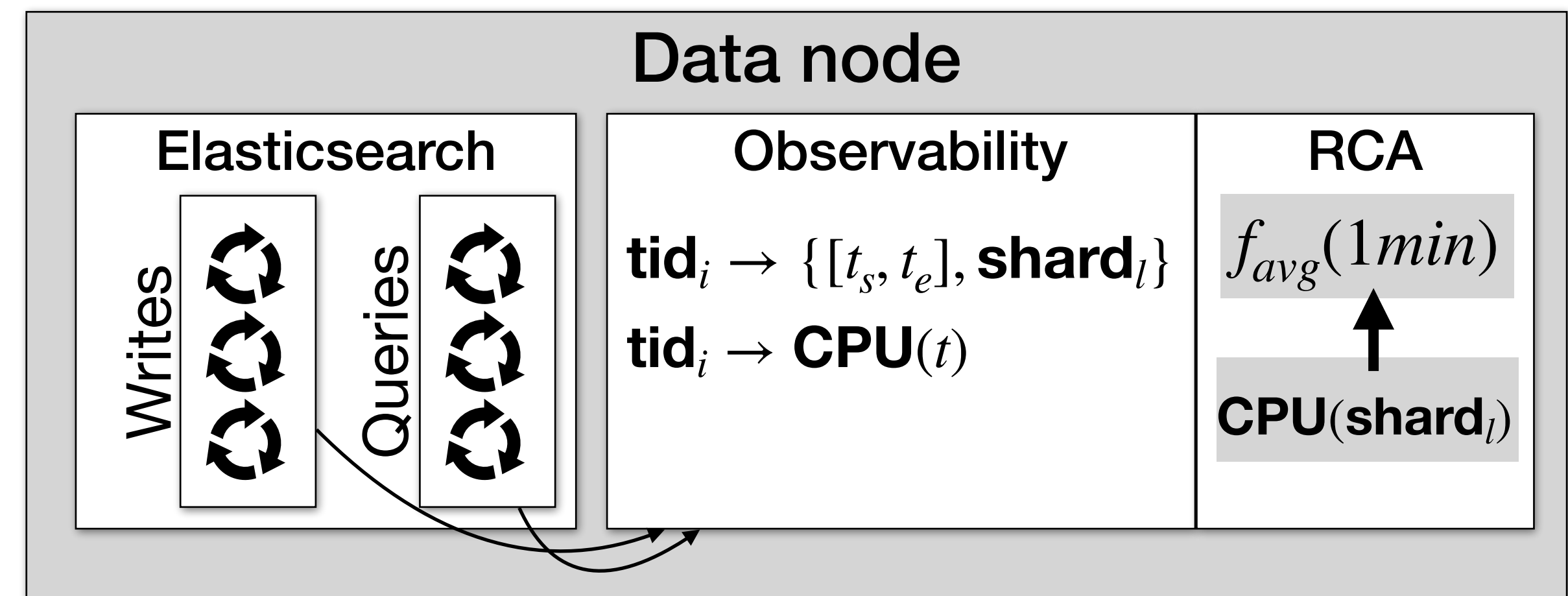
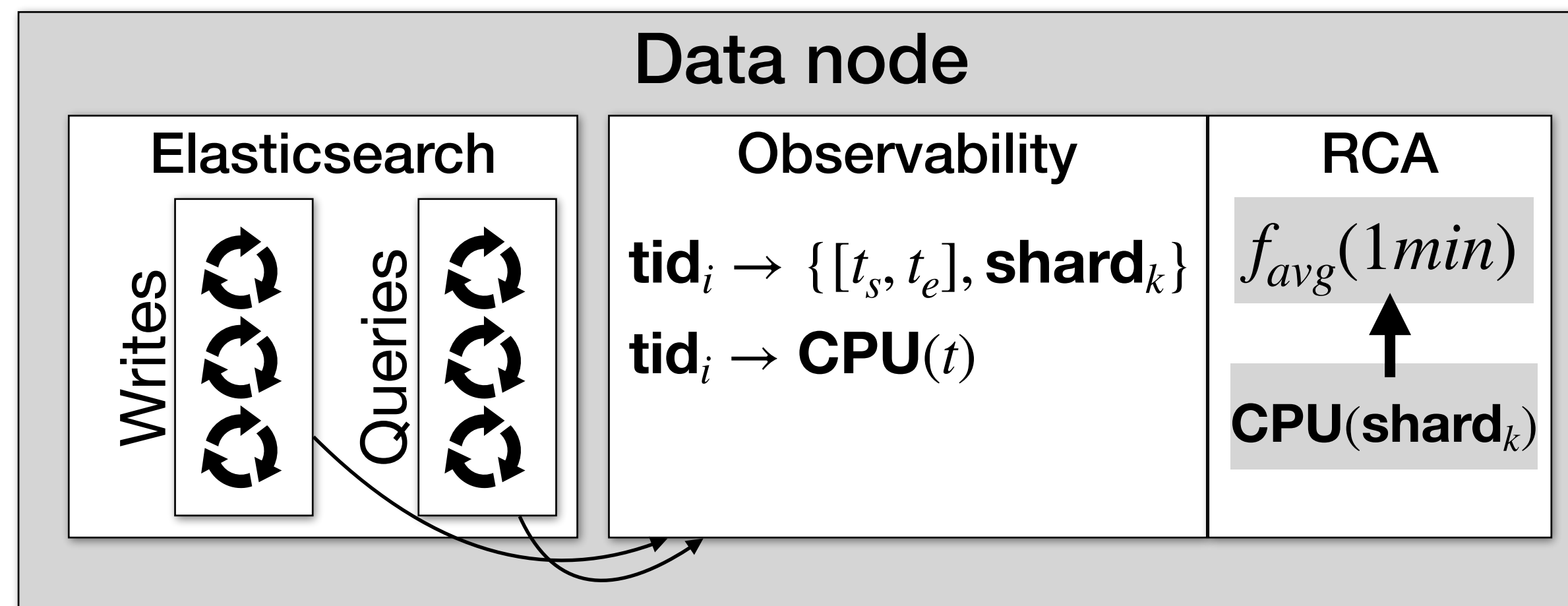
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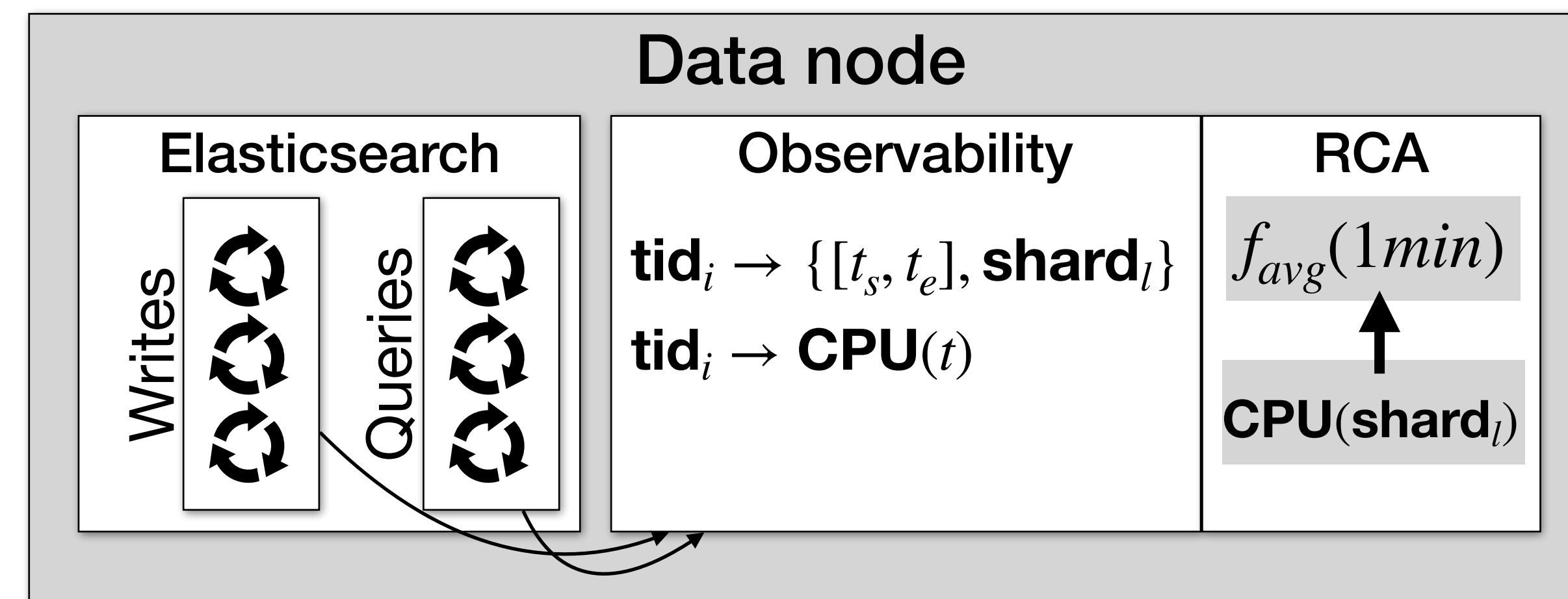
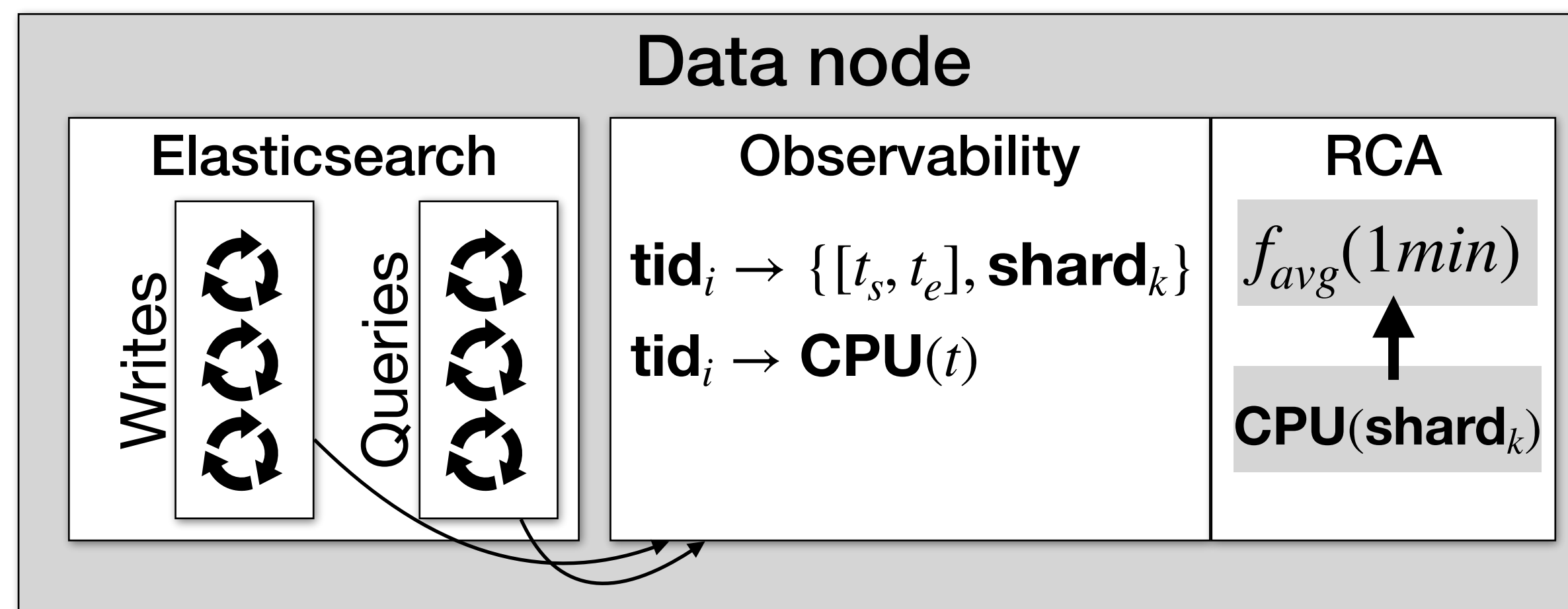
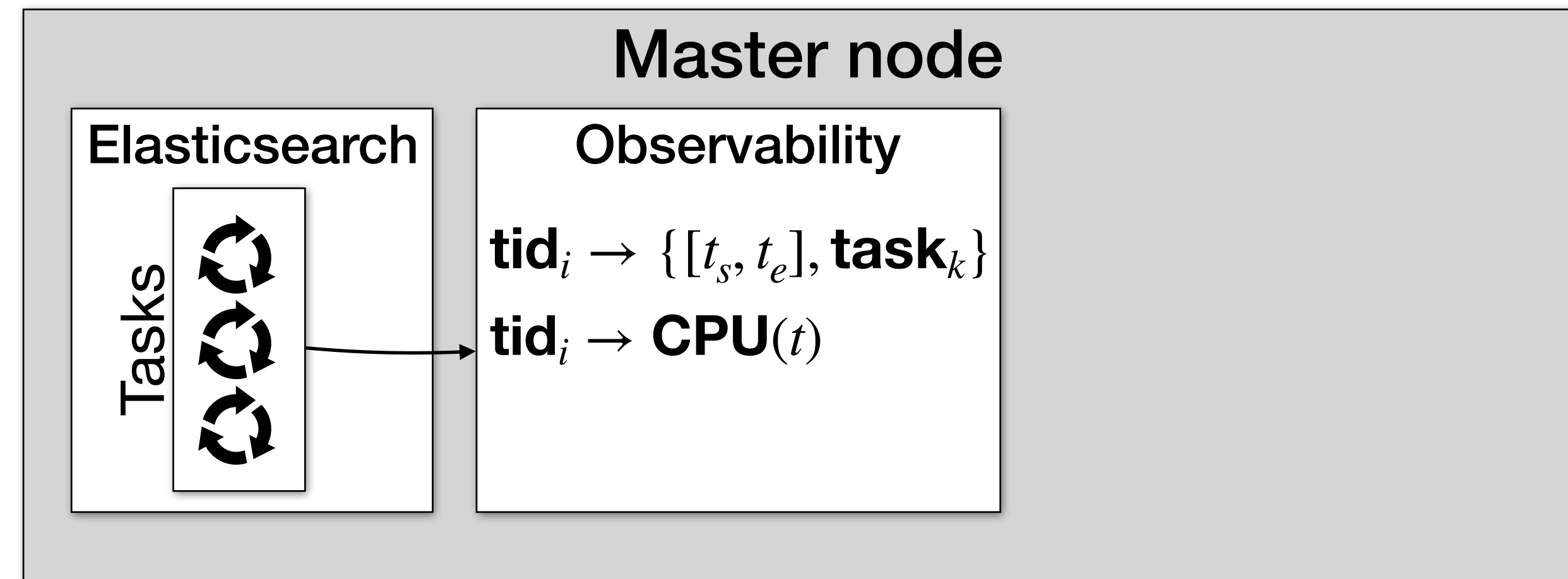
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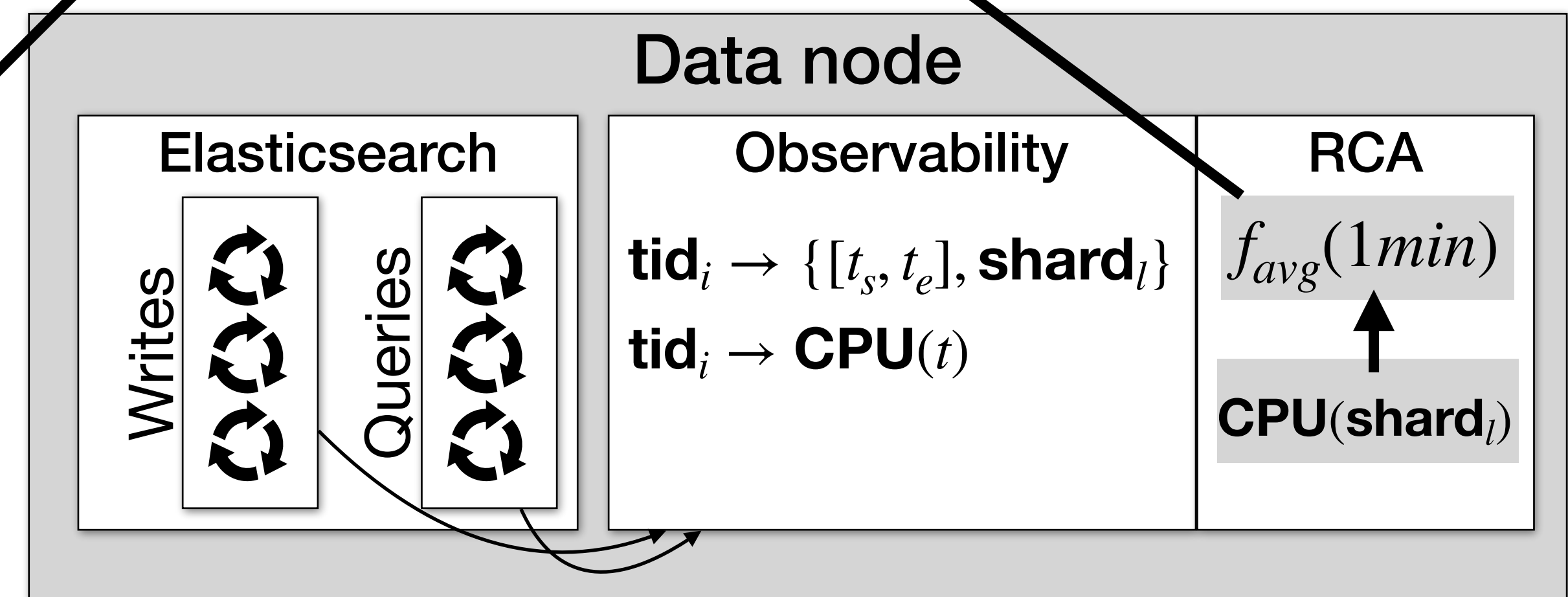
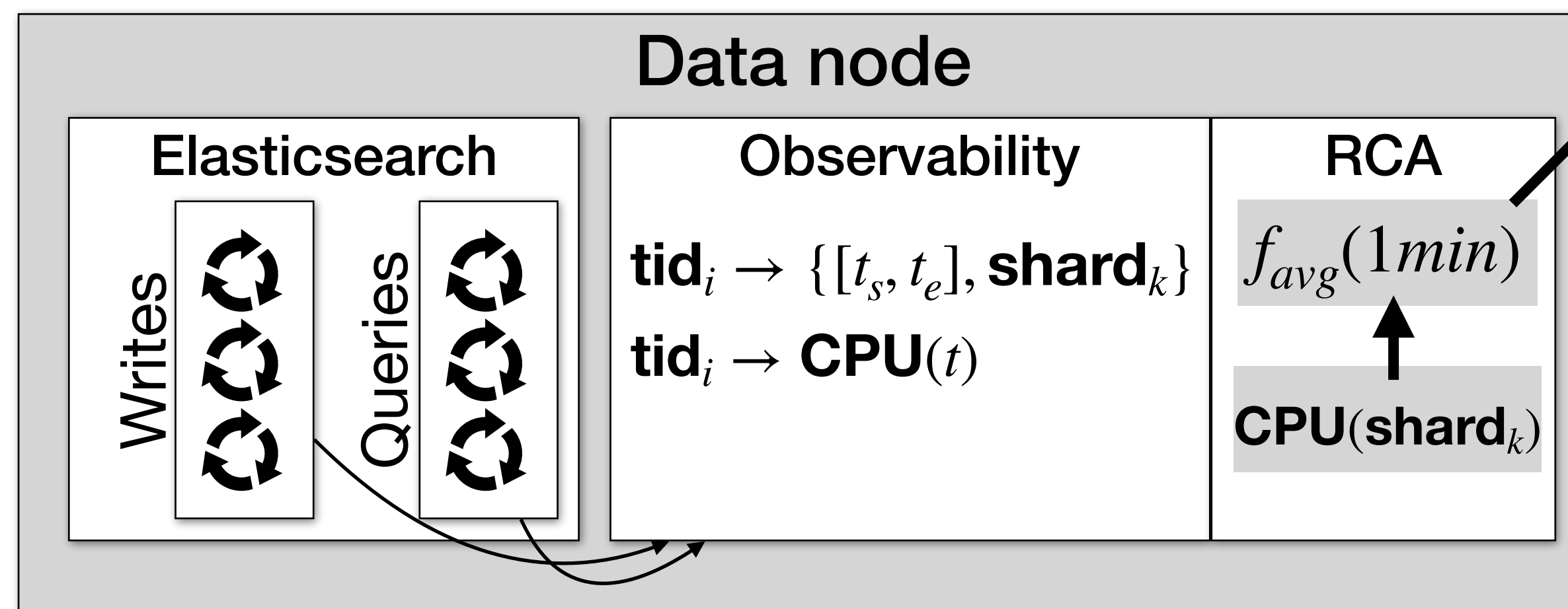
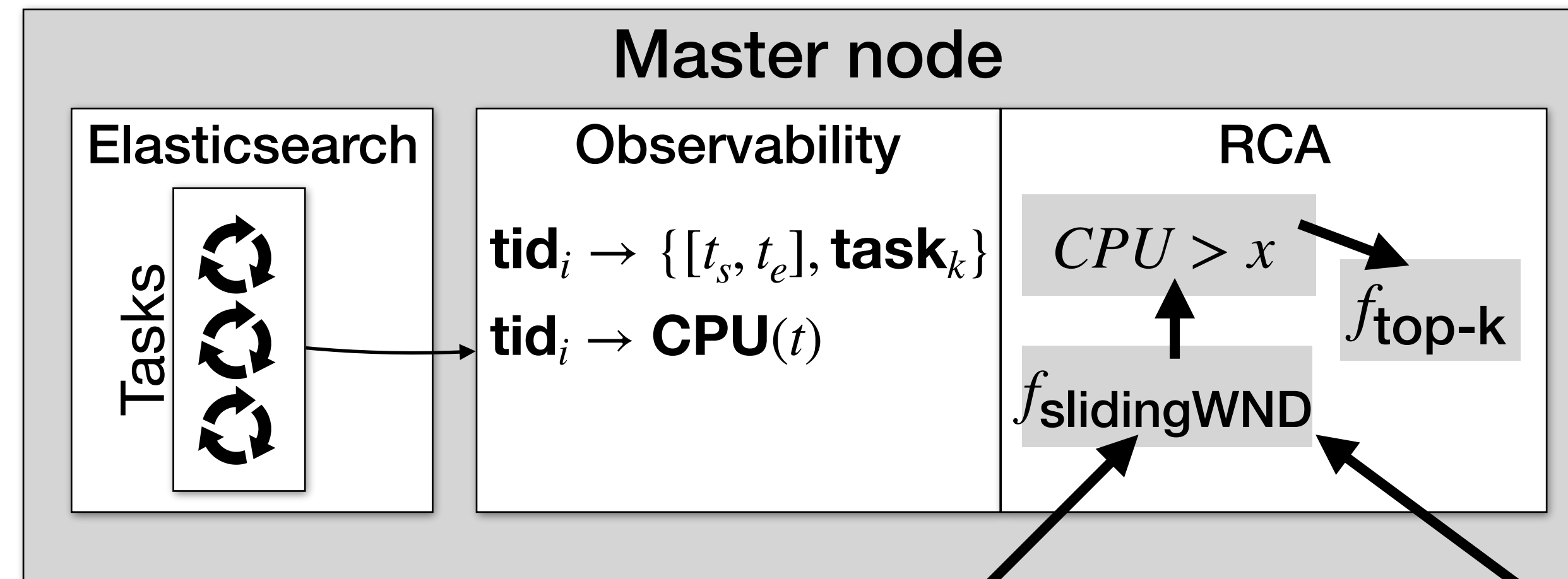
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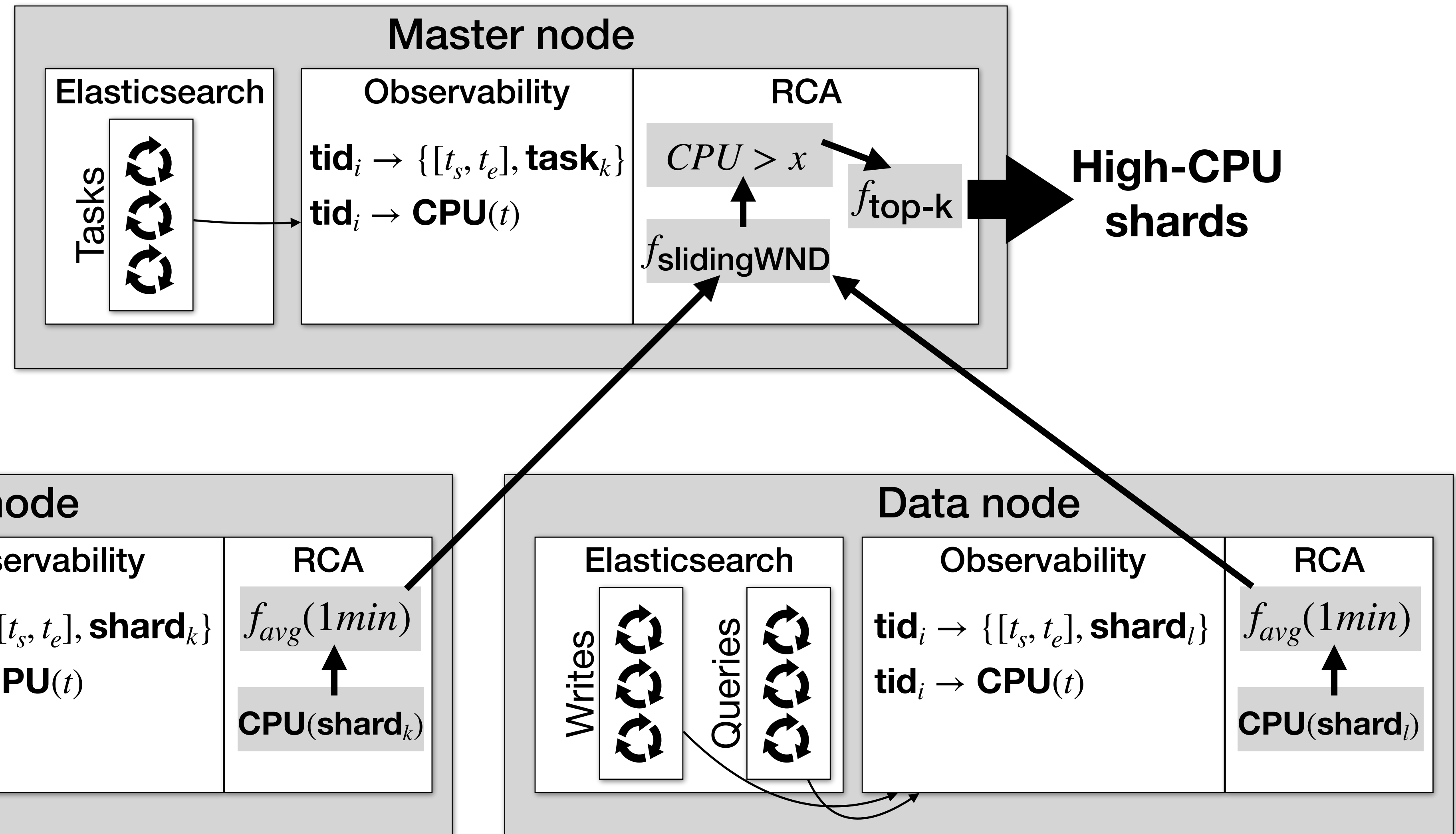
Example RCA: High-CPU Shards



Example RCA: High-CPU Shards



Example RCA: High-CPU Shards



RCA framework enables...

- Distributed ML dataflows for dynamic performance analyses (e.g., anomaly detection front ends)
- Gossip for fault tolerant cluster-wide views
- Centralized and decentralized control

RFC for Root Cause Analysis

- Posted on Open Distro Github
<https://github.com/opendistro-for-elasticsearch/performance-analyzer/blob/master/rca/>
- Contributions and comments welcome!
<https://github.com/opendistro-for-elasticsearch/performance-analyzer/issues/73>

RFC: Root Cause Analysis

SEPTEMBER 2019

Introduction

The Open Distro for Elasticsearch Performance Analyzer captures Elasticsearch and JVM layer activity, and lower-level resource usage (e.g., disk, network, CPU and memory) of these activities. Based on this instrumentation, Performance Analyzer computes and exposes *diagnostic* metrics, with the goal of enabling Elasticsearch users and administrators to measure and understand bottlenecks in their Elasticsearch clusters. The Open Distro for Elasticsearch PerfTop client provides real time visualization of these diagnostic metrics to surface bottlenecks to Elasticsearch users and operators.

This RFC proposes a design for a framework that builds on the Performance Analyzer to support root cause analysis (RCA) of performance and reliability problems for Elasticsearch instances. This framework would conduct real time root cause analysis of such problems using Performance Analyzer metrics. Root cause analysis can significantly improve operations, administration and provisioning of Elasticsearch clusters, and it can enable Elasticsearch client teams to tune their workloads to reduce errors.

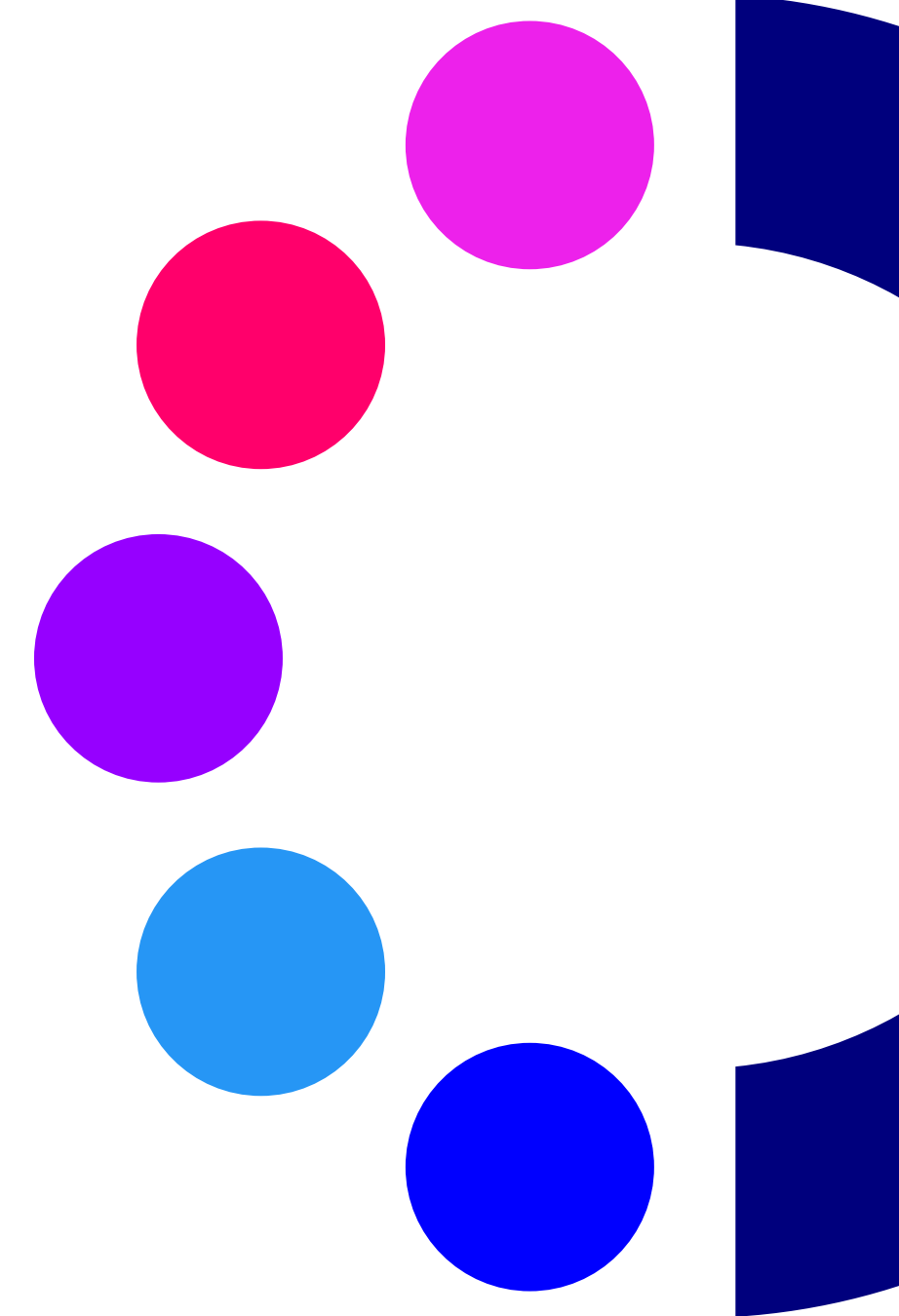
Root Cause Analysis

The first step is to define what we mean by a “root cause”. We define a root cause as a function of one or more *symptoms*. A symptom is an operation over metrics or *other* symptoms. Root causes may also be a function of other root causes. Note that these operations may involve aggregations; for example, a symptom could consume a time average of a metric. In addition, for confidence, a root cause could be a computation over a sufficiently long window of time. This definition does not allow for cycles in the dependency graph between metrics and root causes. The following equations show an example of these relationships:

$$\begin{aligned} \text{root_cause}_i &= f_{\text{RCA}}(\text{symptom}_k, \text{root_cause}_j) \\ \text{symptom}_k &= f_{\text{Symptom}}(\text{metric}_1, f_{\text{Avg}}(\text{metric}_2)) \end{aligned}$$

PART-II:

Real Time ML-based Anomaly Detection for Data Analytics

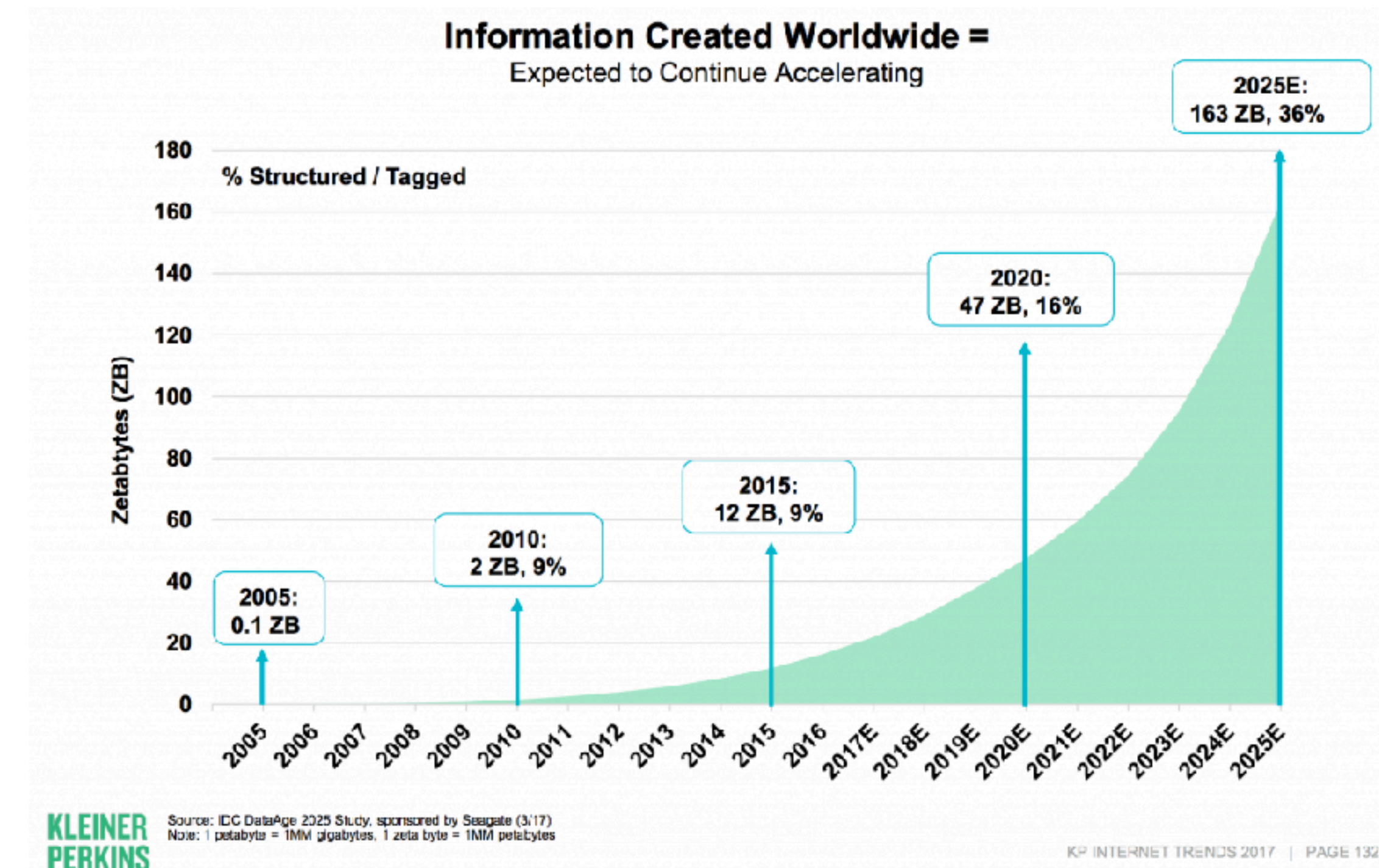


This talk

- Overview of an ML project
 - Interfaces and UX design
 - Algorithm design
 - Systems design

Analytics over-and-above DBs

- Exponential growth in data volume
- Queries need to be revisited often
- DB users need data deep dives
 - frequent... for actionability

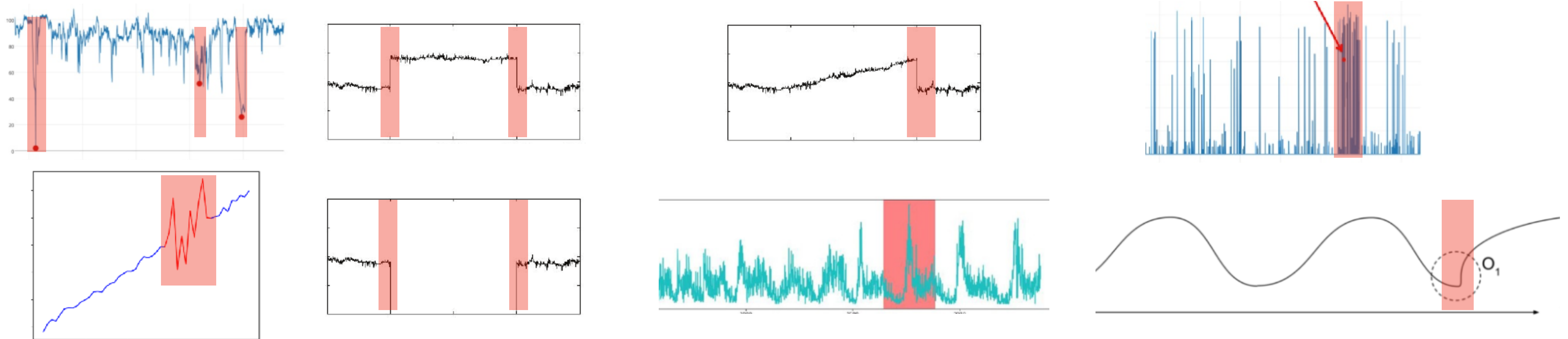


Analytics: Goals

- Goal: automatically mine real time patterns across ingestion streams
 - actionable insights
 - interactive, guided exploration
 - Simple experience (~~labels, algorithms, models, hyperparameters~~)
 - Anomaly detection first step!

Anomaly Detection

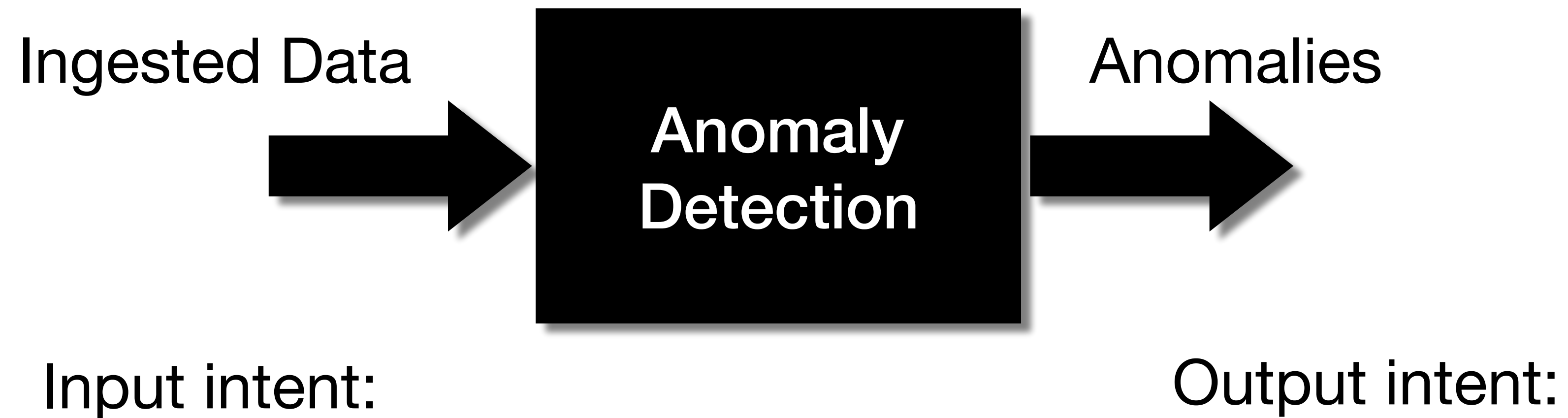
“Anomalies are data patterns that have different data characteristics from normal instances.”



Anomaly Detection: Requirements

- Overall: actionable outputs
- Algorithms aspects:
 - Streaming, unsupervised
 - High accuracy (precision and recall)
 - Low training time: ~ secs.
 - accounts for seasonality
- Systems aspects:
 - zero user cost
 - negligible overhead
 - fully distributed compute and heap
 - elastic

UX design: Actionability



- *features*: functions on fields
- *examples*: labeled data
- *tags* on data
- *validation*: mark anomalies
- *severity*: time spent browsing data

AD methods

- Supervised learning
 - Deep Learning
 - Classification
 - SVMs
- Unsupervised learning
 - statistical (timeseries)
 - nearest neighbor, clustering
 - subspace/spectral
 - density estimation

Anomaly Detection: A Survey

VARUN CHANDOLA, ARINDAM BANERJEE, and VIPIN KUMAR

University of Minnesota

Anomaly detection is an important problem that has been researched within diverse research areas and application domains. Many anomaly detection techniques have been specifically developed for certain application domains, while others are more generic. This survey tries to provide a structured and comprehensive overview of the research on anomaly detection. We have grouped existing techniques into different categories based on the underlying approach adopted by each technique. For each category we have identified key assumptions, which are used by the techniques to differentiate between normal and anomalous behavior. When applying a given technique to a particular domain, these assumptions can be used as guidelines to assess the effectiveness of the technique in that domain. For each category, we provide a basic anomaly detection technique, and then show how the different existing techniques in that category are variants of the basic technique. This template provides an easier and more succinct understanding of the techniques belonging to each category. Further, for each category, we identify the advantages and disadvantages of the techniques in that category. We also provide a discussion on the computational complexity of the techniques since it is an important issue in real applications. This survey will provide a better understanding of the different directions in which research has been done on this topic, and how techniques developed in one area can be applied in domains for which they were not intended to begin with.

6,300 citations
20 pages of refs

Categories and Subject Descriptors: H.2.8 [Database Management]: Database Applications—*Data mining*

General Terms: Algorithms

Additional Key Words and Phrases: Anomaly detection, outlier detection

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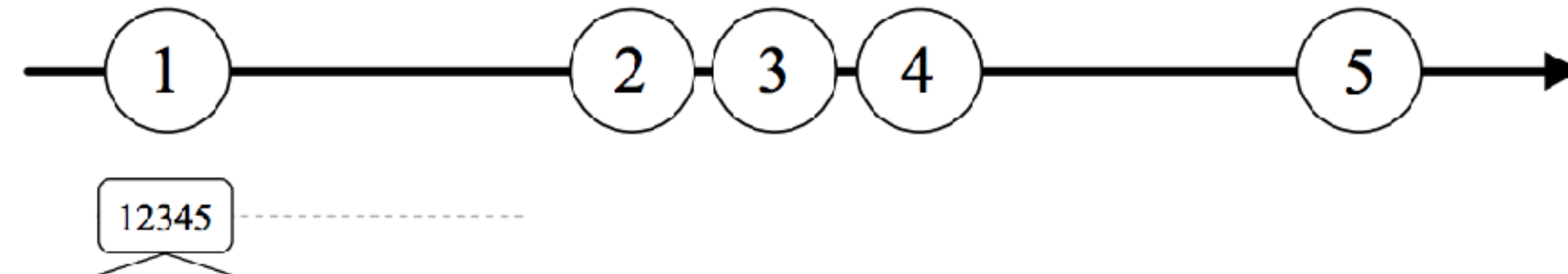
Random Forest

- Ensemble learning method
- Each tree updated probabilistically on input sample
- Output based on all trees

Density estimation by RF, 2003

- Estimates the PDF of data (a.k.a. histogram)

1-D data

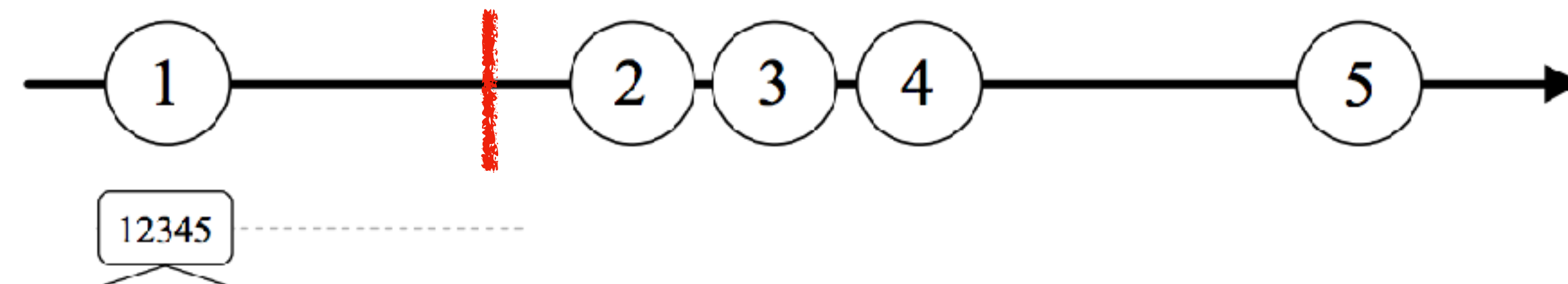


Random
forest via
random
partitioning

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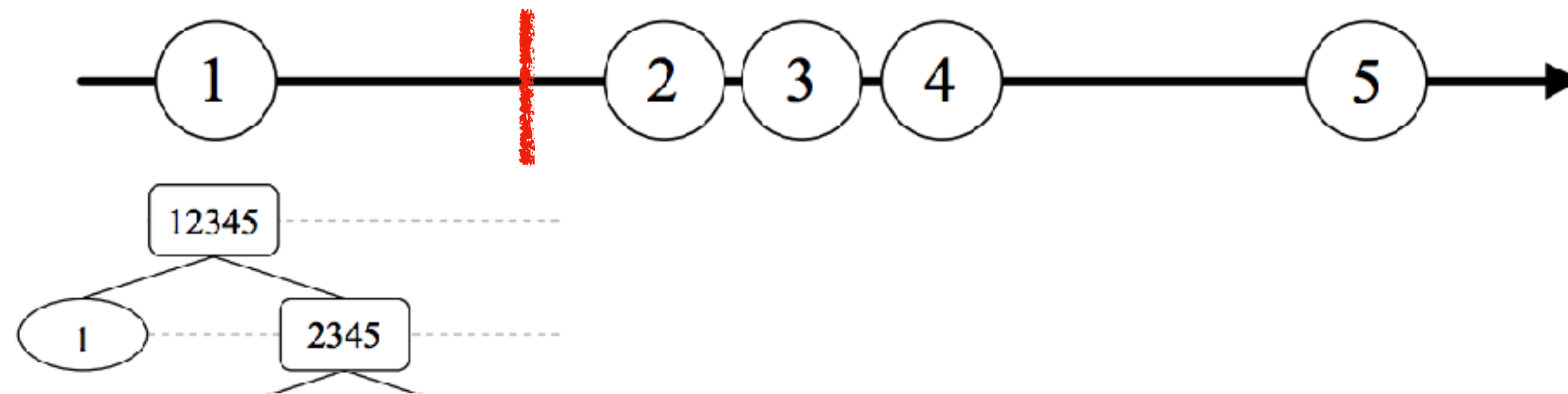


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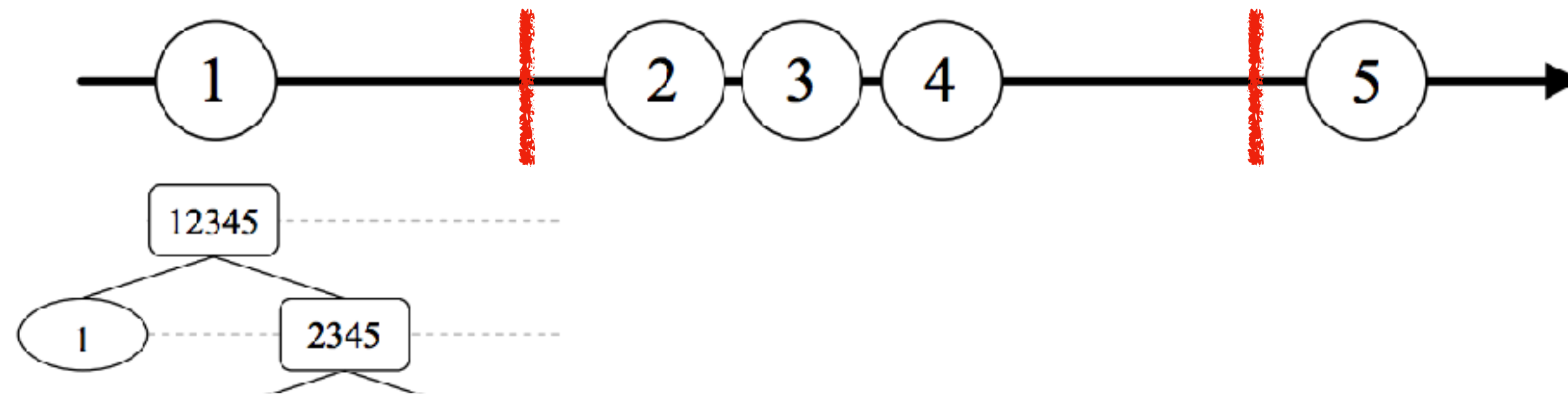


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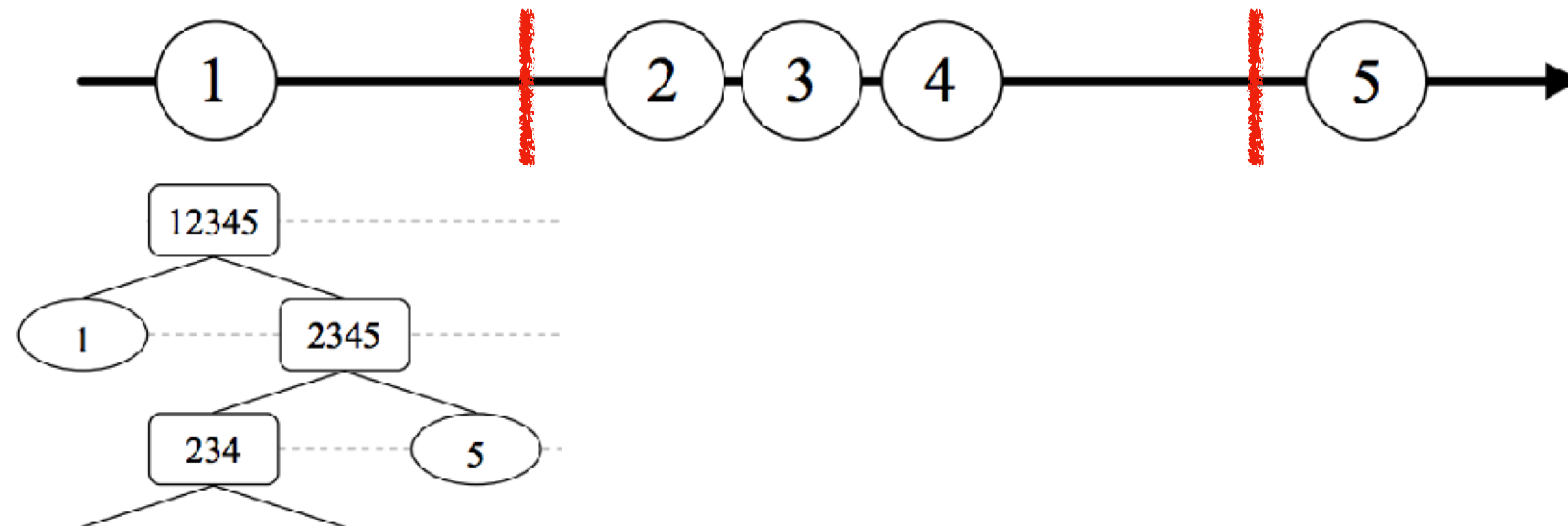
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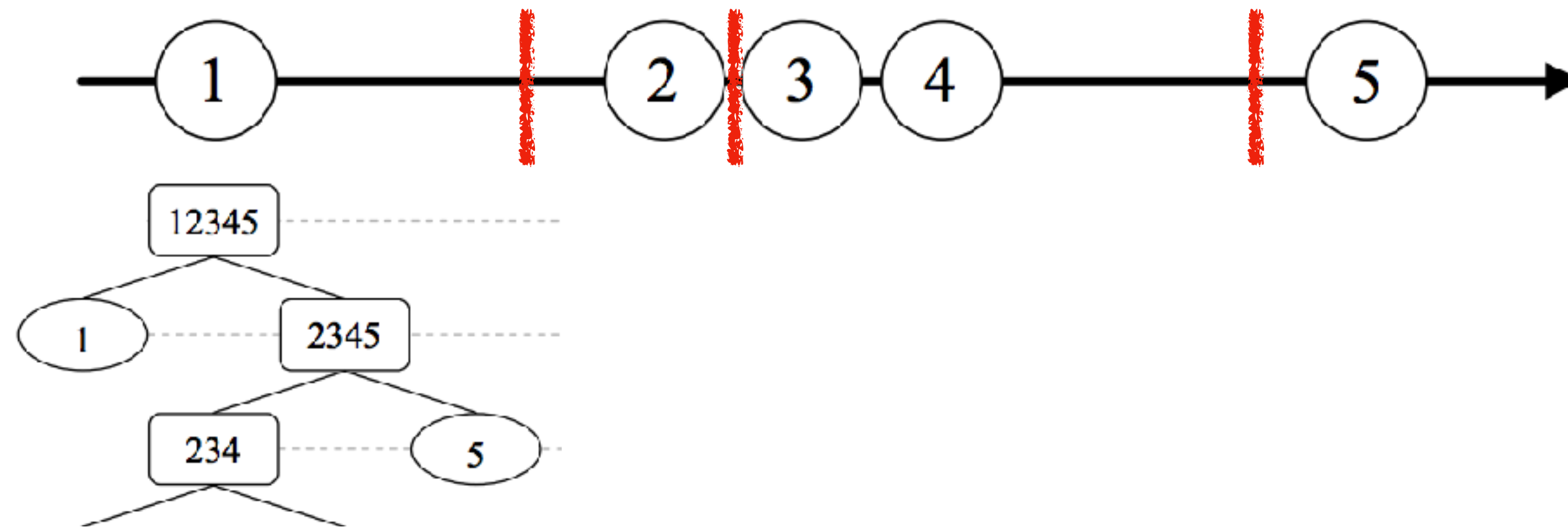


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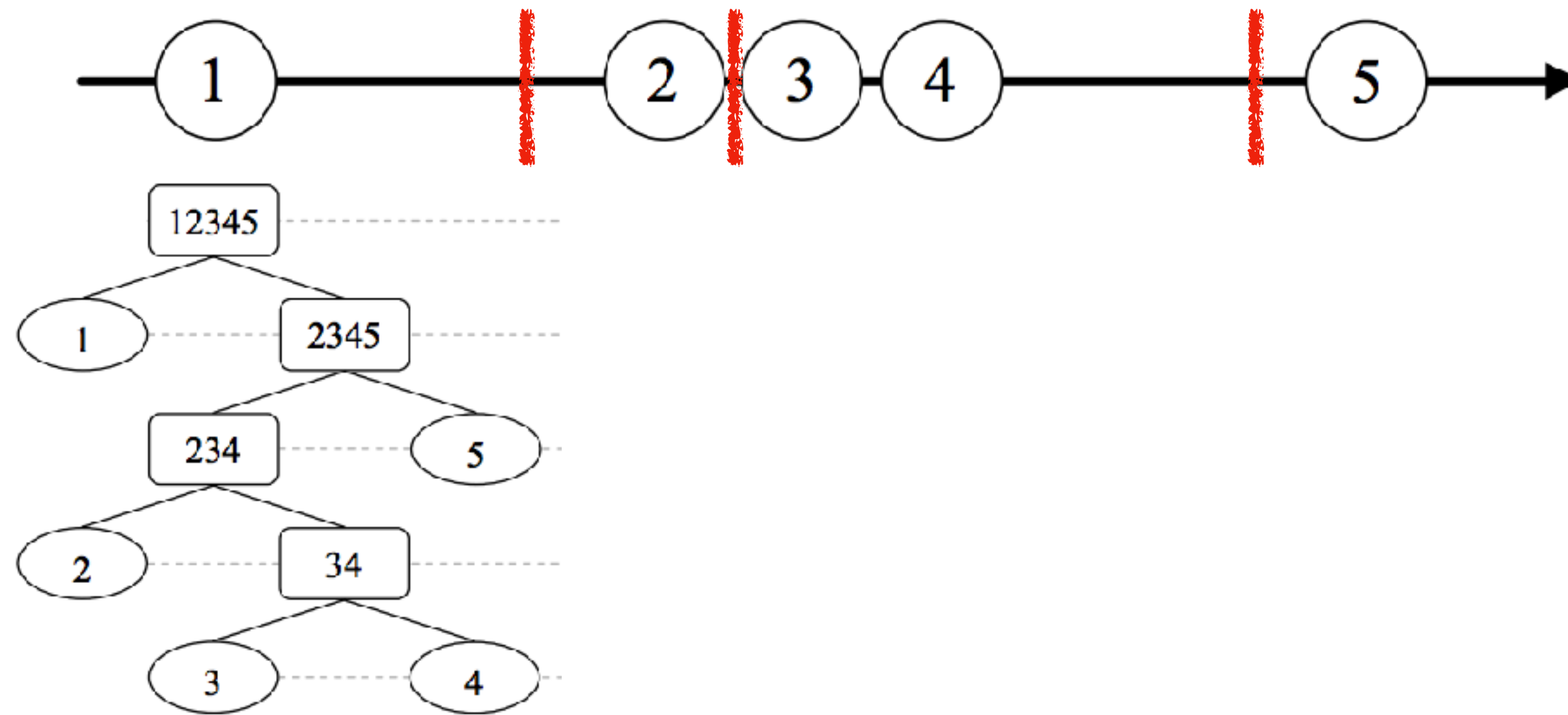


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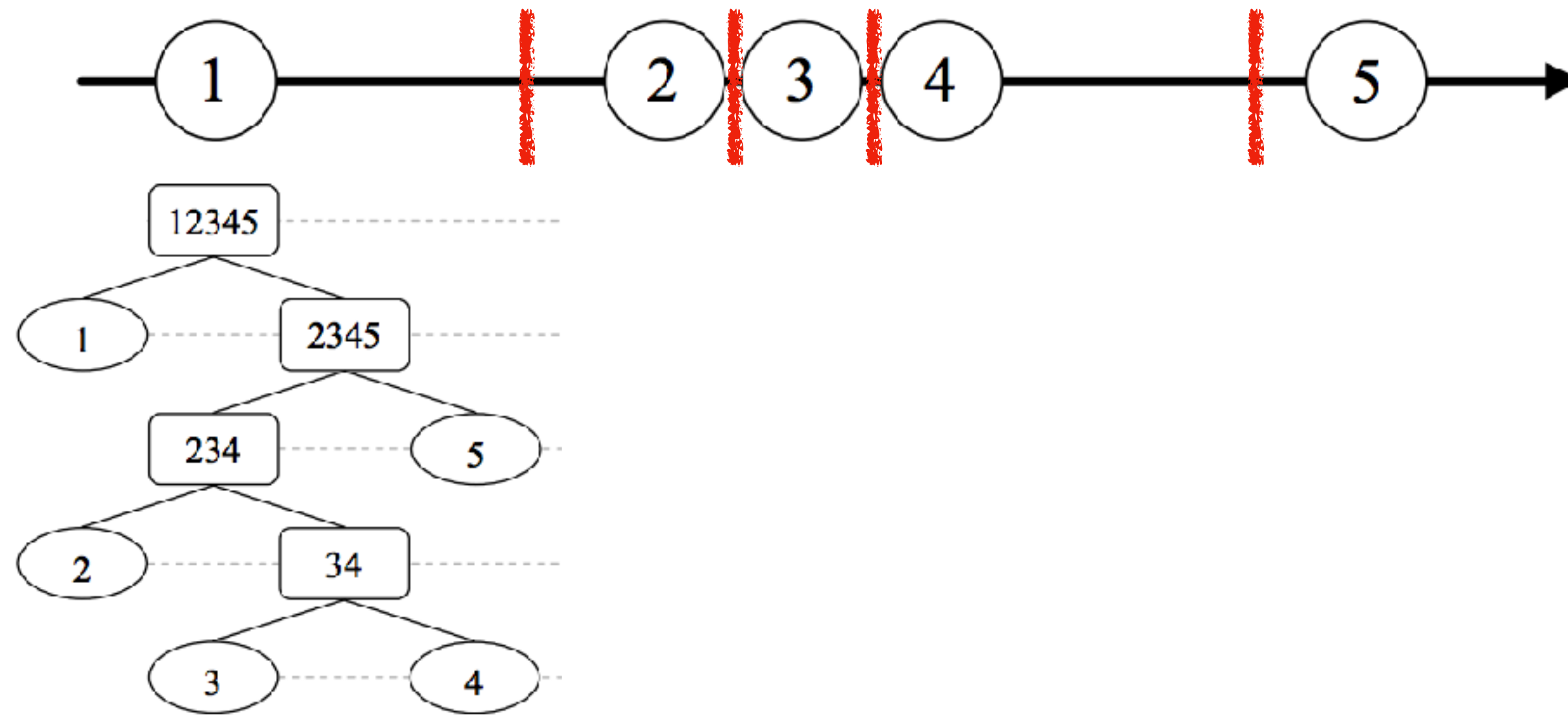


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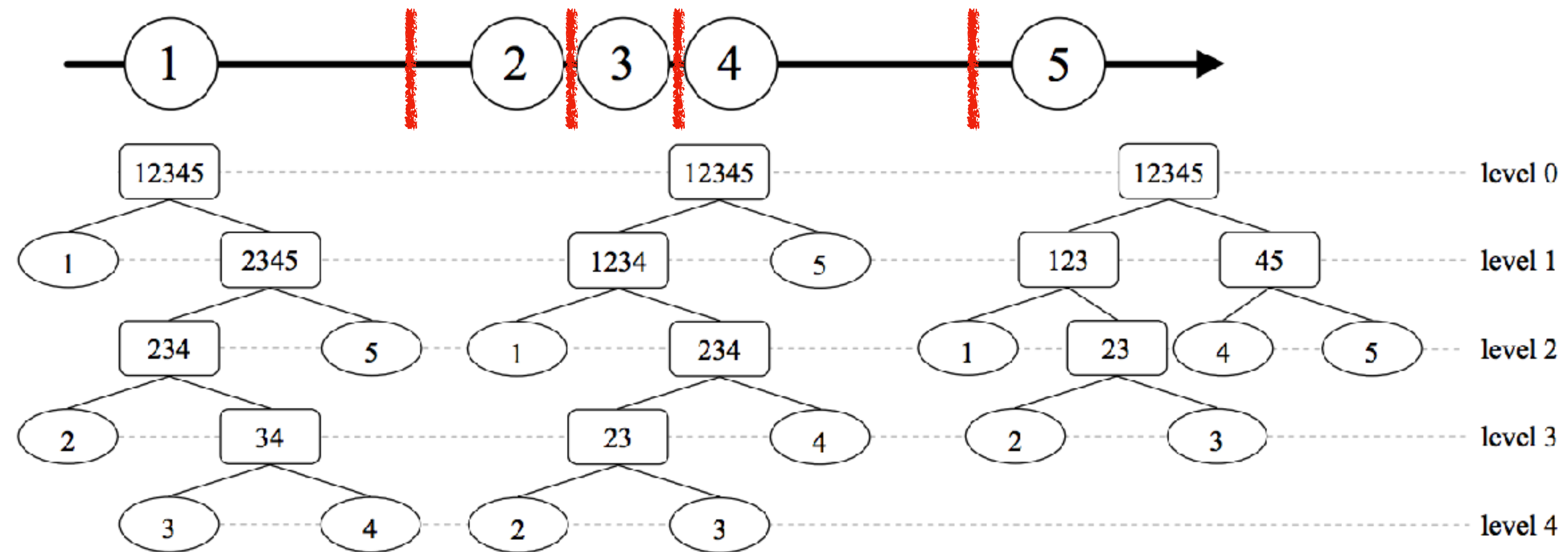


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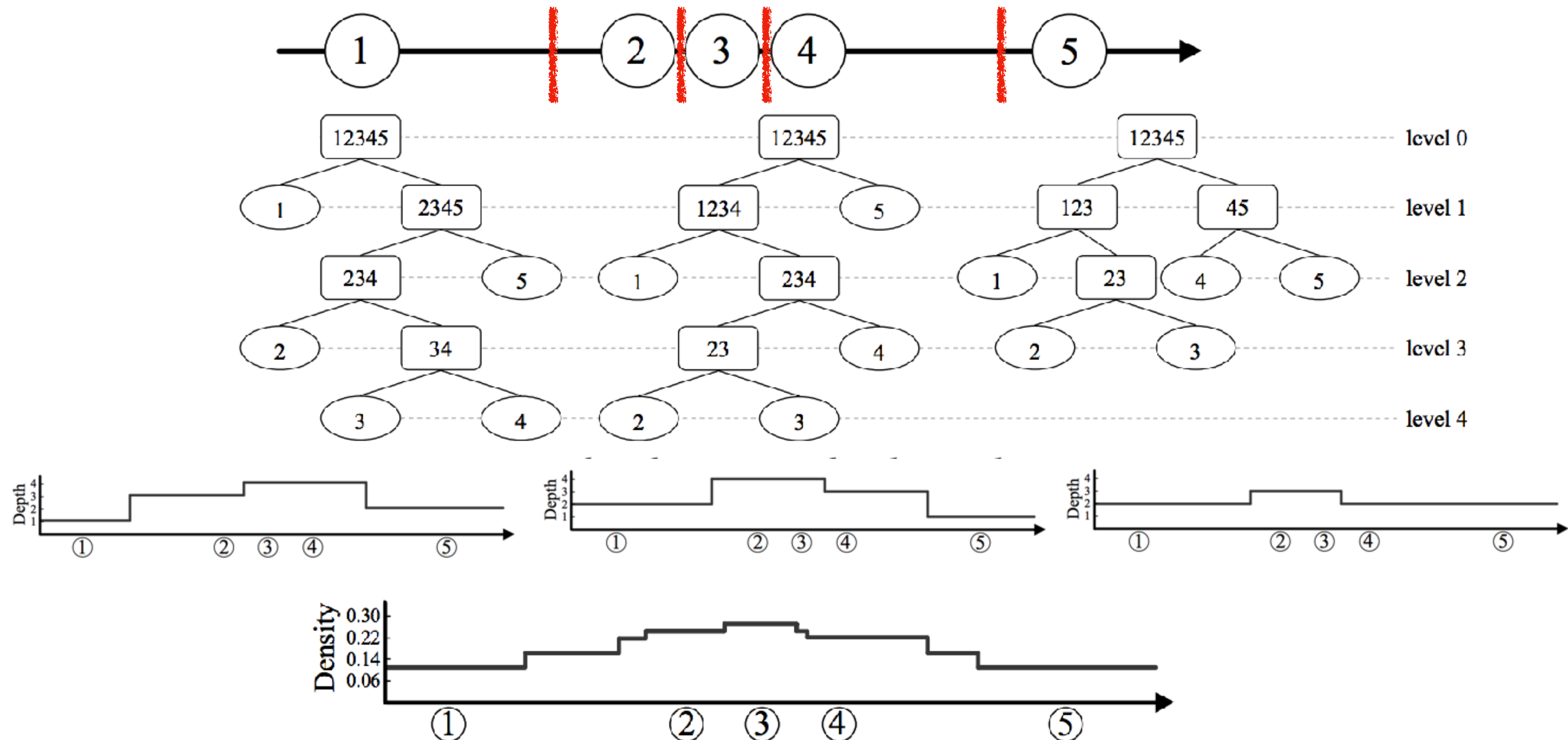
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Random forest via random partitioning

Data depths

PDF estimate



Density estimation by RF, 2003

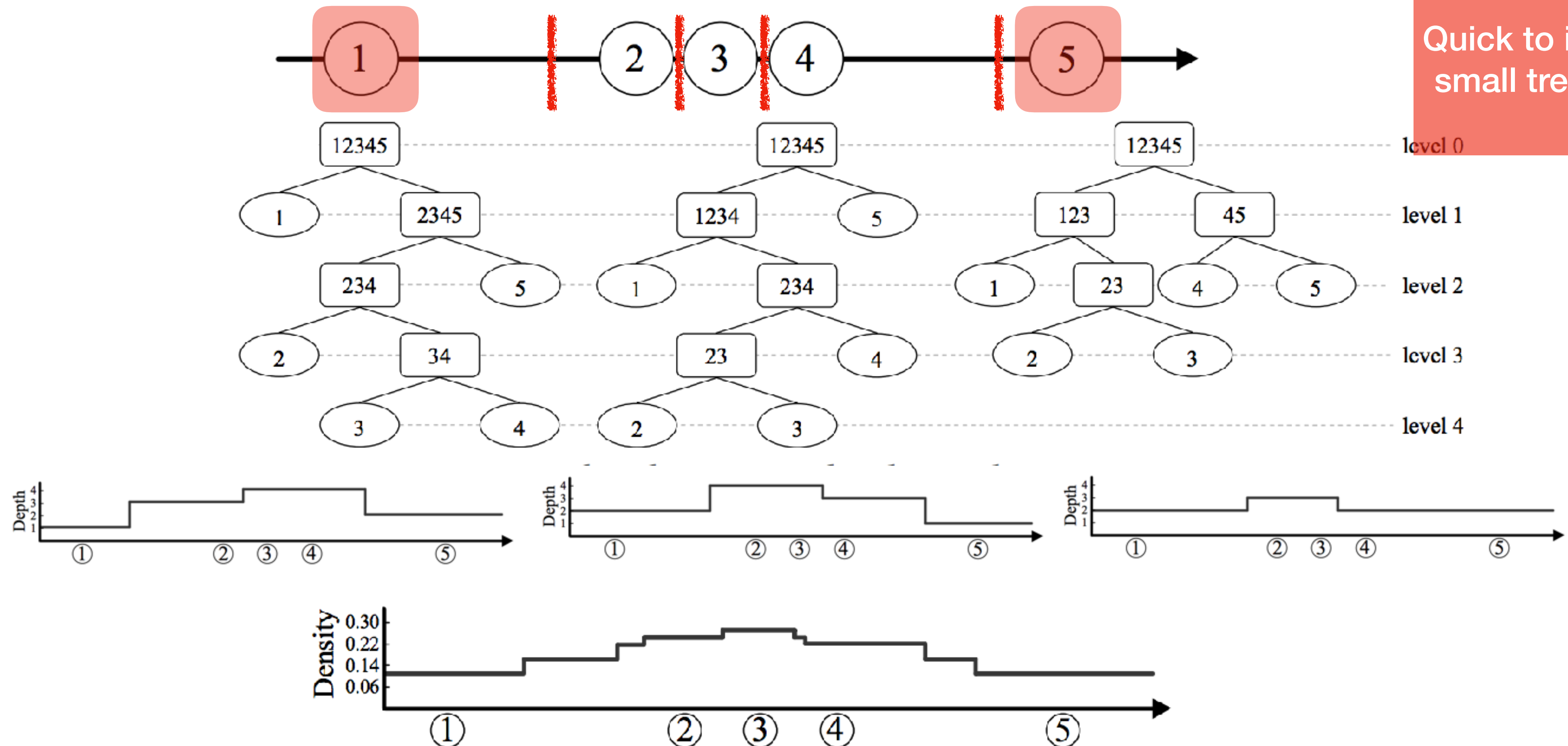
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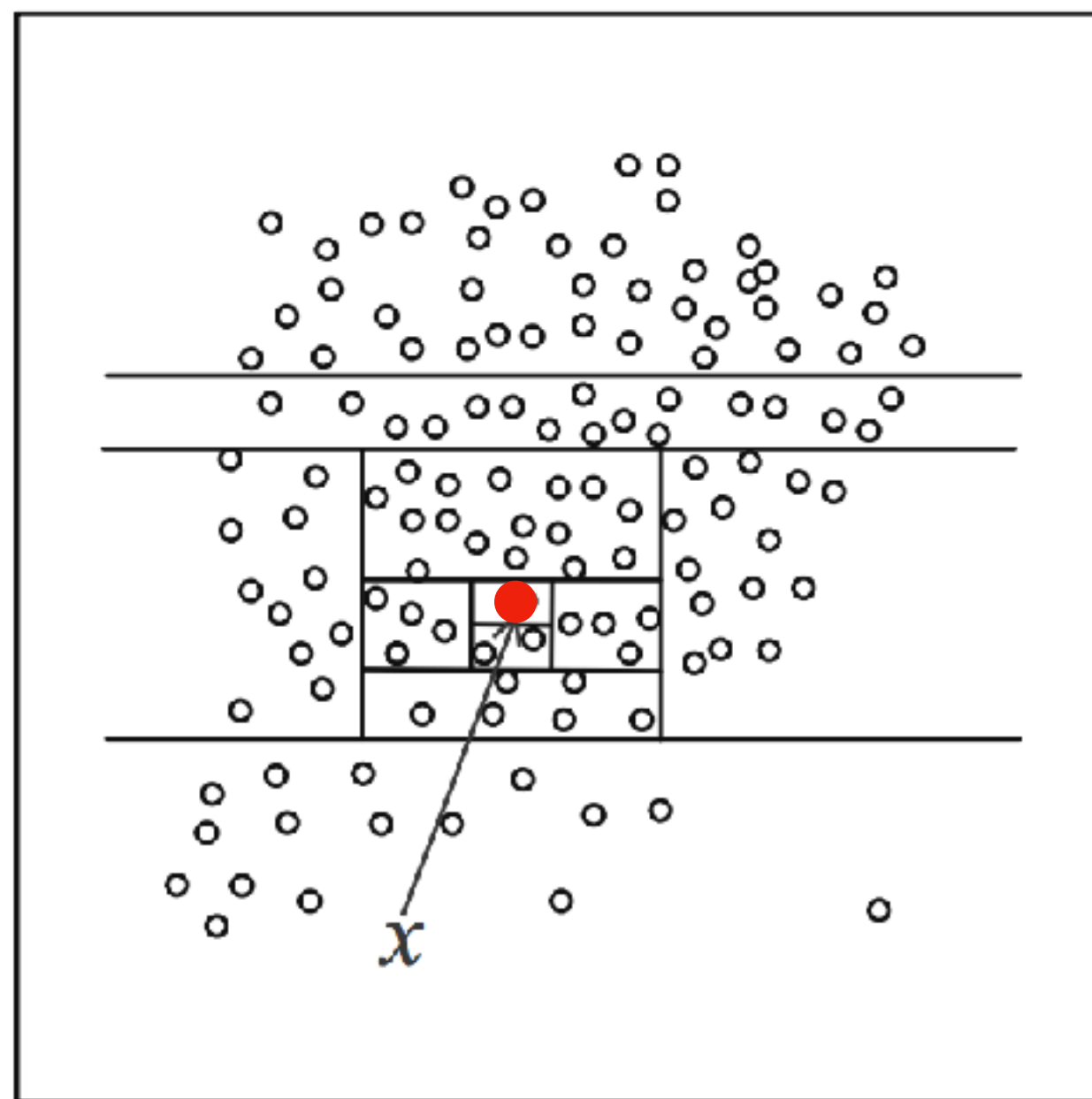
Isolation Forest, 2008

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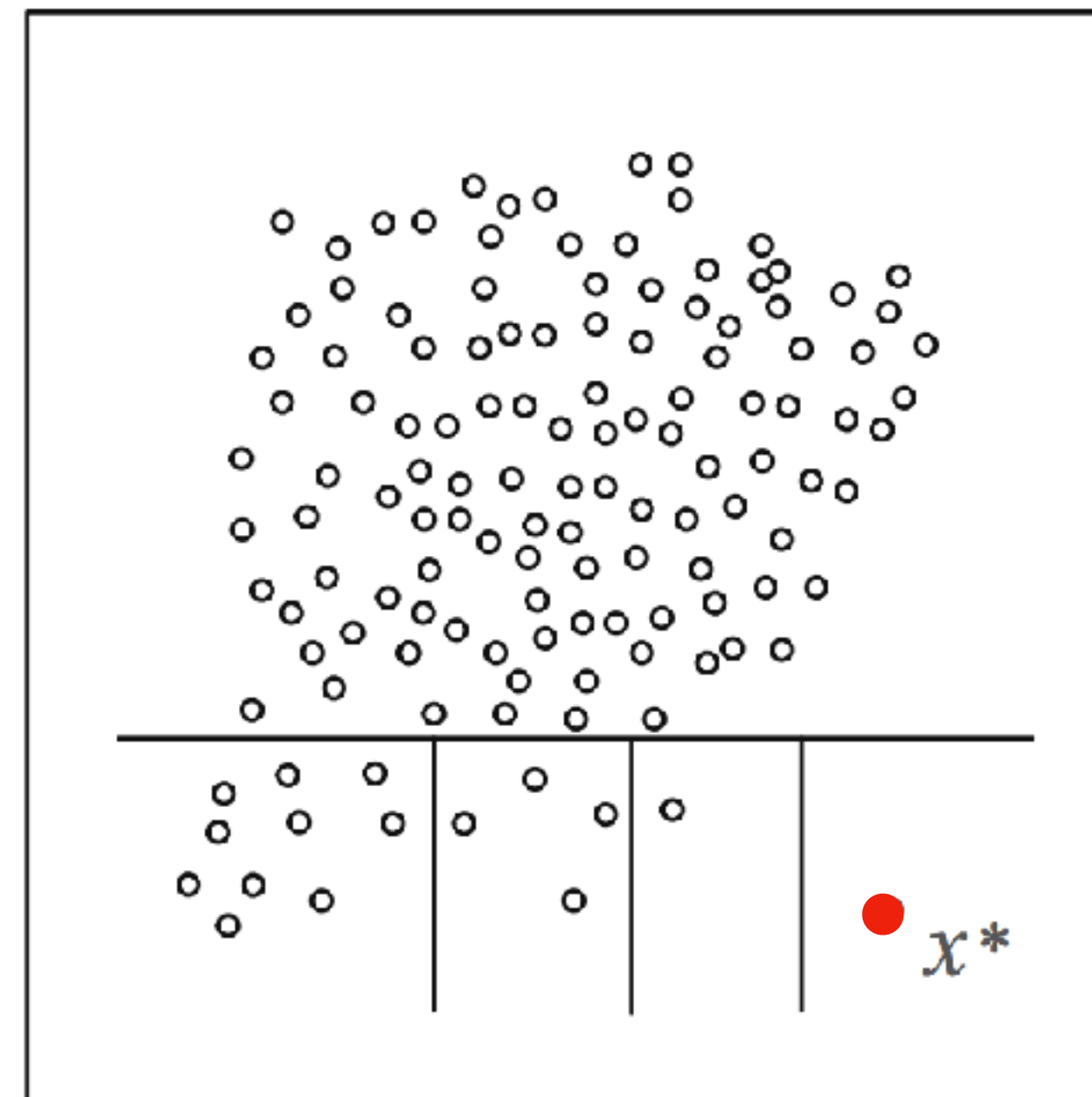
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normal point:
10 cuts



anomalous point:
4 cuts

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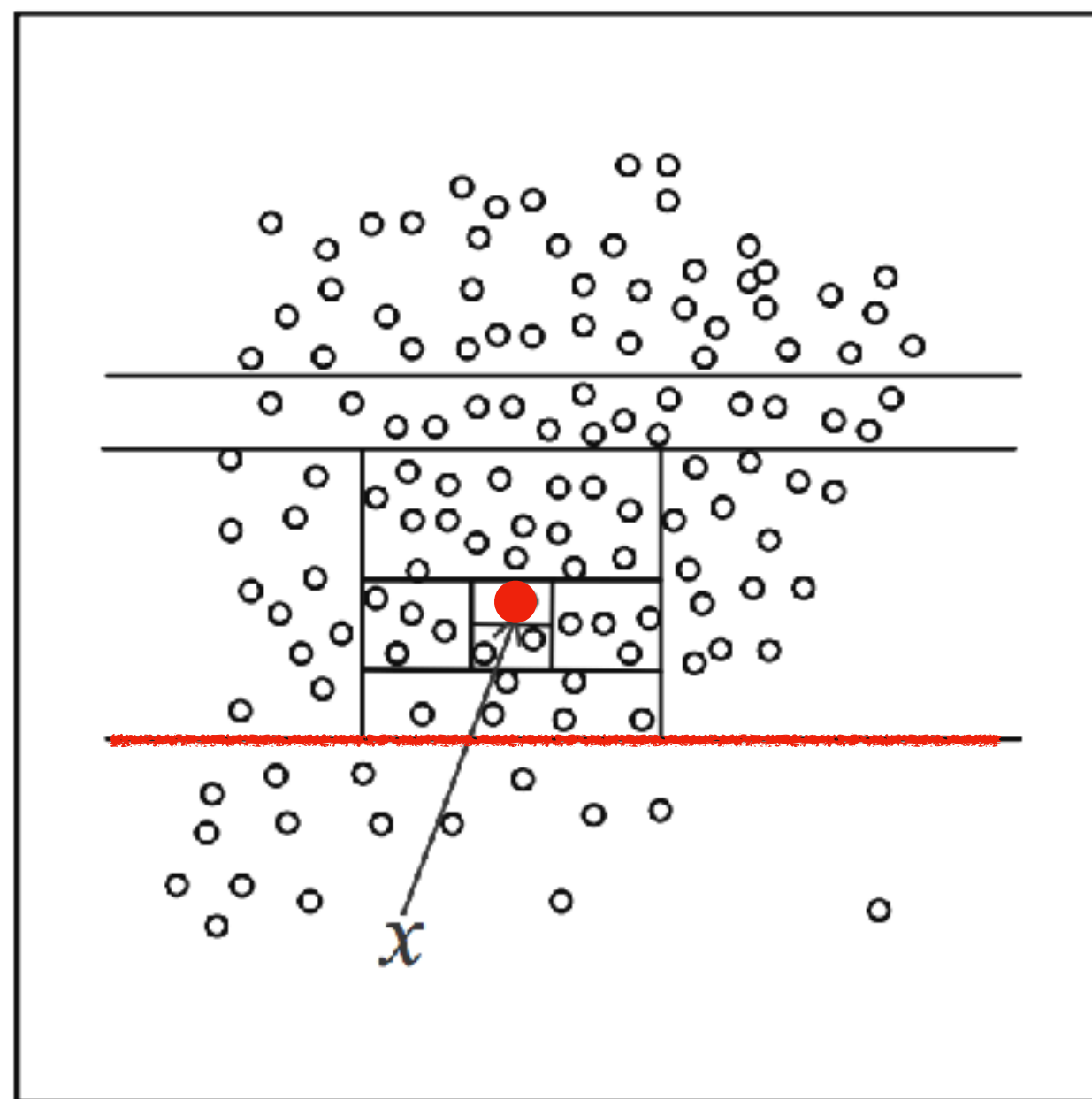
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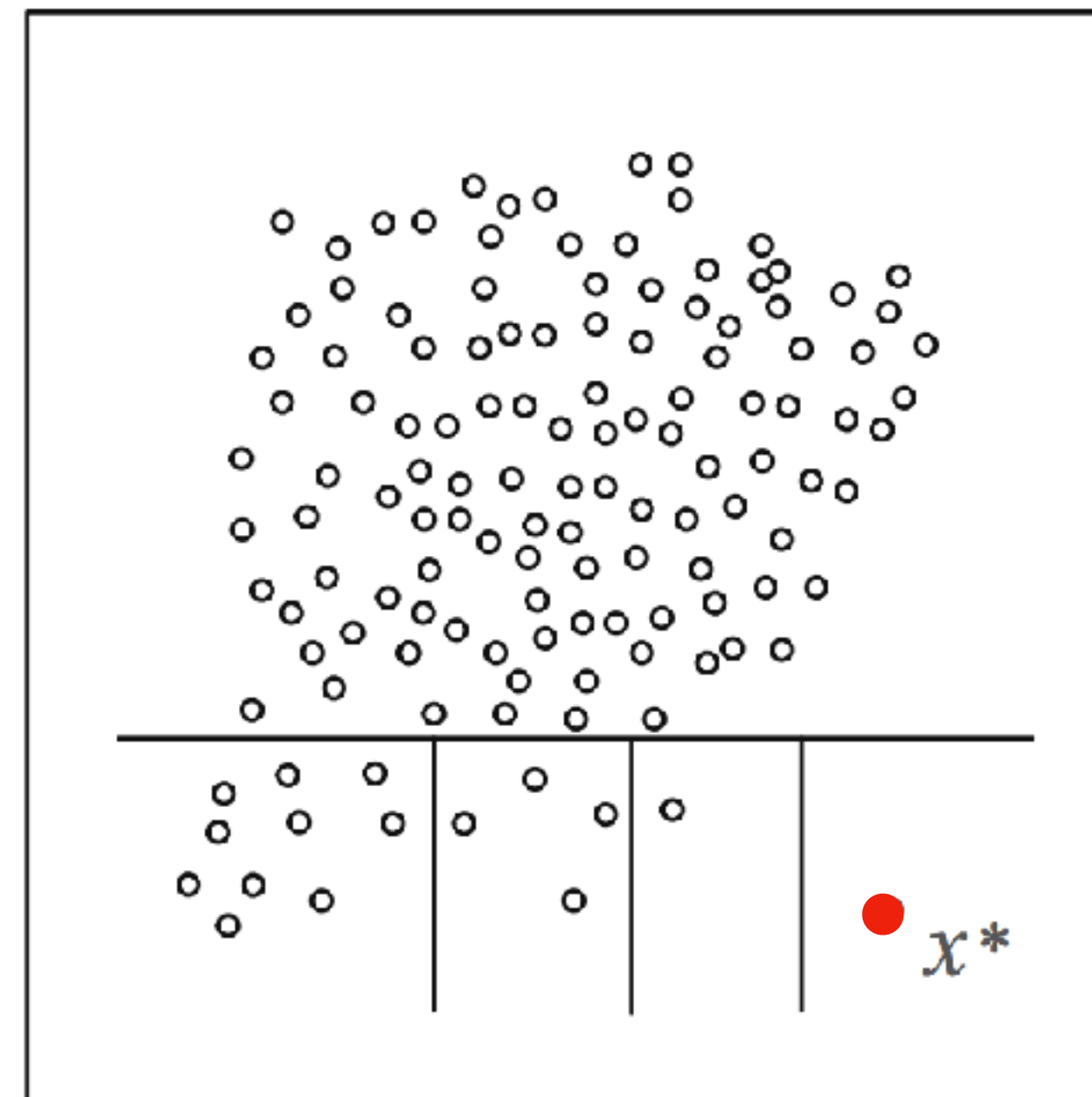
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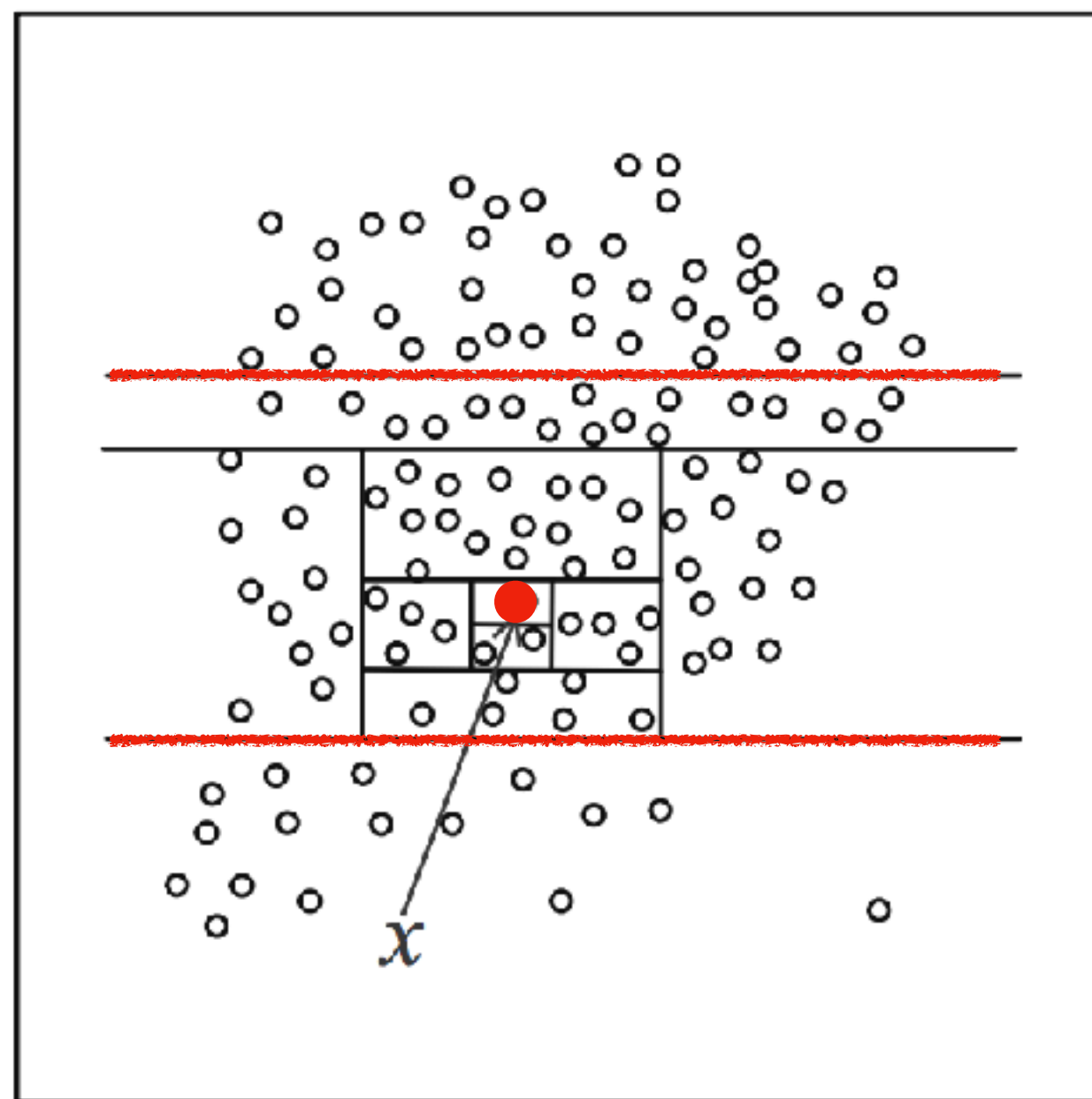
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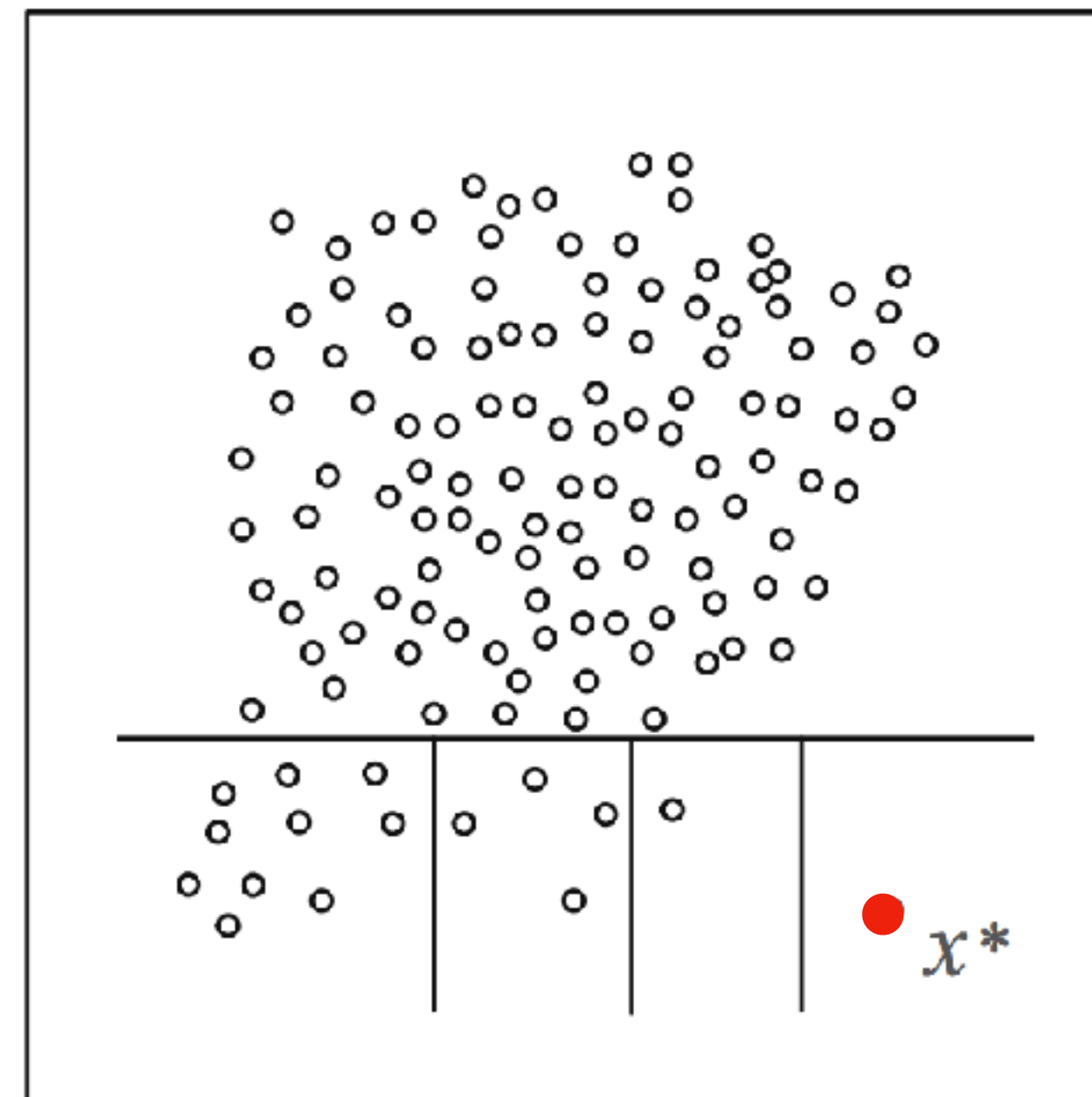
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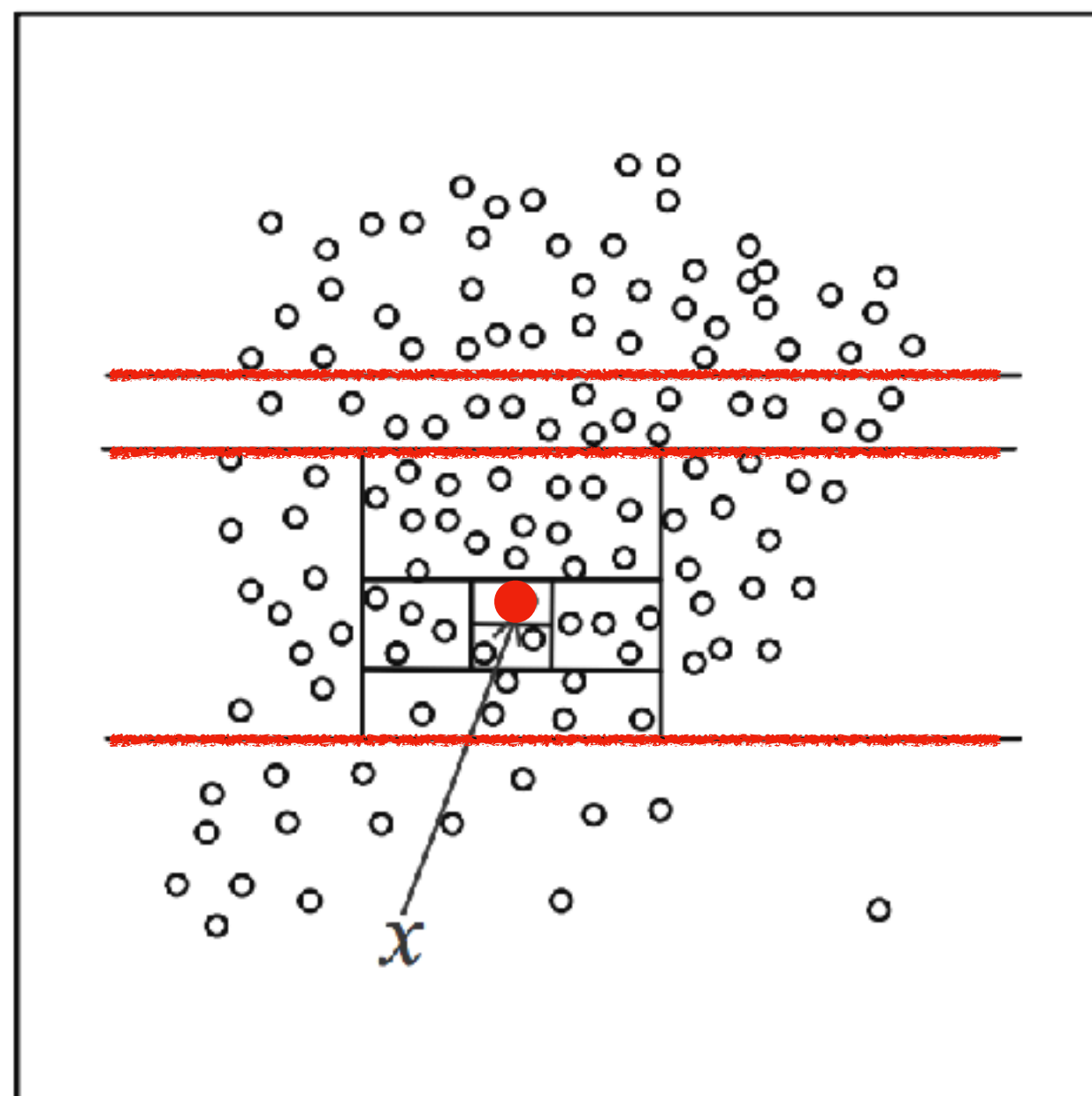
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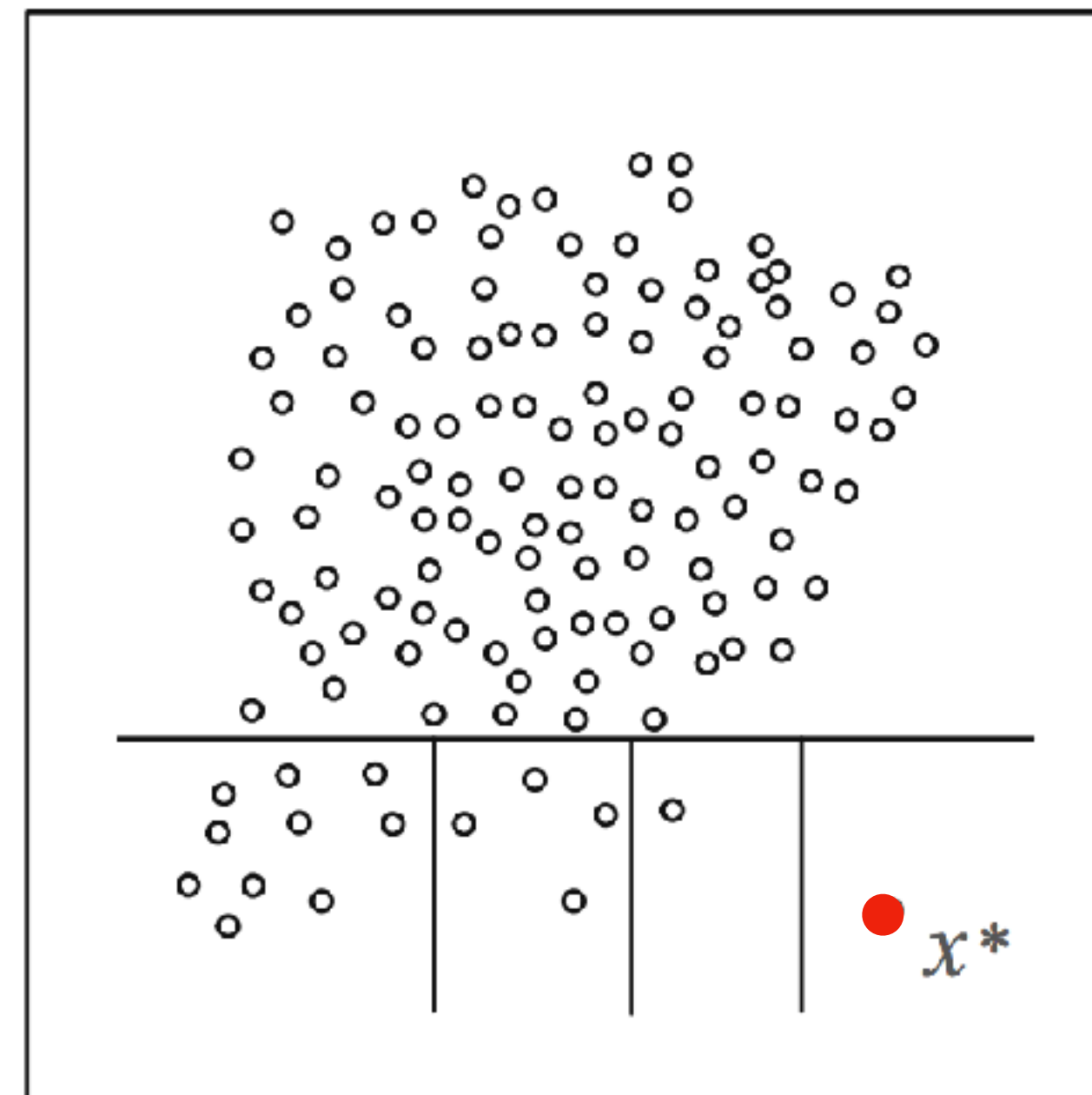
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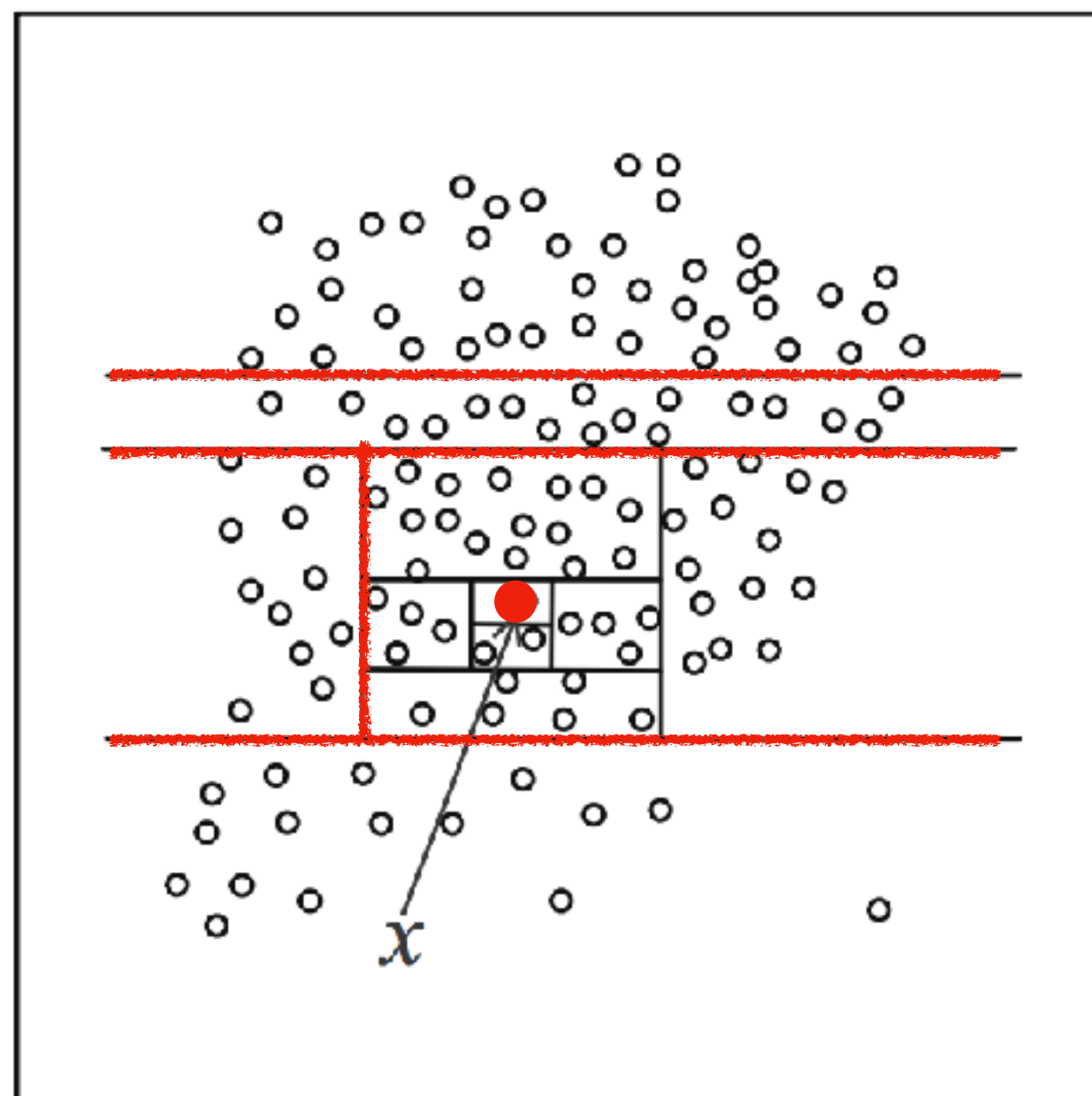
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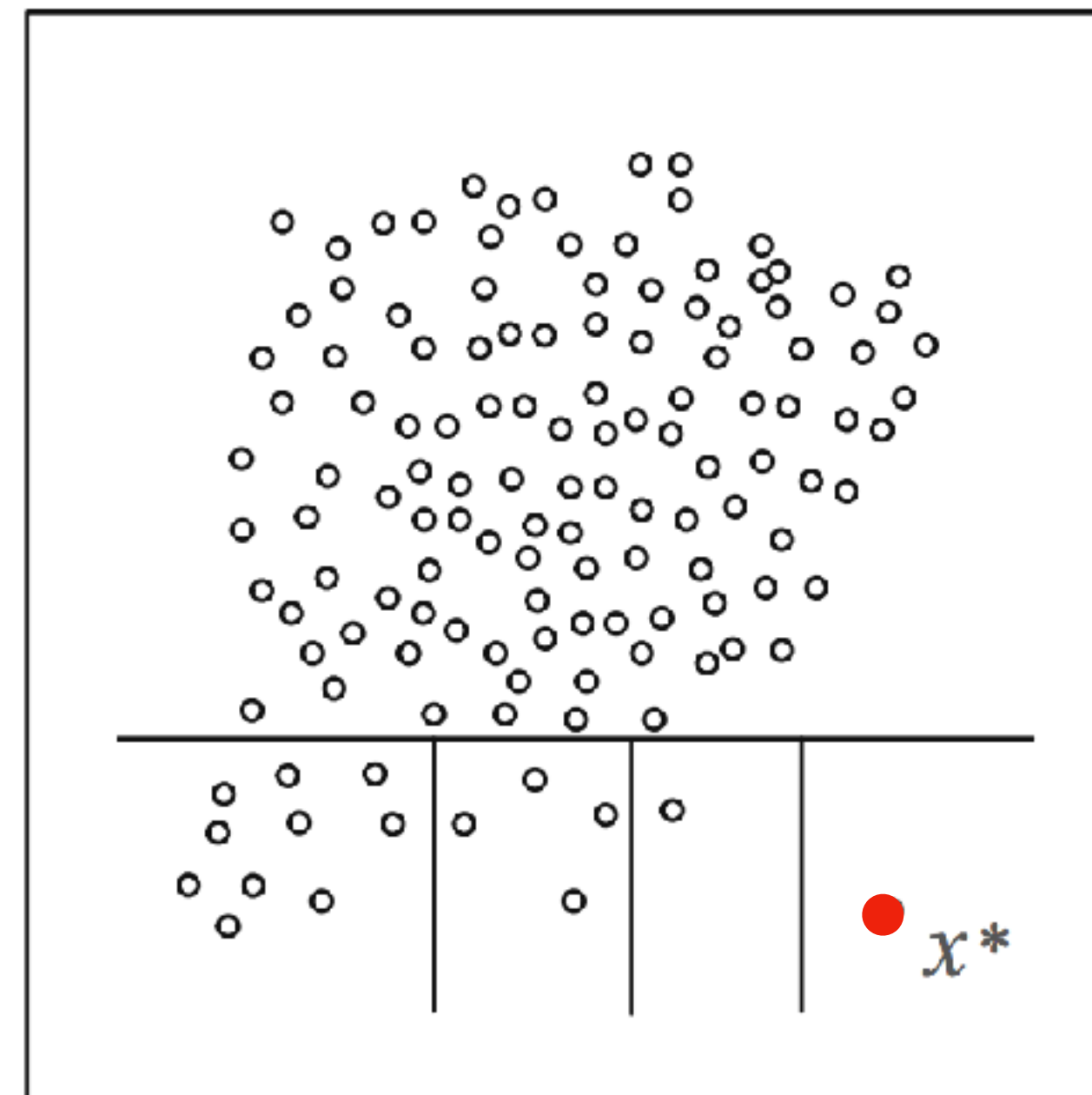
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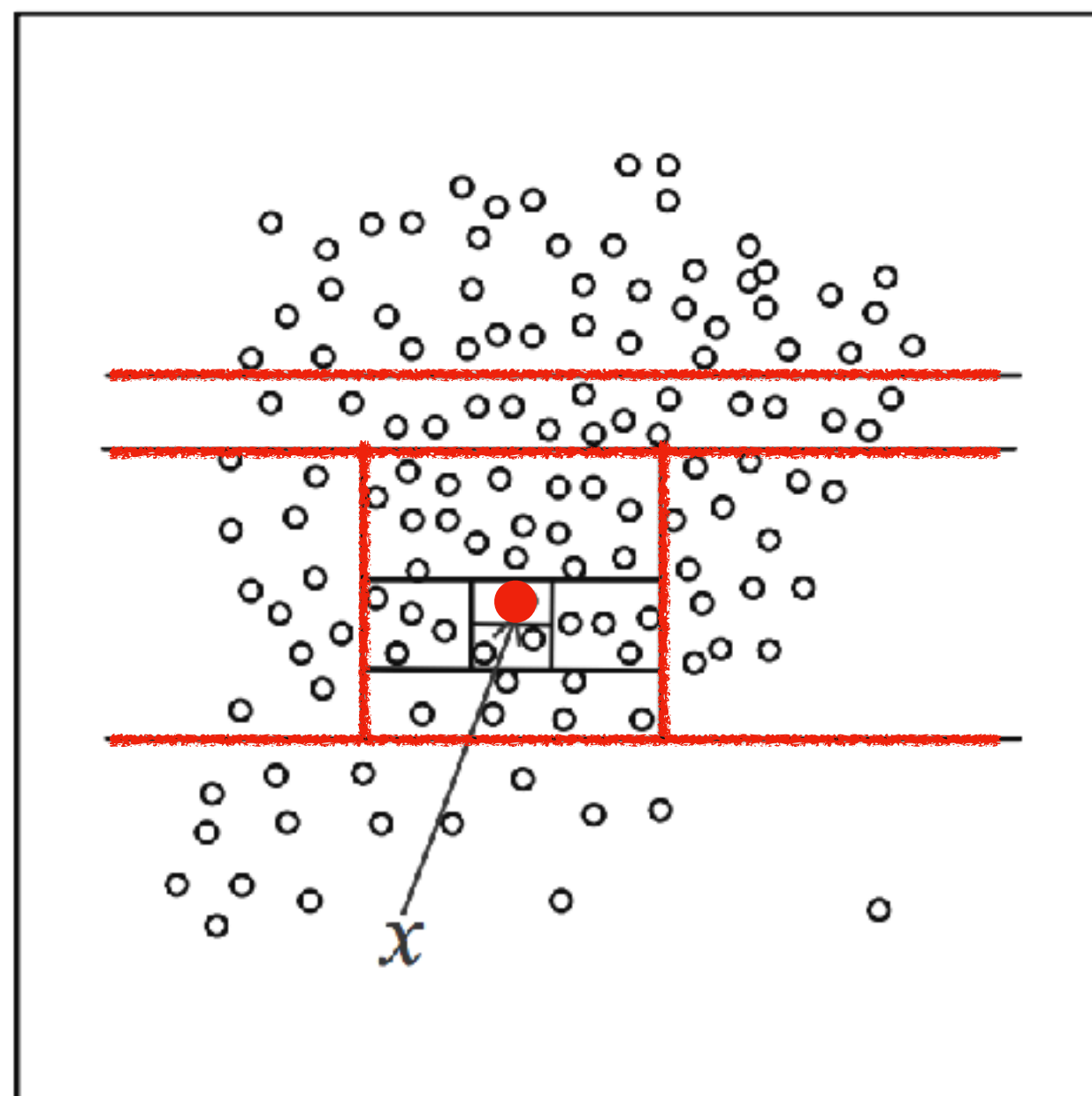
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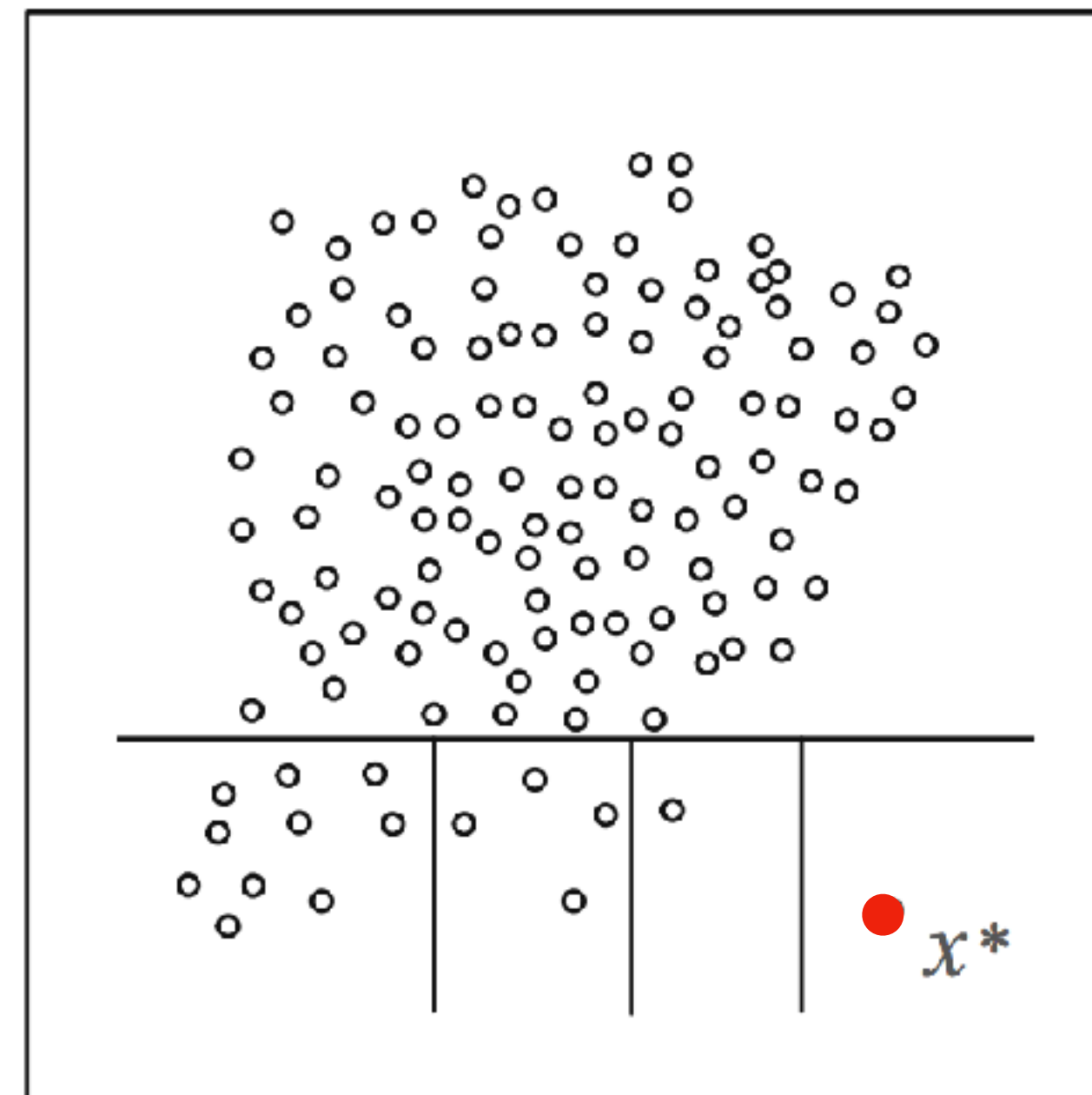
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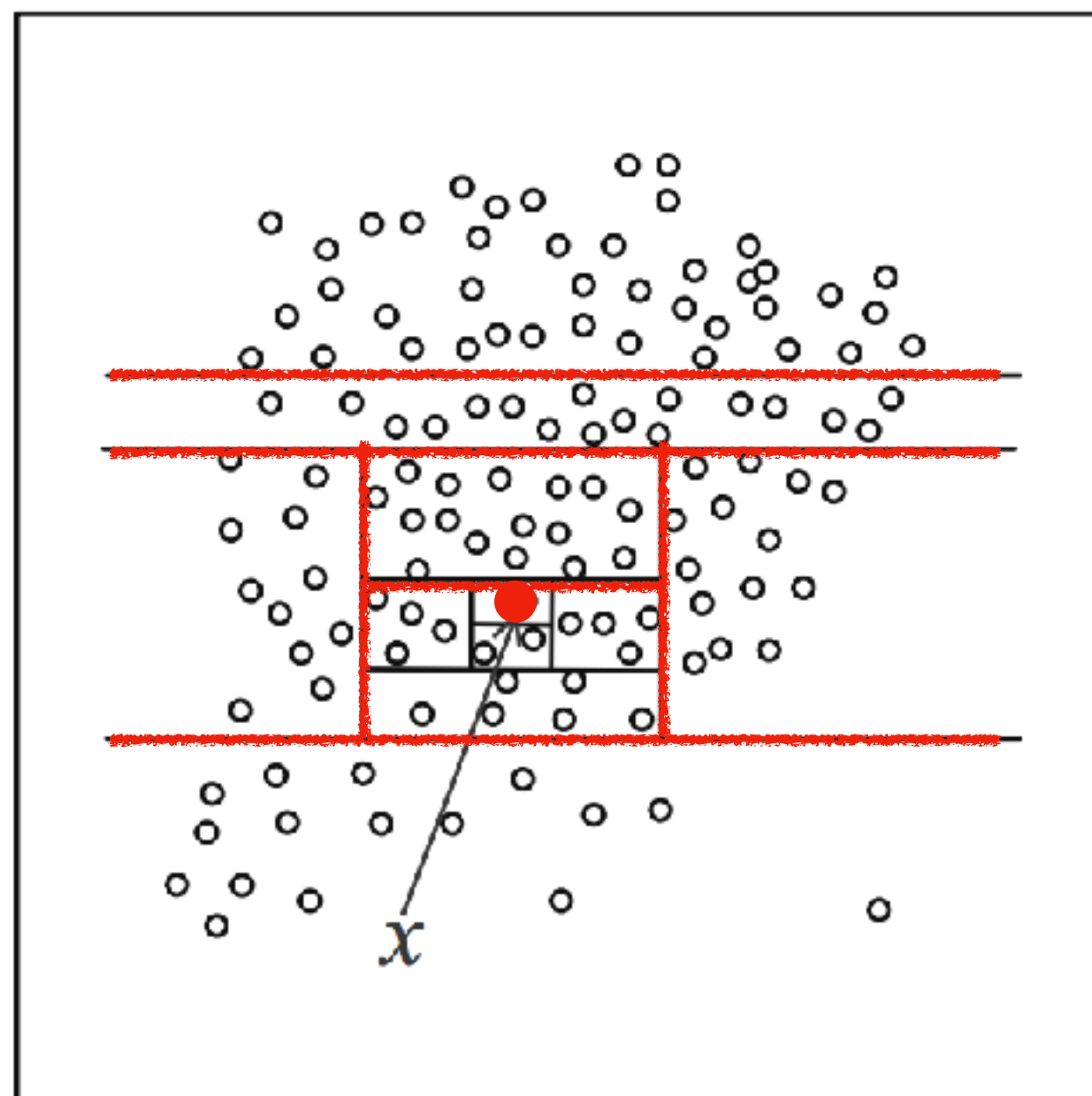
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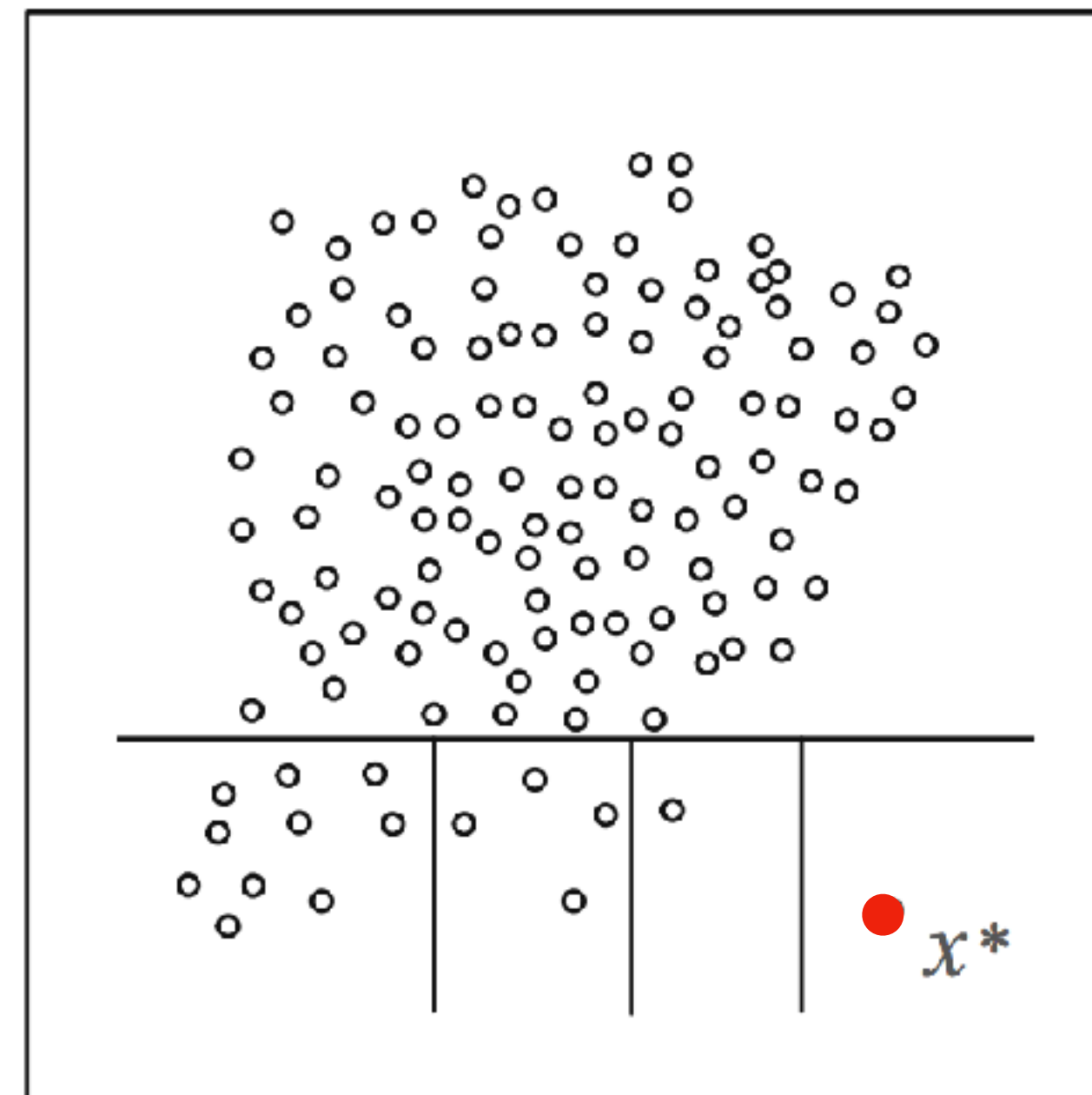
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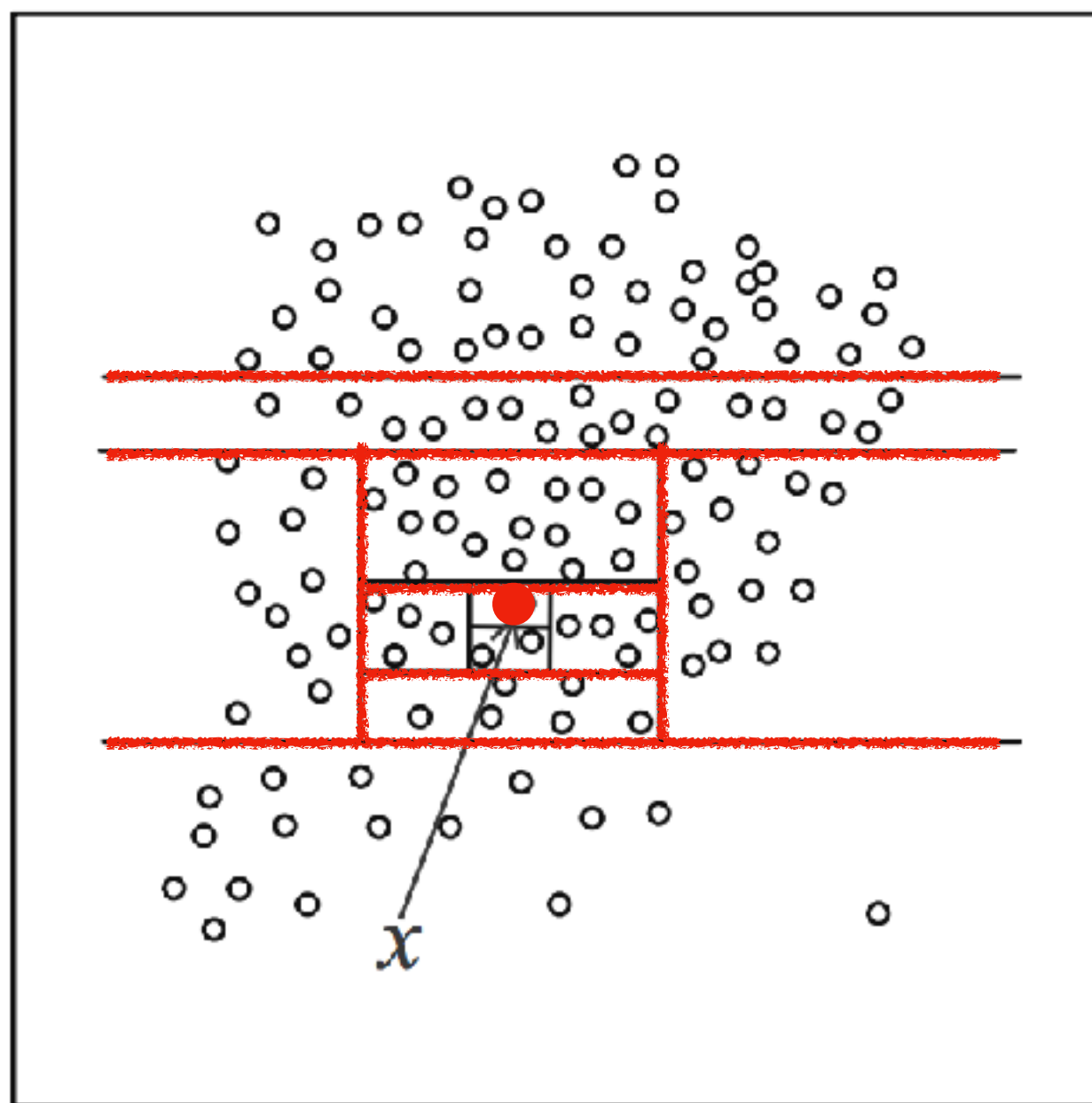
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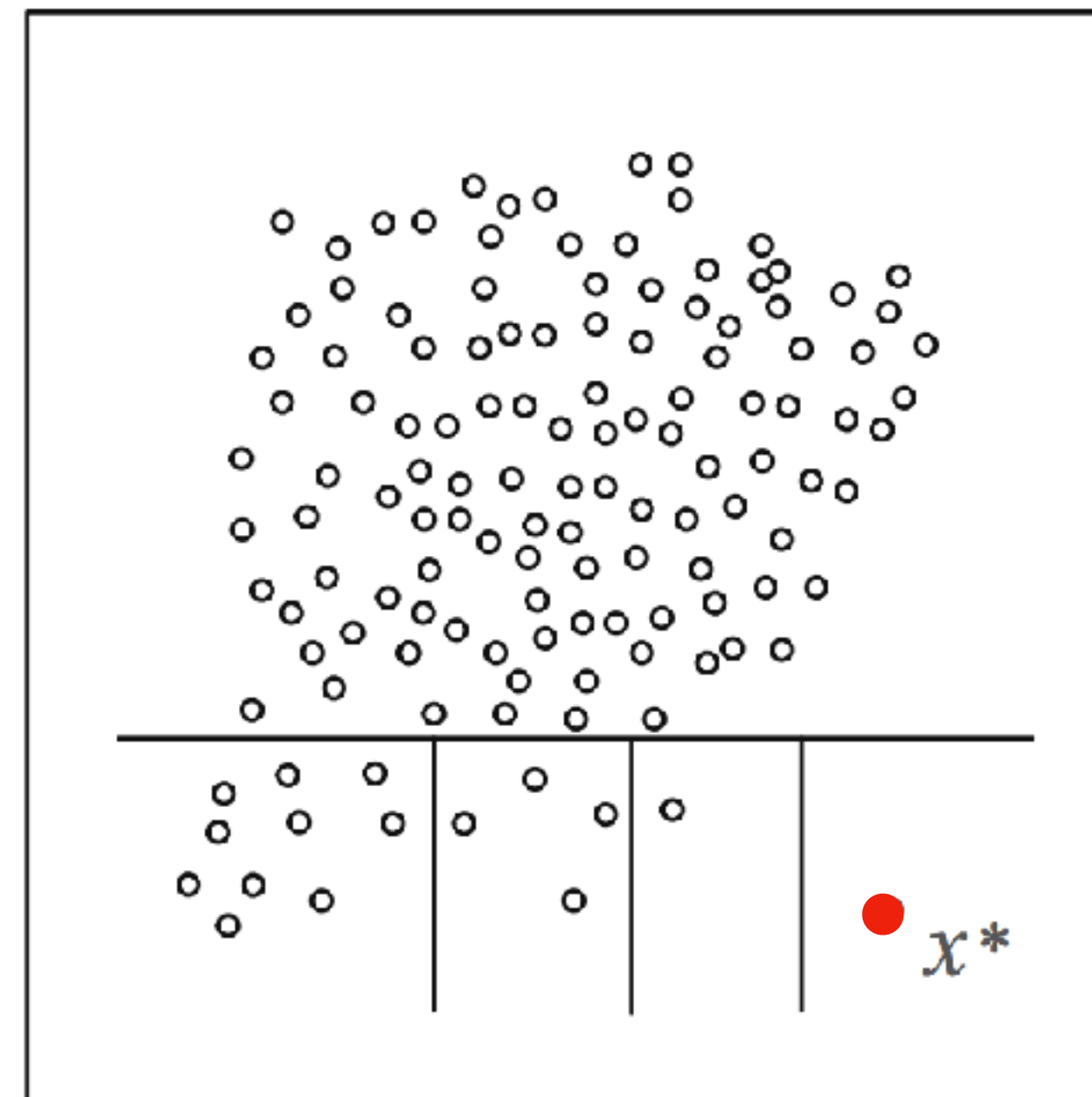
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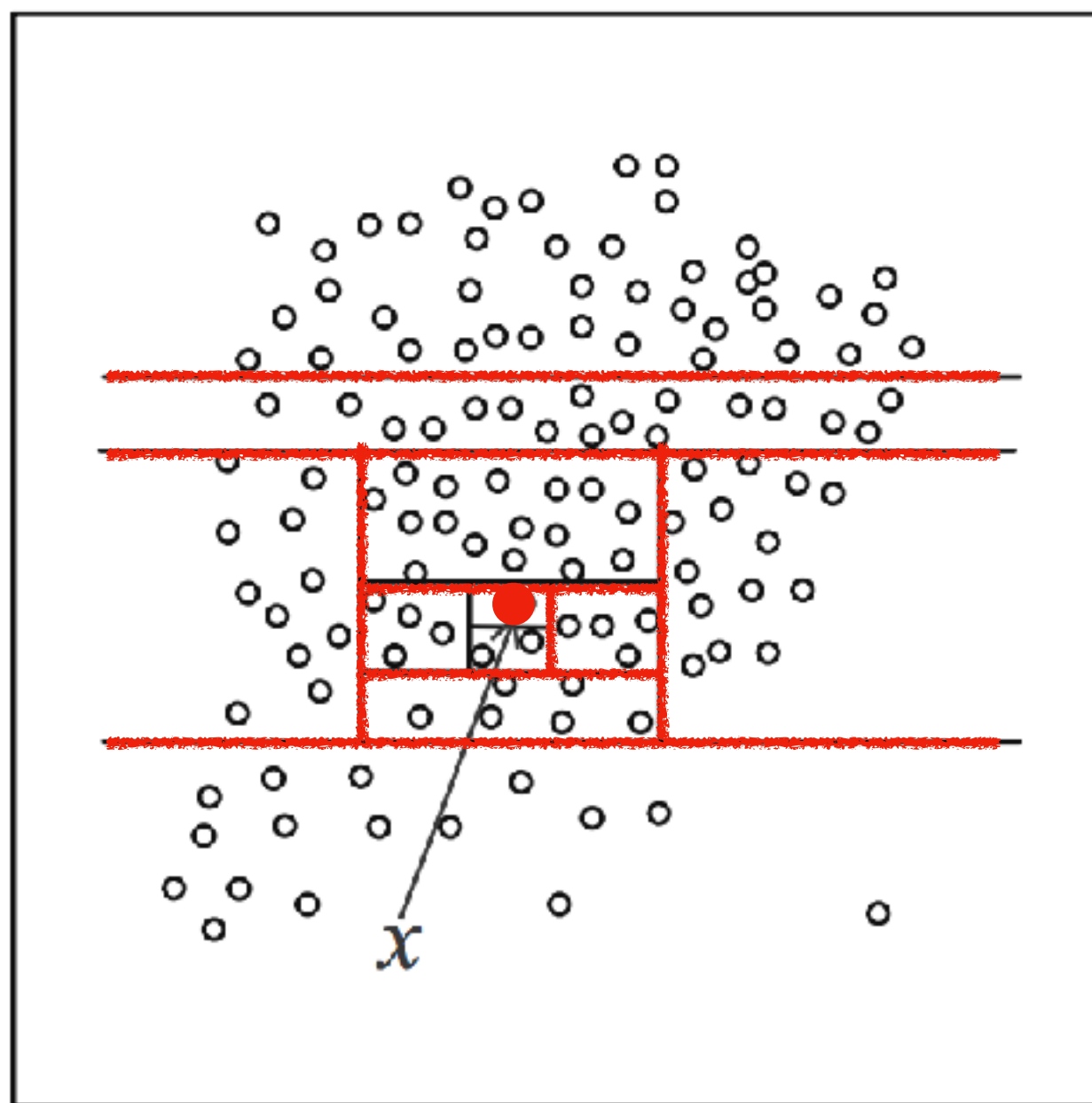
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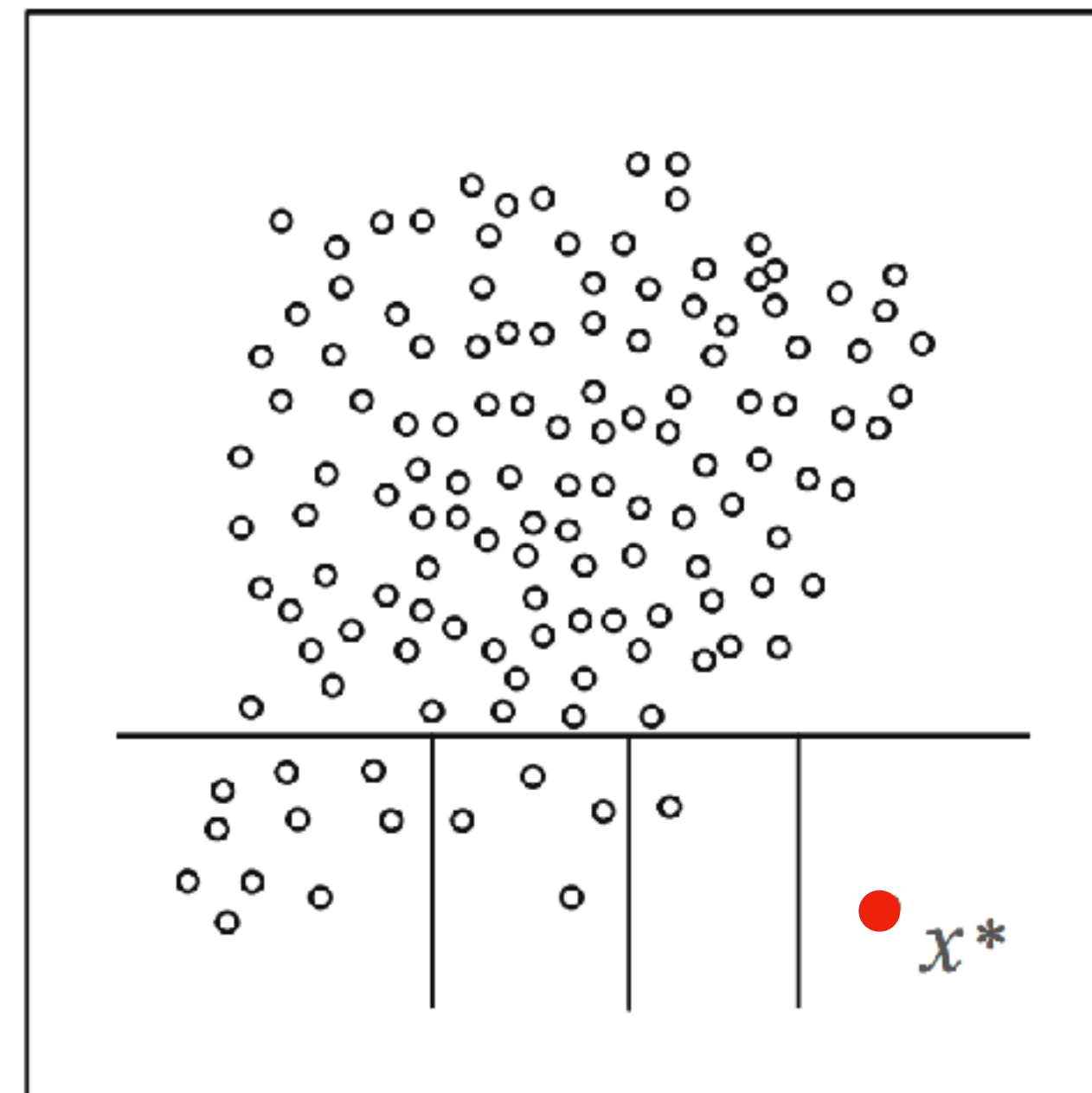
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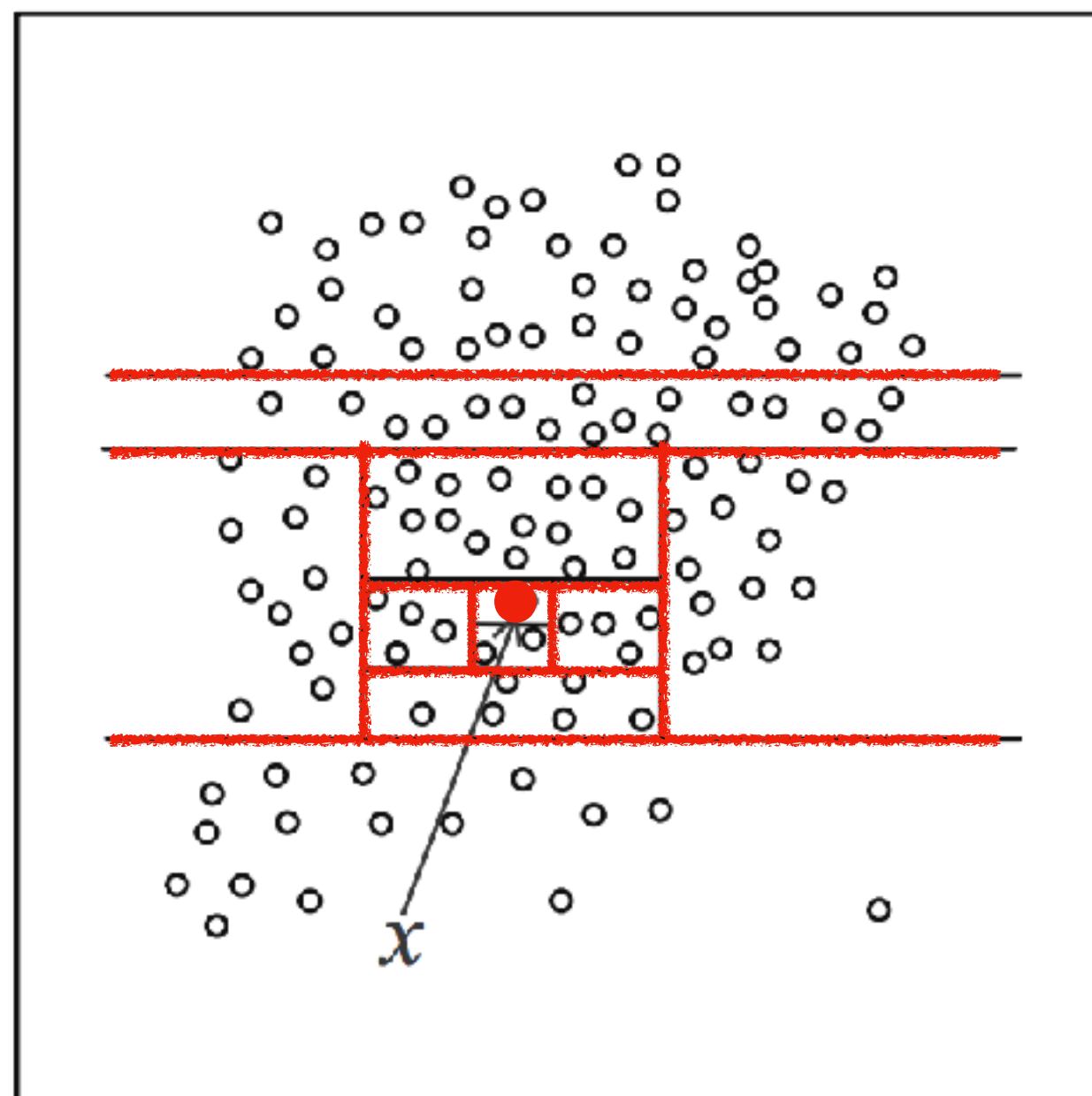
Isolation Forest, 2008

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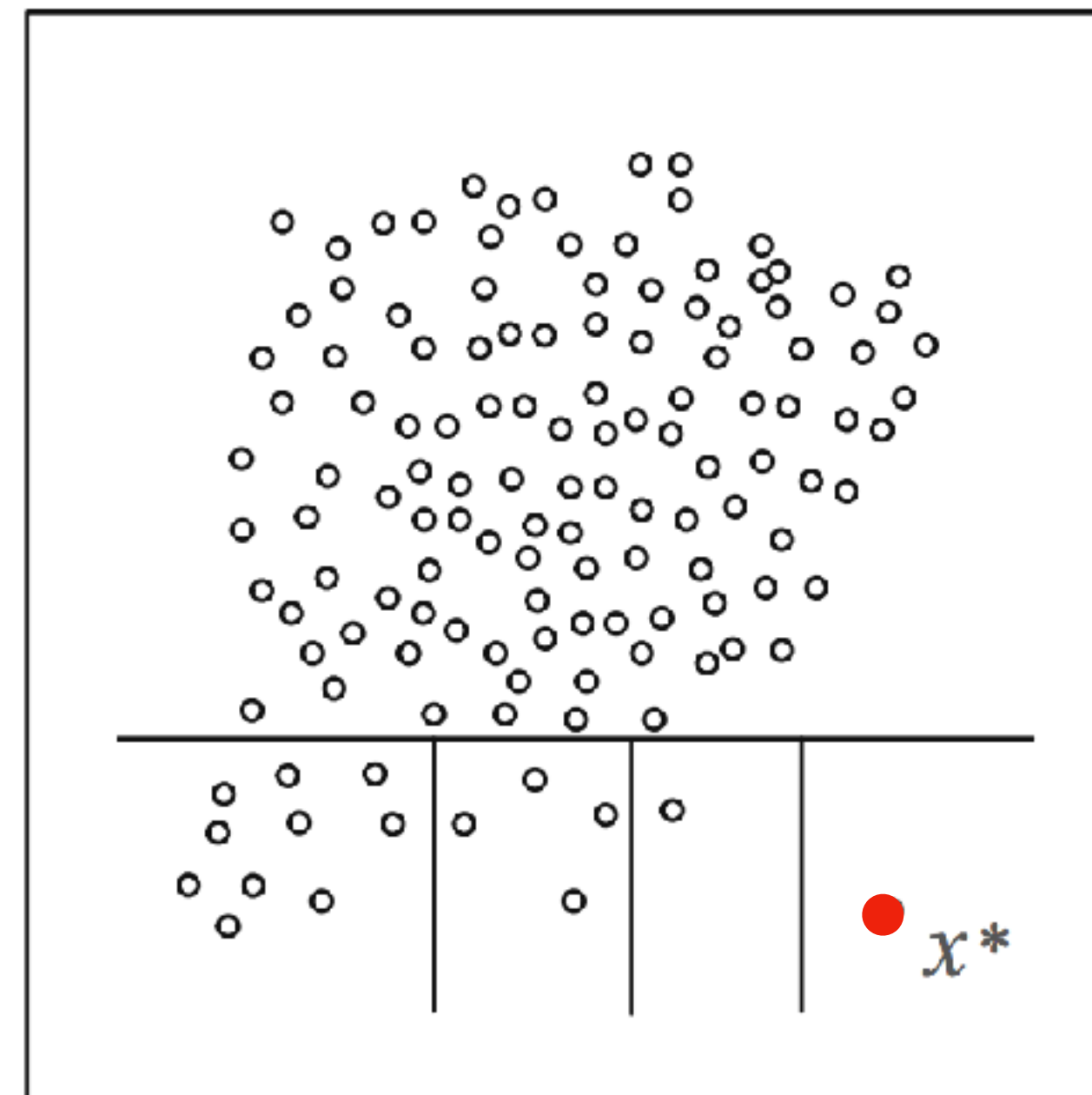
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Fei Tony Liu, Kai Ming Ting
Gippsland School of Information Technology
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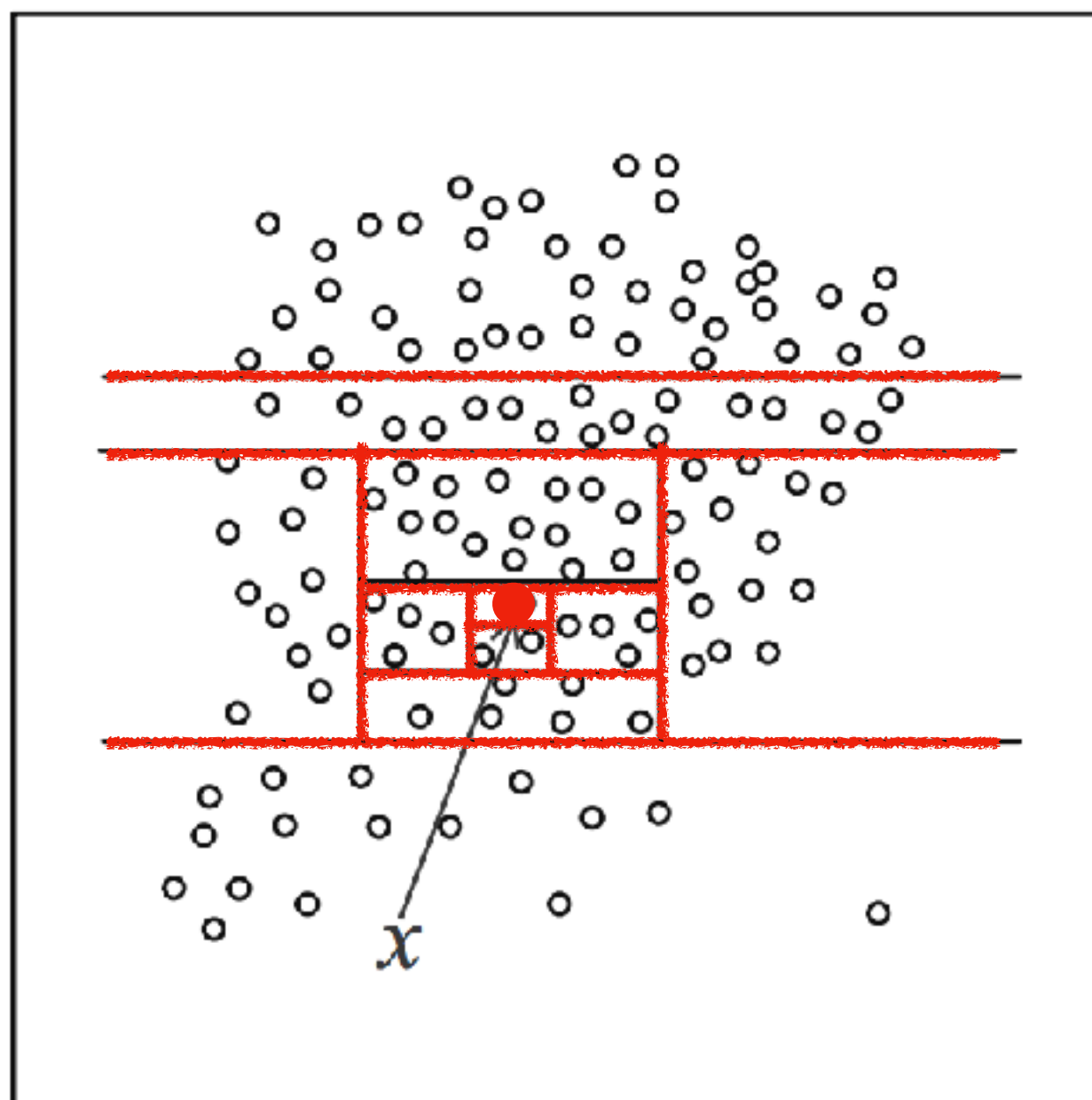
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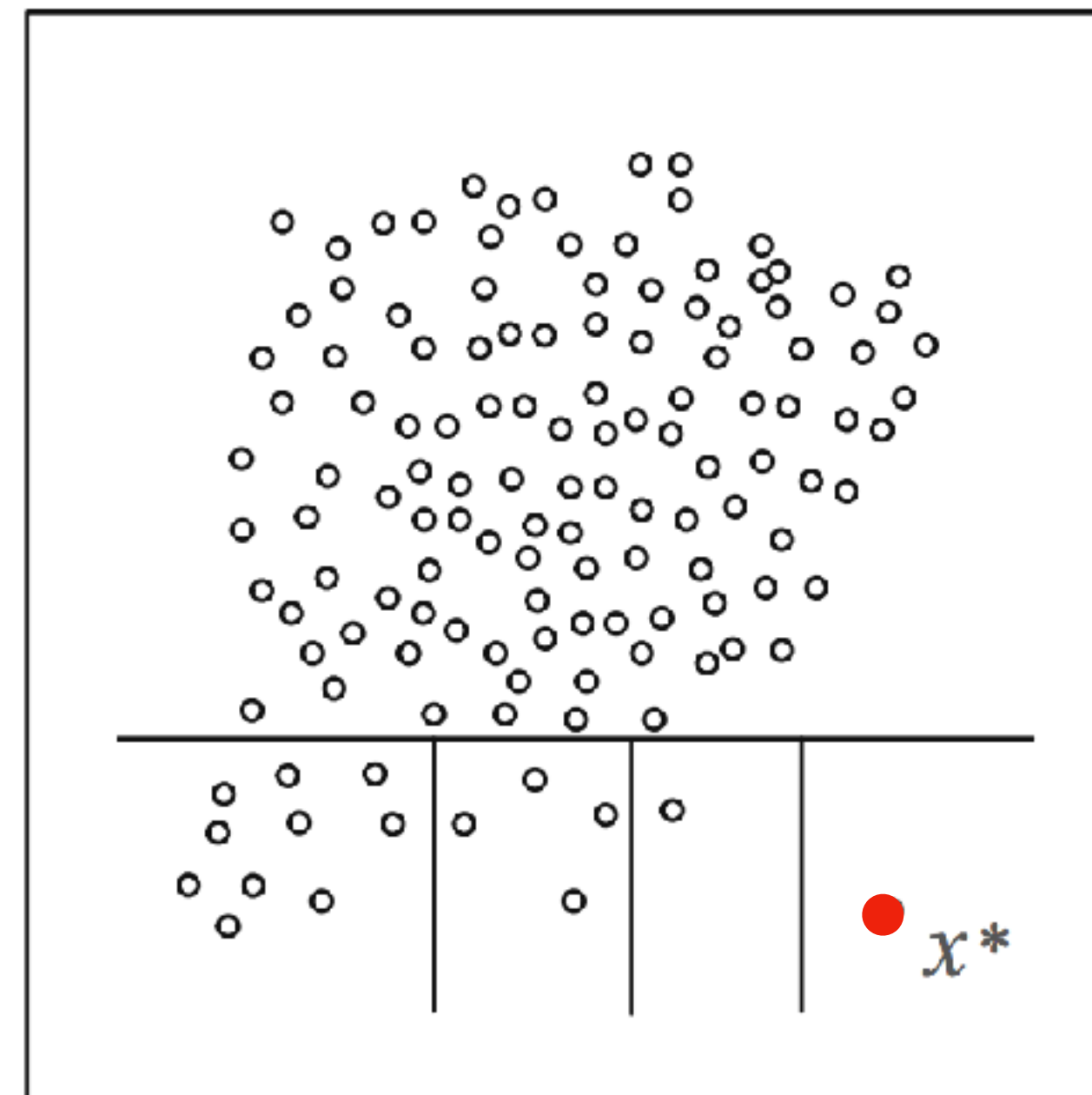
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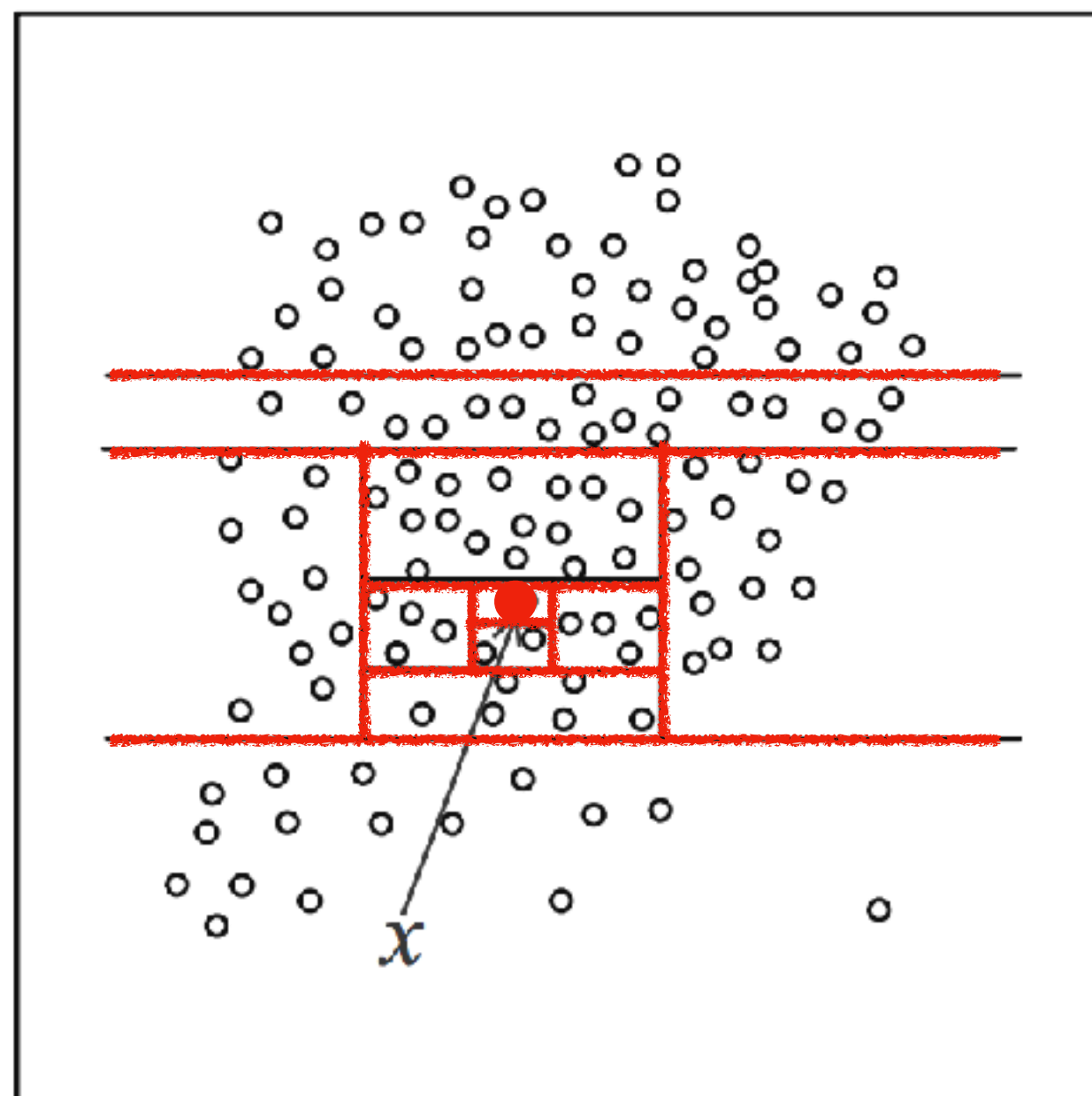
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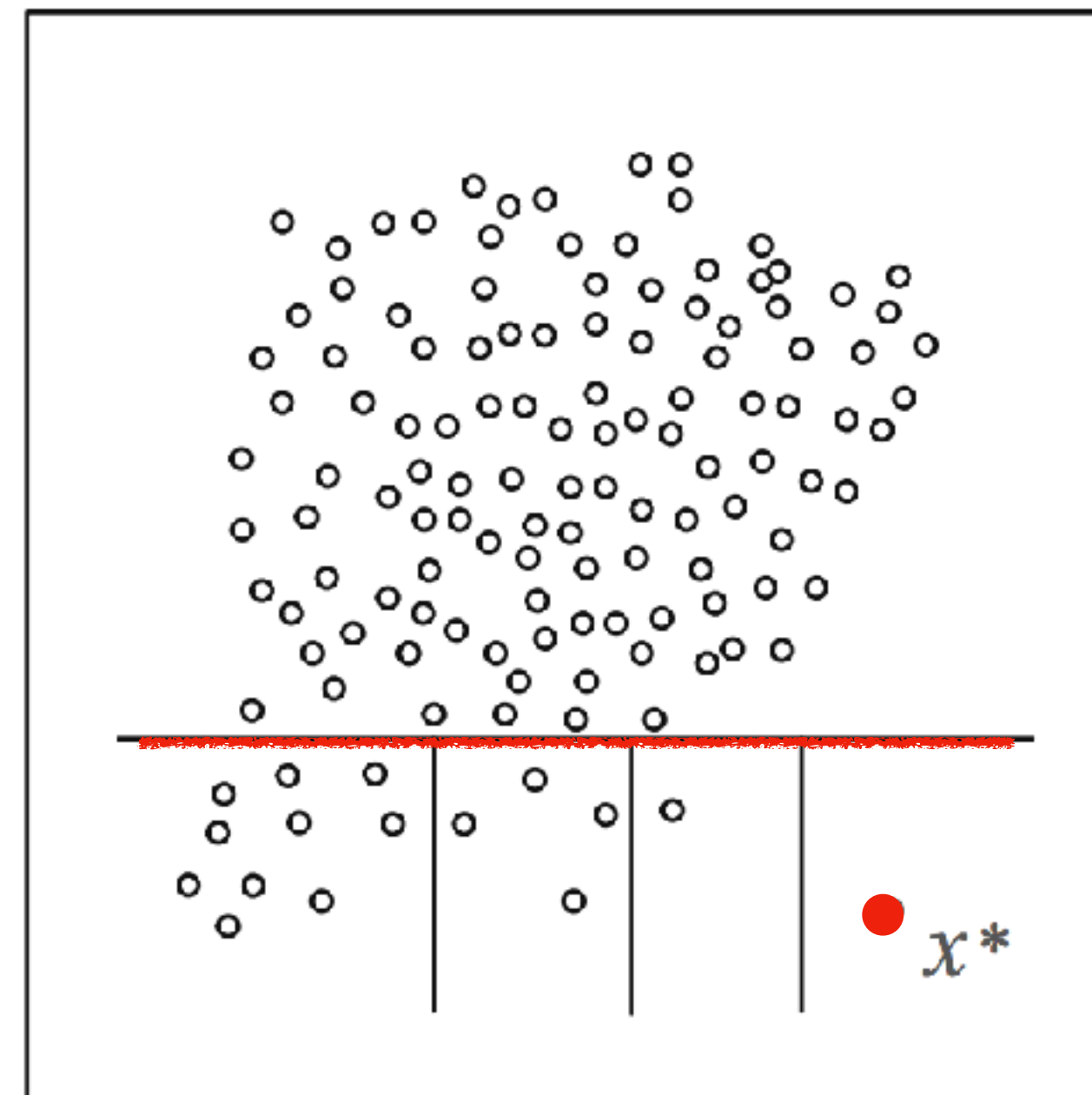
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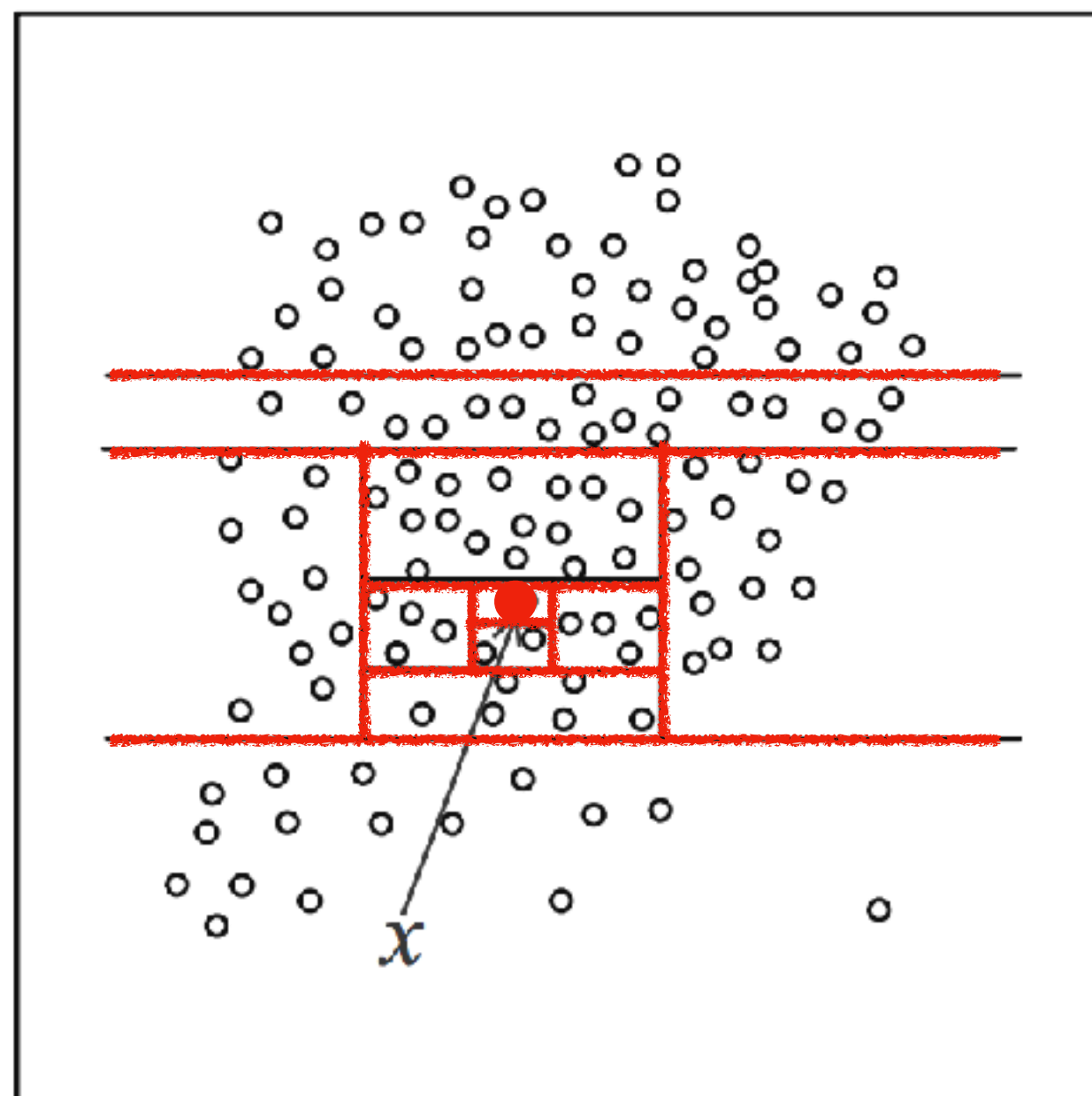
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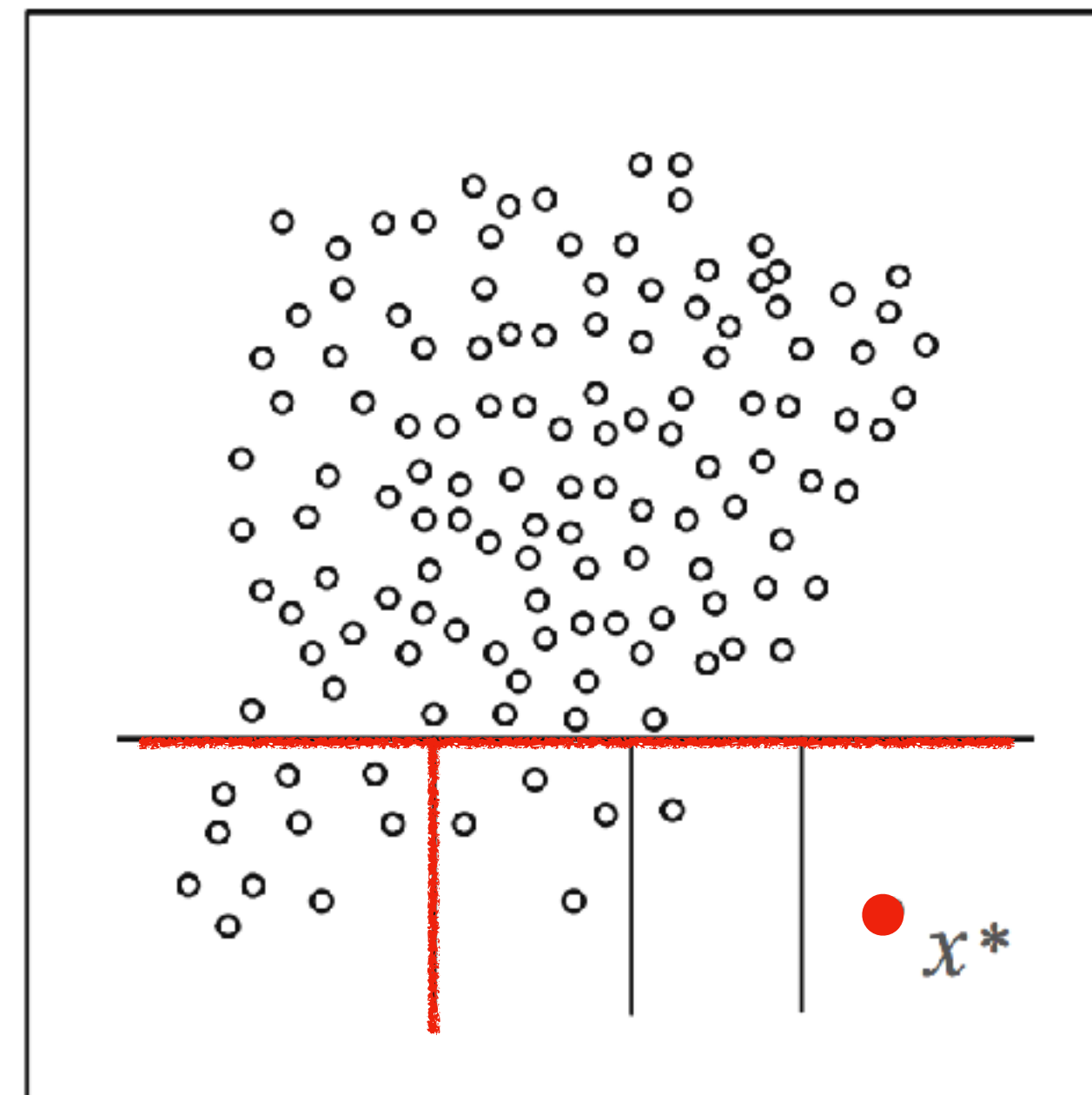
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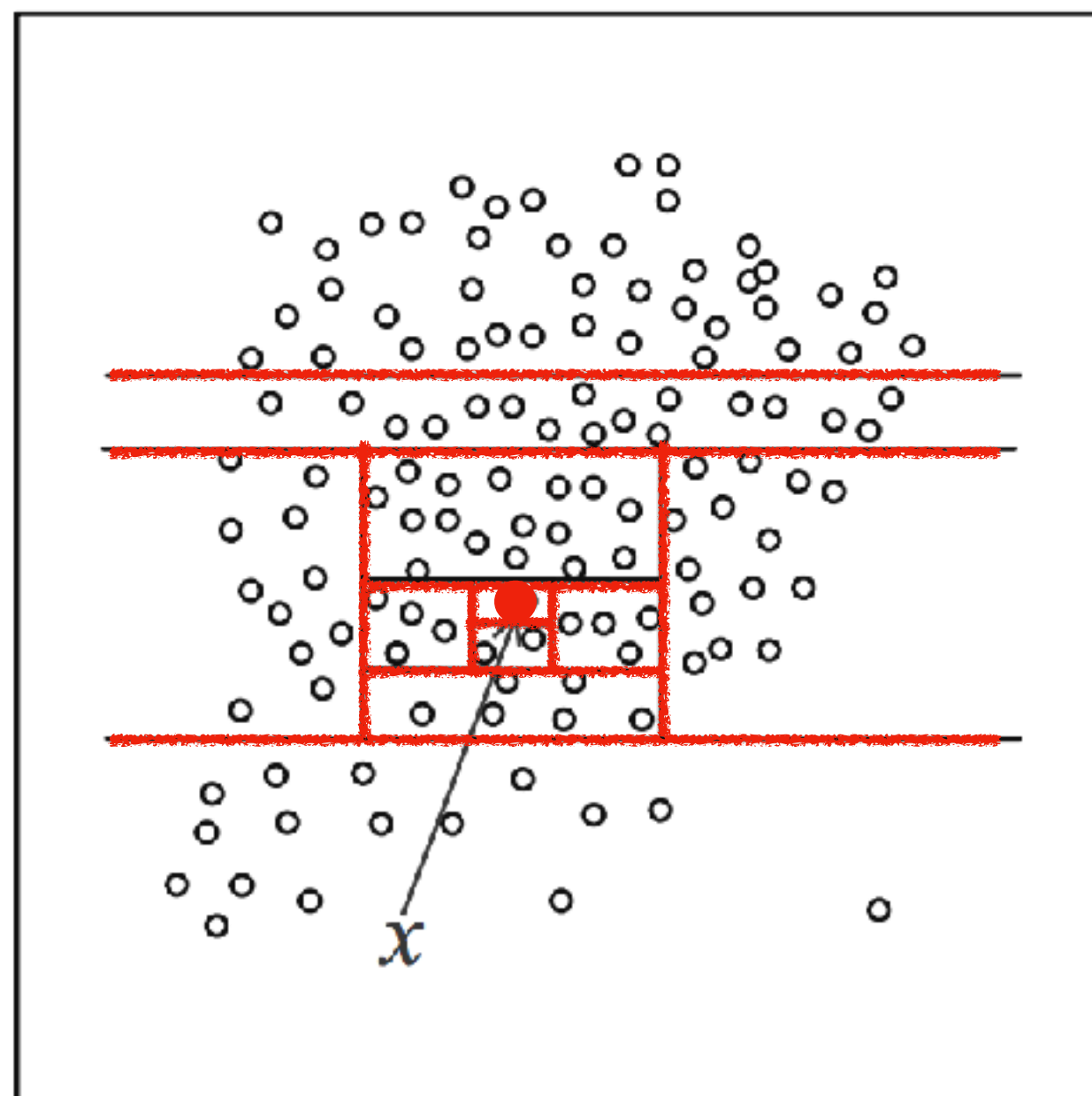
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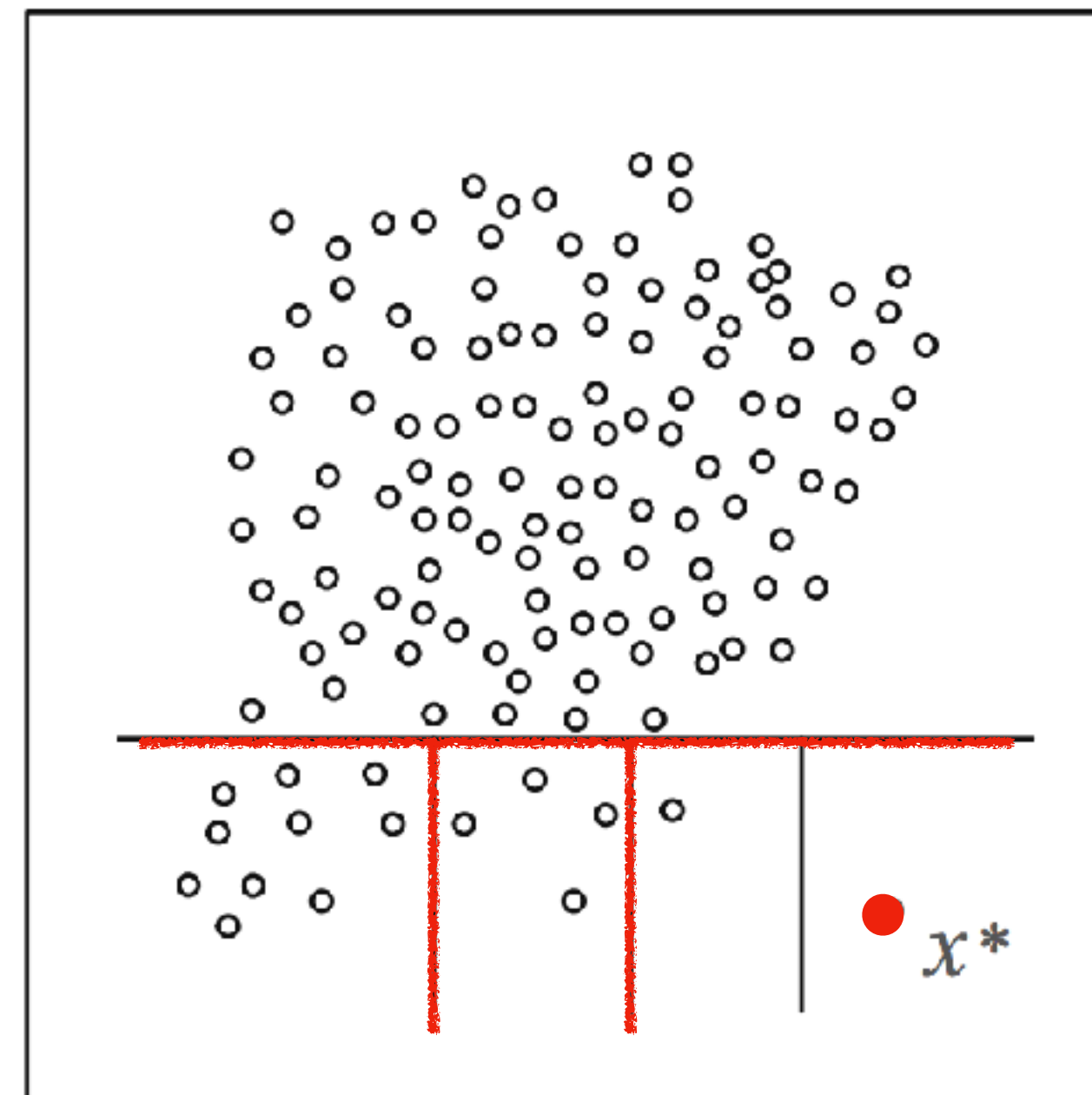
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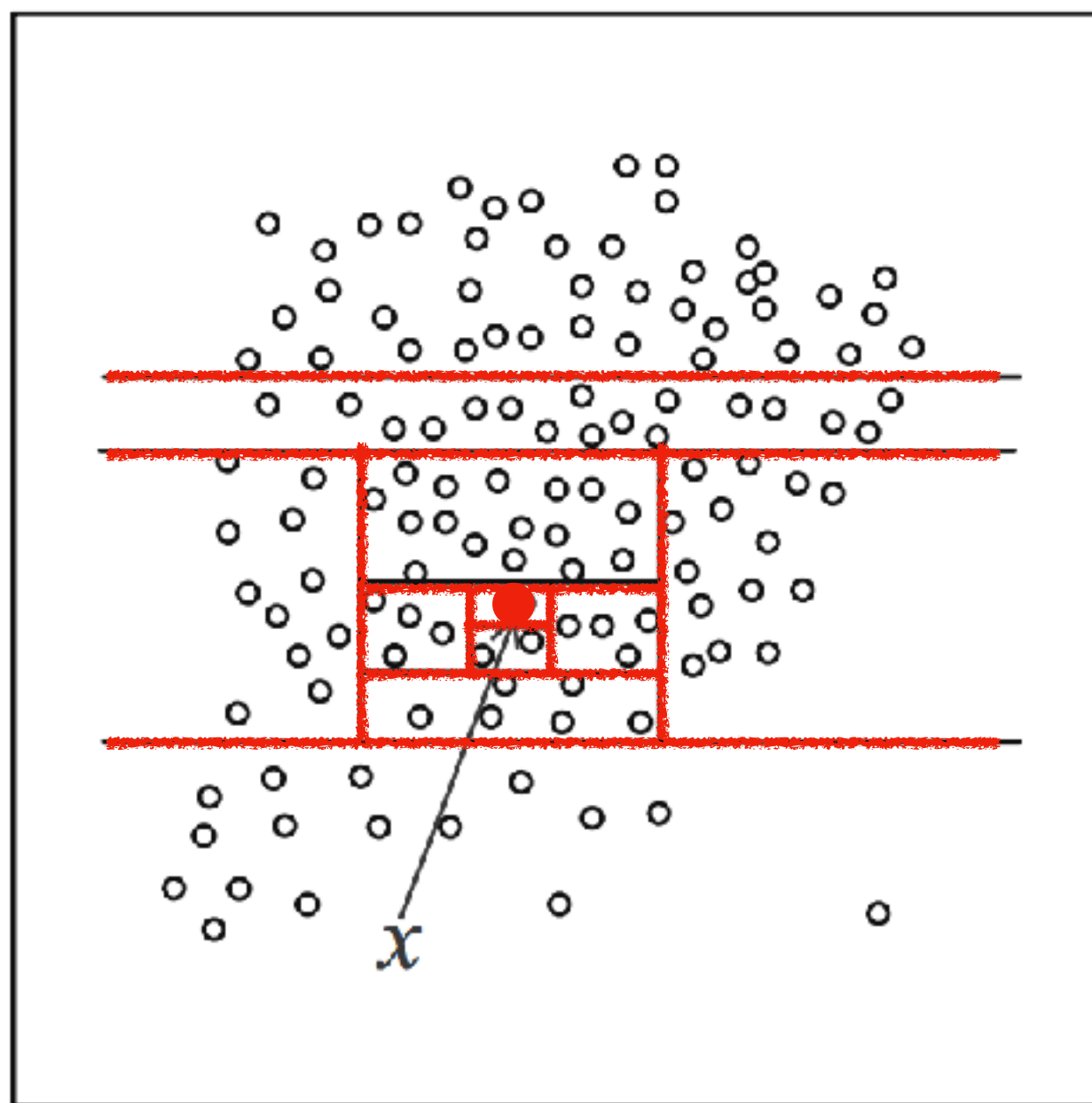
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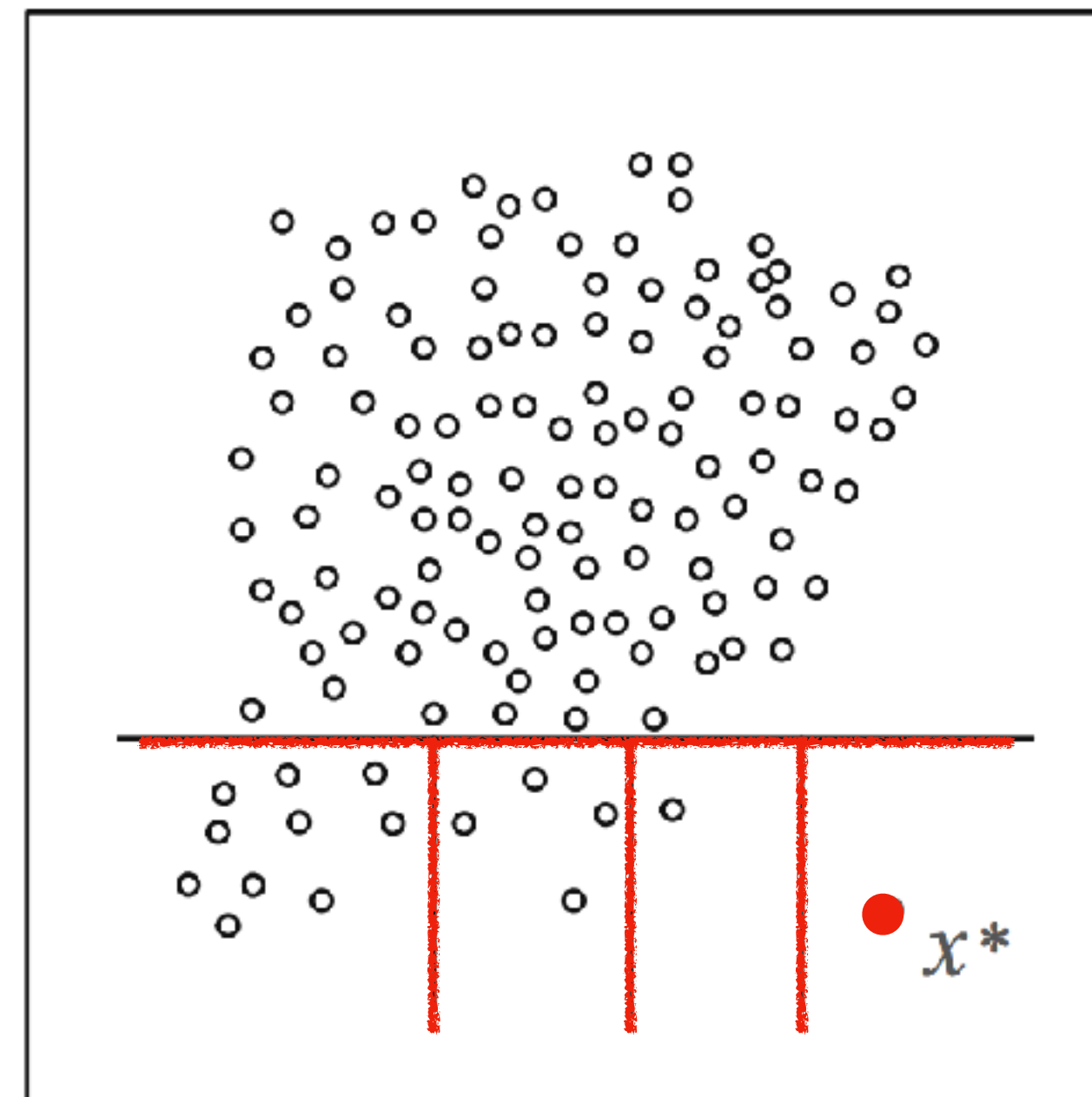
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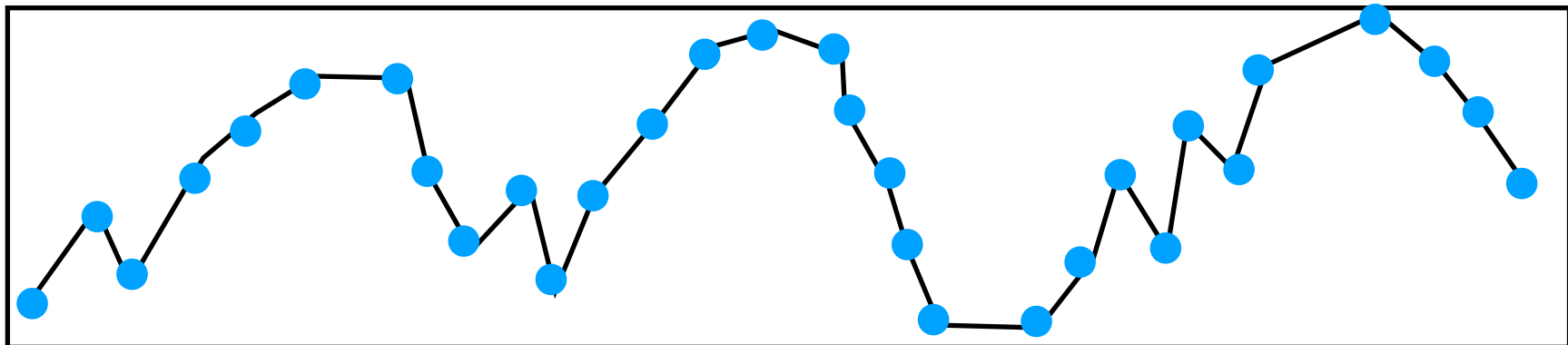
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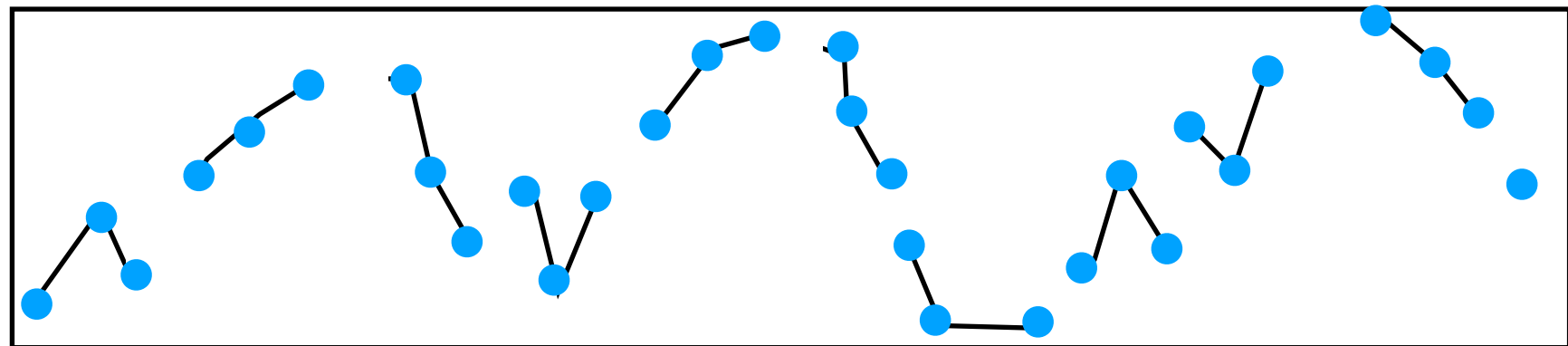
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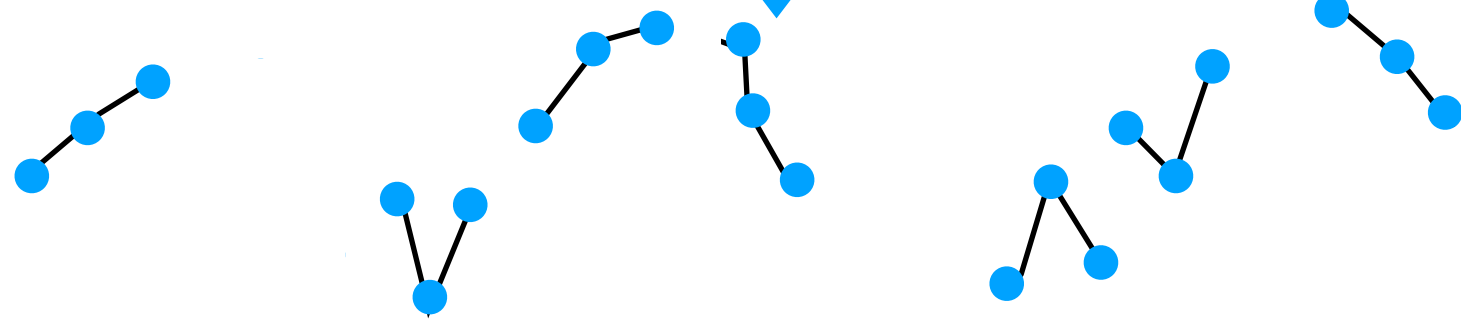
Random Cut Forest, 2016: Streaming



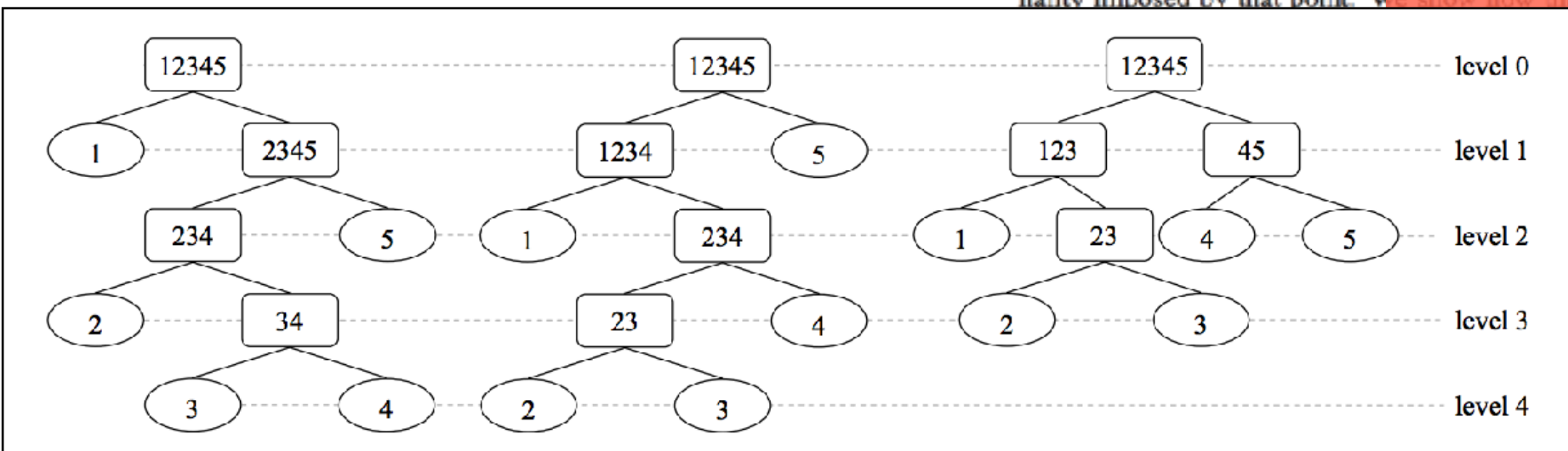
input data stream



3-sized shingles



reservoir sampling



RF training and inference:
incremental

Robust Random Cut Forest Based Anomaly Detection On Streams

Sudipto Guha
University of Pennsylvania, Philadelphia, PA 19104.
SUDIPTO@CIS.UPENN.EDU

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Gourav Roy
Amazon, Bangalore, India 560055.
GOURAVR@AMAZON.COM

Okke Schrijvers
Stanford University, Palo Alto, CA 94305.
OKKES@CS.STANFORD.EDU

Abstract

In this paper we focus on the anomaly detection problem for dynamic data streams through the lens of random cut forests. We investigate a robust random cut data structure that can be used as a sketch or synopsis of the input stream. We provide a plausible definition of anomalies based on the influence point on the remainder of the data, later called *externality*, and show how the

deployed in Kinesis Analytics, Quicksight and SageMaker

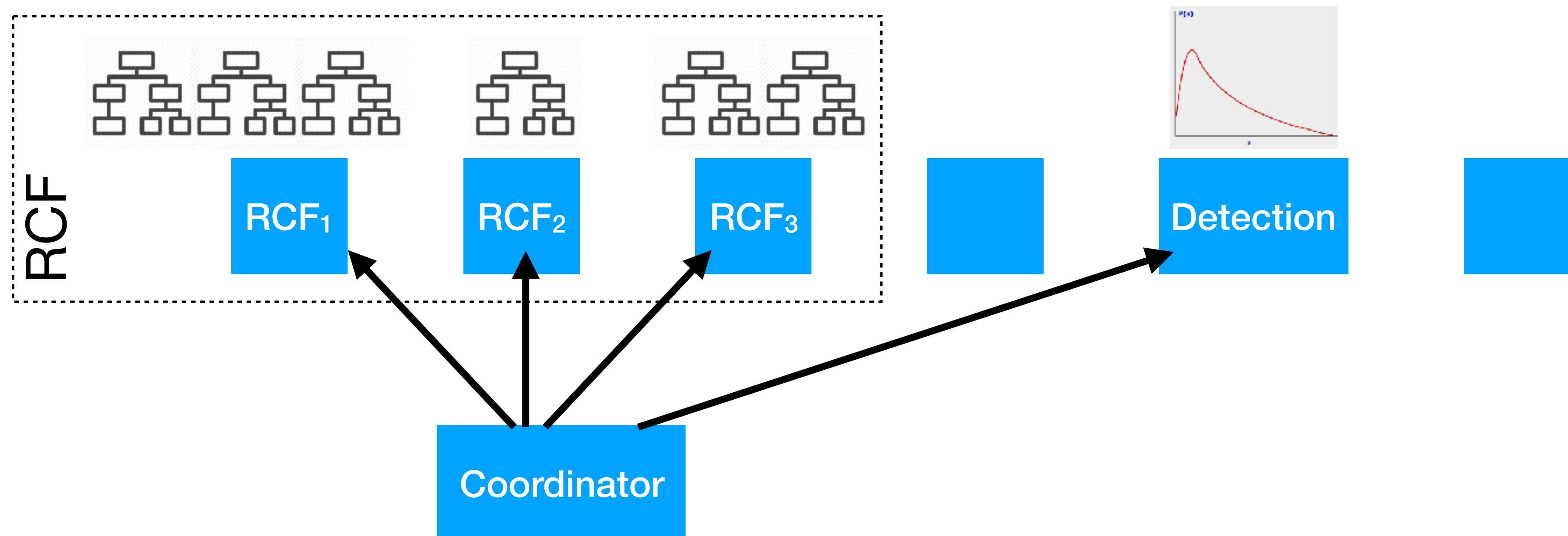
a point is data dependent and corresponds to the externality imposed by the point in explaining the remainder of the data. We extend this notion of externality to handle "outlier masking" that often arises from duplicates and near duplicate records. Note that the notion of model complexity has been a long standing question (1) to question (2) which we discuss in greater detail next. However it is worth noting that the notion of externality is a natural extension of the simpler context of static batch processing and (2) remains relevant in the batch situations well. This paper presents a randomized approach, akin to (Liu et al., 2012), due in part to the practical success reported in (Emmott et al., 2013). Randomization is a powerful tool and known to be valuable in supervised learning (Breiman, 2001). But its technical exploration in the context of anomaly detection is not well-understood and the same comment applies to the algorithm put forth in (Liu et al., 2012). Moreover that algorithm has several limitations in stream processing.

Open Questions

- RCF anomaly score quantifies incoming data
 - learn score distribution to adaptively find anomalies?
 - quantify severity and confidence?
- RCF requires (long) training window (hours with 1min samples)
 - cold start an RCF in seconds?

Model partitioning

- Features computed via queries
- RCF is embarrassingly parallel by definition
- AD coordinator node (same as alerting) uses hash-based partitioning
- elasticity: recompute hash-based assignments on new ring



Model elasticity

- Each node automatically resizes its RCF partition:
 - under heap pressure
 - compute not necessary (RCF is lightweight)
- Node-level circuit breaker (non-ES) shuts down local RCF
- delayed computations:
 - use interpolation when available



**actions impact
AD confidence;**

NOT availability

RFC for Anomaly Detection

- Coming soon!
- Contact: pak@amazon.com

Real Time Performance Diagnosis and Machine Learning

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All Things Open 2019

